Strategic Classification for Non-uniform Preferences using Penalty Labels and Randomisation

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Abstract

Strategic classification is the problem of classifying feature vectors that can be manipulated at the testing phase of the classifier. The classifier aims for its decision rule to be robust against strategic manipulation and also be efficiently learnable. In this paper, we present two main ideas for the classifier to achieve this goal. The first idea is to enrich the classifier's decision-making capabilities in two ways: ignoring feature vectors and making randomized decisions. This approach helps the classifier classify more feature vectors correctly by limiting the sender's options for manipulation. The second idea is to use the strategic structure of the problem during learning. This approach, in certain contexts, can simplify the learning process. Combining these ideas, we propose and compare two learning algorithms, Vanilla ERM and Strategy-aware ERM, and provide a heuristic for selecting the one that yields a classifier both robust to strategic manipulation and efficiently learnable.

Index Terms

Strategic Classification, Penalty Labels, Non-uniform preferences, Stackelberg equilibrium, Screening game

I. Introduction

Consider the following classification situation. A university assigns students to various programs based on their scores in an aptitude test using a decision rule that maps scores to programs. For example, one such rule could be to assign the Physics program to students who have scored well in the physics section of the exam, regardless of their total score. For transparency, the university must declare its decision rule before the students write the exam. Every student taking the exam has a *true aptitude* corresponding it with a *true program*. However, the students may have preferences for programs that may differ from their true programs. Knowing the decision rule, they can seek external help, such as coaching, to improve their scores, incurring some costs for such assistance. Now suppose the university offers two related programs (say theoretical and experimental physics programs), where the students with an aptitude in one program may prefer the other program and, therefore, prepare for the entrance exam accordingly. The university aims to create a question paper for the entrance exam in such a manner as to assign the maximum number of students to their true programs. The challenge is that the university cannot measure the true aptitude of a student and must rely on the exam scores, i.e., its *demonstrated aptitude*, to assign the program. Moreover, the university does not know the *true function* (which maps true aptitude to a true program) and must learn it from past data.

The university admission problem is an example of a *strategic* classification problem. The strategic classification problem extends the classification problem to the regime where, during the classifier's testing phase, a strategic data sender can manipulate feature vectors to obtain a preferred label while incurring some cost. A non-strategic or *naive classifier* places the decision boundary precisely at the class boundary. Since this classifier's decision rule is known to the strategic sender, it is prone to manipulation. A *strategic classifier* is robust to such manipulation and, in general, differs non-trivially from the naive classifier.

To the best of our knowledge, all previous works on the strategic classification problem assume that the sender has a uniform preference for the labels (e.g. [1]). The standard example of this setting is the

email classification problem, where the strategic sender prefers the non-spam label over the spam label, irrespective of the underlying email content. The strategic classifier for this case is the one that shifts the decision boundary into the *non-spam* region of the feature space, thereby making the criterion for classification as the non-spam label more stringent.

This paper considers a binary classification problem where the strategic sender has a possibly non-uniform preference for labels, as illustrated in the above example for admission to theoretical and experimental physics programs. This assumption poses a new challenge as preferred feature space for one class overlaps with the space not preferred by the other. In particular, the structure of the strategic classifier obtained by [1] no longer holds without a universally preferred label. The problem of constructing a strategic classifier with non-uniform preferences results in two subproblems: a *decision problem* and a *learning problem*. In the decision problem, the university knows the *true function* (map between the true aptitude and the true program) and has to calculate the strategic classifier from it. In the learning problem, the university does not know the true function and has to learn the strategic classifier from a finite sample of the true function.

Our main findings are that classification performance improves significantly when the classifier is either allowed to make randomized decisions or allowed to *not* classify certain feature vectors. Moreover, in certain problem instances, particularly those involving complex hypothesis classes, learning can be made easier by incorporating the sender's strategic behaviour in the classifier's learning phase.

A. Main findings

We adopt a game theoretic viewpoint for our analysis. We model the binary classification problem with non-uniform preferences as a screening game[1] between a sender and a receiver (classifier), with the receiver as the leader. The sender wants to maximize the difference between its utility and its cost and the receiver wants to maximize the probability of correctly classifying the feature vectors. The strategic classifier corresponds to the receiver's Stackelberg equilibrium strategy. For most of our results, we work with a scalar feature space. We show that under certain assumptions, the problem with the vector feature space with a linear class boundary can be reduced to the problem we consider.

We introduce two main ideas. The first is to expand the classifier's capabilities in two ways. One way is by introducing a new penalty label Δ, by which means the receiver ignores a feature vector assigned to this label. From the sender's perspective, this situation is equivalent to receiving a $-\infty$ utility for the label Δ. In the context of university admission problems, the student is the sender, and the university is the receiver. The exam paper comprises of a subset of questions from a question bank, with the requirement that the student has to attempt exactly one question. The university assigns a label to each question in the exam paper with the implication that a student who correctly answers that question gets assigned that label or program. Assignment of label Δ to a question amounts to *dropping* that question from the exam paper. This approach is in stark contrast to the uniform preference case, where *all* the questions in the question bank are included in the question paper, and only *cut-off marks* for the preferred program are increased. Ignoring feature vectors can also be considered a generalization of feature selection.

Another way to expand the classifier's action set is by allowing randomized classification. In the university admission problem, this is analogous to saying that solving a particular question is correlated (but not with certainty) with admission to a specific program. These ideas for expanding the classifier's action set are motivated from mechanism design, where penalties influence a strategic sender's behavior, and game theory, where randomization allows for richer outcomes.

The second main idea involves incorporating the underlying game-theoretic structure of the problem by modifying the hypothesis class used for learning. Learning from the original hypothesis class will be inefficient if the university's inductive bias does not consider students' strategic nature. We show that

[1]In a screening game, the receiver commits to a strategy first; the sender then observes this commitment and chooses a strategy in response, after which the receiver plays the committed strategy.

incorporating the game's strategic structure through hypothesis class modification can make an unlearnable hypothesis class learnable.

Using these two ideas, we address the decision and learning problems of strategic classification in the scalar feature space setting. For the decision problem, our first finding is that a deterministic strategic classifier using a penalty label makes fewer classification errors than any classifier that does not use a penalty label, at least strictly fewer errors in some instances of the problem. Furthermore, under the adversarial assumption, i.e., when the feature vector always prefers the label opposite to its true label, and assuming a uniform distribution over the feature space with the Euclidean cost function, a stochastic strategic classifier makes fewer errors than any deterministic classifier, including those that use a penalty label, and at least strictly fewer errors than a classifier with a penalty label in some instances of the problem.

However, the stochastic classifier uses only randomization on the original label set; using the penalty label does not increase its accuracy. Nonetheless, the penalty label becomes relevant when it is constrained to use a deterministic strategic classifier, a common situation encountered in real-life applications. Hence, the penalty label can be thought of as partially compensating the classifier for the absence of randomization.

The classifier described above assigns the label Δ to feature vectors near the class boundary. Indeed, multiple such classifiers exist, all with the same classification error, but they differ in the following three aspects. The first aspect is classifier complexity, measured by the number of decision boundaries. A decision boundary is a feature vector at which the classifier changes its assigned label. Thus, the number of decision boundaries corresponds to the number of times the classifier switches labels as the feature vector varies. We show that classifiers with fewer decision boundaries are easier to learn due to their lower VC dimension. The second aspect is classifier fairness, which refers to fair distribution of errors across classes. If in addition to classification performance, fairness is also in consideration, then the receiver can pick among the many classification rules, the one which proportionately distributes the error across classes. The third aspect is the sender's payoff under a strategic classifier. This aspect becomes relevant if a sender can opt-out and obtain a reservation payoff. Thus, a receiver may select a classifier that gives the sender a payoff higher than its reservation payoff. Although we work with scalar features, our problem suffices to demonstrate the structural aspects of strategic classification and the importance of a penalty label. Moreover, in many university admission problems like those we consider, the final admission decision has to be done based on a single scalar score, whereby the scalar problem is practically relevant.

We introduce two new algorithms called vanilla ERM and strategy-aware ERM to address the learning problem. In vanilla ERM, the classifier first uses ERM to learn the true function from the original hypothesis class. The decision problem is then applied to the learned hypothesis. In the strategy-aware ERM approach, the hypothesis class is modified to incorporate the underlying game-theoretic structure. This modification involves a two-step process. First, each hypothesis in the original class is replaced with its corresponding strategic classifier, which is obtained by applying the decision problem to each hypothesis. Second, the strategic classifier obtained in the previous step is further replaced with a new hypothesis, formed by composing the strategic classifier with the sender's worst-case best response. The classifier then uses the Empirical Risk Minimization (ERM) algorithm to learn from this modified hypothesis class. What makes strategy-aware ERM novel is its ability to transform an otherwise unlearnable problem into a learnable one. We show that there exist hypothesis classes that are unlearnable under vanilla ERM but become learnable with strategy-aware ERM, illustrating how incorporating game-theoretic structure can fundamentally improve learnability.

While strategy-aware ERM can learn from a broader range of hypothesis classes, its advantage in the ease of learning over vanilla ERM only sometimes holds. This ease of learning is controlled by the VC dimension of the hypothesis class, which in turn determines the sample complexity in PAC learning. The comparison, therefore, becomes context-dependent if a hypothesis class is learnable via vanilla ERM. We provide a heuristic argument that strategy-aware ERM generally makes learning easier if the original hypothesis class is too complex, as it can reduce the effective VC dimension. On the other hand, vanilla ERM proves more efficient if the original hypothesis class is simple. We show this with the help of an

example. A formal result, even for scalar features, remains elusive.

B. Related Works

The strategic classification problem has been a topic of interest in recent years. Works such as [2], [3], [4], [5], [6], [7] explicitly formulate the strategic classification problem as a game between a strategic sender and a classifier. In our work, we model the interaction between the strategic sender and the classifier as a sender-receiver screening game. We introduce a penalty label, which expands the receiver's strategy space and gives rise to equilibrium strategies that are robust against the sender's manipulation. To the best of our knowledge, no such penalty label has been considered in the literature so far.

Works such as [1], [8], [9], [10] consider the impact of a strategic sender on the PAC learnability of the hypothesis class. In [1], [8], [9], the sender is assumed to have a uniform preference for positive labels, whereas in [10], the sender is assumed to have a non-uniform preference for labels for each feature vector. In [1], [8], [10], strategic manipulation incurs a cost, whereas in [9], strategic manipulation does not incur a cost. The works [1], [8], [9], [10] do not explicitly utilize the underlying sender-receiver game in the learning process. The learning process in these works can be equivalently described as using the ERM algorithm to optimize over a modified hypothesis class, obtained by replacing each hypothesis with the composition of the hypothesis and the sender's best response to that hypothesis. In contrast, in our work, the sender has a non-uniform preference over labels for each feature vector, and the structure of this preference differs from that in [10]. Furthermore, in our work, there is a cost incurred for modifying a feature vector. We explicitly make use of the underlying sender-receiver game in the learning process by allowing the ERM algorithm to optimize over a modified hypothesis class, which is obtained by replacing each hypothesis with its corresponding Stackelberg equilibrium composed with the sender's best response.

There are other interesting formulations of the problem that we do not consider in our work. For instance, some research explores scenarios where the sender lacks knowledge of the classifier. Notable works in this area include [3], [4], and [5]. Additionally, other studies introduce more structure to the sender by incorporating user types, as seen in [6] and [11].

Finally, the social welfare implications of strategic classification have been explored in works such as [12], which analyze the social costs imposed on the sender. In our work, we briefly address related issues such as fairness and participation constraints in Section III-F.

II. Problem Formulation

In this section, we formulate the strategic classification problem. Recall that in strategic classification, the sender can alter the feature vector at a cost during the classifier's testing phase to obtain a preferred label.

We formalize the strategic classification problem within this framework. Let $\mathcal{X}$ be the set of feature vectors; for simplicity we take $\mathcal{X} := [0, 1]$. Consider an abstract probability space and let X be a $\mathcal{X}$-valued random variable defined on that space. Let $\mathcal{L}(X)$ denote the probability law of X on $\mathcal{X}$. Let $\mathfrak{D}$ be the collection of laws of X such that the corresponding cumulative distribution function, denoted as $\psi(\cdots)$, is differentiable and has a differentiable inverse except at finitely many points. We use D to represent an element of $\mathfrak{D}$. Let $\mathcal{Y} := \{-, +\}$ be the set of labels. Let $h : \mathcal{X} \to \mathcal{Y}$ be a *true function* that maps each feature $x \in \mathcal{X}$ to its corresponding *true label* $y \in \mathcal{Y}$. We assume the true function to be given by

$$h(x) := \begin{cases} -, & x \in [0, \mathbf{h}] \\ +, & x \in (\mathbf{h}, 1] \end{cases}, \tag{1}$$

where $0 \leq \mathbf{h} \leq 1$. We refer to $\mathbf{h}$ as the class boundary. We call a label a *majority class* label if the corresponding feature vectors whose true label is the said label has a probability (under probability law $D \in \mathfrak{D}$) greater than or equal to half. The other label is called the *minority label*.

We model the strategic classification problem as a game between two players, a sender and a receiver (classifier), who engage in a sender-receiver game with the receiver acting as the leader. The receiver

is interested in correctly classifying the feature vectors, whereas the sender is interested in obtaining a preferred label for each feature vector by potentially misreporting the feature vector. The gameplay proceeds as follows. The sender observes a realized feature vector $x \in \mathcal{X}$ drawn according to a probability law $D \in \mathfrak{D}$ and modifies it to $s(x) \in \mathcal{X}$ according to a rule s. It then communicates $s(x)$ to the receiver, who then classifies it with a label $r(s(x))$ according to a rule r.

The sender's strategy set for modifying feature vectors is given by

$$\mathsf{S} := \{s \mid s : \mathcal{X} \to \mathcal{X}\}$$

and the receiver's strategy set for classification is given by

$$\mathsf{R} := \{r \mid r : \mathcal{X} \to \mathcal{Y} \cup \Delta\}.$$

Here, label Δ is an additional label, which we call the penalty label, whose interpretation will be provided later in this section. We refer to the set $\{x : r(x) = \Delta\}$ of feature vectors with label Δ as the Δ-region.

We analyze the above game as a screening game; the receiver first commits a strategy $r \in \mathsf{R}$, and then the sender, who knows the committed strategy, plays a strategy $s \in \mathsf{S}$, and finally the receiver plays his committed strategy r. Notice the asymmetry in the role of the sender and the receiver. The receiver can perform classification but cannot access the private information of the sender (the feature vector x). In contrast, the sender has access to private information but can not do classification and has to depend on the receiver to get his (preferred) label.

The sender's preference for the label for each feature vector is represented by a utility function $U : (\mathcal{Y} \cup \Delta) \times \mathcal{X} \to \mathbb{R}$. We assume it to be given by

$$U(y, x) := \begin{cases} \mathbf{u}_+ & x \in [0, \mathbf{h}], y = + \\ \mathbf{u}_- & x \in (\mathbf{h}, 1], y = - \\ 0 & y = h(x) \\ -\infty & y = \Delta \end{cases}, \tag{2}$$

where $\mathbf{u}_-, \mathbf{u}_+ \in \mathbb{R}$. Note that our choice of the utility function is more general than the one considered in [1], as it allows the sender to have a non-uniform preference for labels at each feature vector, making the receiver's task more difficult. The cost incurred by the sender when modifying x to x' is given by $C(x, x') = |\phi(x') - \phi(x)|$, where the function $\phi : \mathcal{X} \to [0, 1]$, called cost kernel, is a monotonically increasing function such that $\phi(0) = 0$ and $\phi(1) = 1$.

In the strategic classification problem, for a strategy (r, s) and a feature vector $x \in \mathcal{X}$, the sender's payoff at x is given by

$$\mathbb{U}(x; r, s) := U(r(s(x)), x) - C(s(x), x).$$

For a probability law $D \in \mathfrak{D}$ on $\mathcal{X}$, the sender overall payoff, which is wants to maximize, is given by

$$\mathcal{U}(r, s) := \mathbb{E}_{x \sim D}[\mathbb{U}(x; r, s)].$$

For a strategy (r, s) and a function $f : \mathcal{X} \to \mathcal{Y}$, the error set $E(r, s; f)$ be defined as

$$E(r, s; f) := \{x \in \mathcal{X} : r(s(x)) \neq f(x).\} \tag{3}$$

For a probability law $D \in \mathfrak{D}$ with a differentiable distribution function $\psi(x)$, the receiver's error, which it wants to minimize, is given by

$$\mathcal{E}(r, s; h, D) = D(E(r, s; h)) = \int_0^1 \mathbb{1}_{E(r,s;h)} \psi'(x) dx.$$

Here $\mathbb{1}_A$ denotes an indicator function on the set A and $\psi'(x) = \frac{d\psi(x)}{dx}$

Remark II.1. *We assume the set $\{x \in \mathcal{X} | r(s(x)) \neq h(x)\}$ is measurable w.r.t. D.*

A. Interpretation of the penalty label

We now provide an interpretation of the penalty label. If the receiver classifies a feature vector x as Δ, i.e., $r(x) = \Delta$, then the receiver ignores the feature vector x. We assume that ignoring x results in an error for the receiver. From (2), $U(\Delta) = -\infty$, which implies that the penalty label is the least preferred label for the sender.

The receiver can exploit this fact to improve its classification accuracy. This property, where the receiver ignores certain feature vectors, is one of the novel contributions of this paper. It is not considered in [1], and we later show that it helps the receiver reduce overall misclassification.

B. Solution concept

We now describe the solution concept of our problem. For a given receiver strategy r, the best response strategies of the sender are defined as follows.

Definition II.1. Let r be a receiver strategy. Let $B(r)$ be the set of sender strategies s which satisfy

$$s(x) \in \arg\max_{x' \in \mathcal{X}} U(r(x'), x) - C(x', x) \; \forall x \in \mathcal{X}.$$

The set $B(r)$ is called the best response set.

There can be multiple best responses. We assume that the receiver anticipates that the sender will play a worst-case best response strategy, which is defined as follows.

Definition II.2 (Worst case best response)**.** A sender strategy s_r is called a worst-case best response to a receiver strategy r if

$$s_r \in \arg\min_{s' \in B(r)} D(\{x : r(s'(x)) = h(x)\}) \qquad \text{and}$$
$$\{x : r(s'(x)) = h(x)\} \not\subset \{x : r(s_r(x)) = h(x)\} \qquad \text{for all} \quad s' \in B(r).$$

Remark II.2. *We always assume the receiver's strategy set* R *is such that* $B(r)$ *is non-empty and* s_r *is well defined.*

Notation. We will generally drop various arguments in $\mathcal{E}(\cdot, \cdot, \cdot, \cdot)$ when they are apparent from the context. Specifically, when h and ψ are clear from the context, will sometimes be denoted as $\mathcal{E}(r) := \mathcal{E}(r, s_r; h, \psi)$.

The sender-receiver pair of strategies $(r, s_r), r \in \mathsf{R}$ which minimises the receiver's error is called a Stackelberg equilibrium strategy or strategic classifier and is defined as follows.

Definition II.3 (D-Stackelberg equilibrium)**.** Let D be a probability law on $\mathcal{X}$ and $h(x)$ be a true function. A receiver strategy $r^*_{h,D} \in \mathsf{R}$ is called a D-Stackelberg equilibrium strategy if

$$r^\star_{h,D} := \arg\min_{r \in \mathsf{R}} \mathcal{E}(r, s_r; h, D)$$

and $\mathcal{E}(r, s_r; h, D)$ is called the Stackelberg error.

Notation. Sometimes, we will drop the subscript from $r_{h;D}$ if it is clear from the context.

Notation. For a Stackelberg equilibrium (r, s_r), since the sender strategy is directly derived from the receiver strategy r, we sometimes refer to r alone as the Stackelberg equilibrium.

We now consider the two subproblems of strategic classification: the *decision problem* and the *learning problem*. The setup discussed so far in this section remains common to both of them. In the decision problem, both the sender and the receiver know the true function h, and the receiver has to calculate the Stackelberg equilibrium from it. In the learning problem, the receiver does not know the true function h. However, the receiver has access to a hypothesis class $\mathcal{F}$ and a sample $S = \{(x_i, h(x_i))\}_{i=1}^m$ from which it has to estimate the Stackelberg equilibrium. Here, x_i's are the samples which are drawn in an iid manner from a probability law D on $\mathcal{X}$ and $h(x_i)$'s are the corresponding true labels.

III. Decision Problem

In this section, we will characterise the Stackelberg equilibrium for the decision problem.

A. Motivating example: discrete feature space

To motivate the solution of the decision problem, we digress and recall a related problem from [13]. The problem formulation is almost the same as described in Section II, and we only highlight the differences. Let $\mathcal{X} = \{1, 2, \cdots, n\}$ be a finite set, and $h : \mathcal{X} \to \mathcal{Y}$ be the true function. Let $U : \mathcal{Y} \times \mathcal{X} \to \mathbb{R}$ be the utility function such that $U(h(x), x) = 0$ and $C : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be the cost function such that $C(x, x) = 0$. We now characterise the Stackelberg equilibrium strategy in terms of the independence number of a sender graph. The *sender graph* is defined as follows.

Definition III.1. Let $G_h = (V, E)$ be a graph whose node set V is given by $V = \mathcal{X}^2$. The edge set $E \subset \mathcal{X}^2 \times \mathcal{X}^2$ is described as follows: There is an edge between $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ if $h(\mathbf{i}) \neq h(\mathbf{j})$ and one of the following conditions holds:

(a) $\mathbf{k} = \mathbf{l}$.

(b)
$$U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0 \quad \text{or}$$
$$U(h(\mathbf{i}), \mathbf{j}) - C(\mathbf{k}, \mathbf{j}) + C(\mathbf{l}, \mathbf{j}) \geq 0.$$

Recall that a subset of nodes in a graph is called an independent set if no edge connects any two nodes in that subset. The next theorem characterises the Stackelberg equilibrium strategy in terms of the maximum independent set of the sender graph.

Theorem III.1. *Let I be an independent set of the graph G_h and $r(\cdot; I)$ be a receiver strategy defined as*

$$r(\mathbf{k}; I) := \begin{cases} h(\mathbf{i}) & (\mathbf{i}, \mathbf{k}) \in I \\ \Delta & (\mathbf{i}, \mathbf{k}) \notin I. \end{cases}$$

The following statements hold.

(a) If $I^\star$ is a maximum independent set of G_h, then $r^\star(\cdot) := r(\cdot, I^\star)$ is a Stackelberg equilibrium.

(b) Conversely, if $r^\star(\cdot)$ is a Stackelberg equilibrium, then $I^\star := \{(x, s_{r^\star}(x)) : r^\star(s_{r^\star}(x)) = h(x)\}$ is a maximum independent set in G_h, and furthermore, $r^\star(\cdot) = r(\cdot, I^\star)$.

We require the following lemma to establish the relationship between an independent set of graph G_h and the correct classification of the feature vector.

Lemma III.2. *Let I be an independent set of the graph G_h and $r(\cdot; I)$ be a receiver strategy defined as*

$$r(\mathbf{k}; I) := \begin{cases} h(\mathbf{i}) & (\mathbf{i}, \mathbf{k}) \in I \\ \Delta & (\mathbf{i}, \mathbf{k}) \notin I. \end{cases}$$

By the definition of I, $r(\cdot, I)$ is well defined. The following statements hold:

(a) If I is an independent set of G_h, then for a receiver strategy $r(x) := r(x, I)$, the set $\{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ contains I.

(b) If $r(x)$ is a receiver strategy, then $I := \{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ is an independent set in G_h.

Now, we proceed with our proof.

Proof. (a) Let $(\mathbf{i}, \mathbf{k}) \in I$. If $s_r(\mathbf{i}) = \mathbf{l} \neq \mathbf{k}$, then it follows that $r(\mathbf{l}) \neq \Delta$. Hence their exists a $\mathbf{j}$ such that $(\mathbf{j}, \mathbf{l}) \in I$. Since $s_r(\mathbf{i}) = \mathbf{l}$, it implies $U(h(\mathbf{i}), \mathbf{i}) - C(\mathbf{k}, \mathbf{i}) \leq U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i})$. But from Definition III.1, this implies that there is an edge between $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ which contradicts the independence of the set I. Hence $s_r(\mathbf{i}) = \mathbf{k}$ and therefore $r(s_r(\mathbf{i})) = h(\mathbf{i})$. Since $\mathbf{i}$ was an arbitrary element in I, therefore $\{(x, s_r(x)) : r(s_r(x)) = h(x)\} \supseteq I$.

(b) Let $\mathbf{i}, \mathbf{j}$ be such that $r(s_r(\mathbf{i})) = h(\mathbf{i})$ and $r(s_r(\mathbf{j})) = h(\mathbf{j})$. Let $\mathbf{k} := s_r(\mathbf{i})$ and $\mathbf{l} := s_r(\mathbf{j})$. We now show there is no edge between $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$. If $h(\mathbf{i}) = h(\mathbf{j})$, then we are trivially done with the proof. Therefore, assume $h(\mathbf{i}) \neq h(\mathbf{j})$. Now, we check for the first edge conditions. If $\mathbf{k} = \mathbf{l}$, then $r(\mathbf{k}) = r(\mathbf{l})$ which implies $h(\mathbf{i}) = h(\mathbf{j})$ which is a contradiction. Therefore, the first edge condition does not hold. Next, we check for the second edge condition. If $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0$ (same reasoning holds for the other equations in the second edge condition), then it implies that the sender receives greater payoff when it modifies $s_r(\mathbf{i})$ to $\mathbf{l}$ instead of $\mathbf{k}$ which is a contradiction since we assumed $s_r(\mathbf{i}) = \mathbf{k}$. Therefore, the second edge condition does not hold. Hence there is no edge between $(\mathbf{i}, \mathbf{k}) = (\mathbf{j}, \mathbf{l})$. Since $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ were arbitrary elements of G_h, therefore the set $\{(x, s_r(x) : r(s_r(x)) = h(x)\}$ is an independent set in G_h.

■

We now conclude the proof of Theorem III.1.

Proof. (Theorem III.1).

(a) If $I^\star$ is a maximum independent set, then $r(x; I^\star)$ is a Stackelberg equilibrium because if not, then there exists another strategy $r^\star$ such that $\{(x, s_{r^\star}(x)) : r^\star(s_{r^\star}(x)) = h(x)\}$ contains $\{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ which in turn contains $I^\star$. Since $\{(x, s_{r^\star}(x)) : r^\star(s_{r^\star}(x)) = h(x)\}$ and $\{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ are both independent set in G_h from Lemma III.2 b, therefore this contradict the maximality of $I^\star$.

(b) If $r(x)$ is a Stackelberg equilibrium, then $I := \{(x, s_r(x))\}$ is a maximum independent set because if not, then there exists another independent set J such that $|J| < |I|$. This implies the for the receiver strategy $\tilde{r}(x; J)$, $\{(x, s_{\tilde{r}}(x))\} \supseteq J \supset I$. This equation contradicts the assumption that $r(x; I)$ is a Stackelberg equilibrium.

■

We have shown that for the decision problem for a discrete feature vector set, the Stackelberg equilibrium strategy r is better than the naive classifier h. One of the reasons for this is that r can strategically choose to ignore feature vectors that are manipulation-prone.

B. Deterministic strategies for the receiver

In this subsection, we address the decision problem formulated in Section II. We analyze the problem for various cases of utility functions.

1) $\mathbf{u}_-, \mathbf{u}_+ \leq 0$: We show that this case is uninteresting as the objectives of the sender and the receiver are aligned as seen in the next proposition.

Theorem III.3. *Let $\mathbf{u}_+, \mathbf{u}_- \leq 0$. Then, $r(x) \equiv h(x)$ is a Stackelberg equilibrium.*

Proof. First, we will show that $s_h(x) = x$. Let $x \leq \mathbf{h}$, then $h(x) = -$. Since $\mathbf{u}_+ \leq 0$, it follows that

$$U(h(x), x) = 0 \geq \max_{x' \in \mathcal{X}} U(h(x'), x) - |\phi(x') - \phi(x)|.$$

Therefore $s_h(x) = x$. Similar reasoning holds for $x > \mathbf{h}$. Now, the set of feature vectors whose label is correctly classified by h is given by the set

$$E(h, s_h; h) = \{x \in \mathcal{X} | h(s_h(x)) = h(x)\} = \mathcal{X}.$$

Since h correctly classifies all the feature vectors, it is a Stackelberg equilibrium. ■

2) $\mathbf{u}_+ \geq 1 \geq \mathbf{u}_-$*:* For this case we show that any receiver r whose range contains the $+$ label, can at most only correctly classify feature vectors whose true labels is $+$.

Theorem III.4. *The receiver strategy $r(c) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label is a Stackelberg equilibrium.*

Proof. Let r be the receiver Stackelberg equilibrium. Then, either there does not exist any $x \in \mathcal{X}$ such that $r(x) = +$, or there exists some $t \in \mathcal{X}$ such that $r(t) = +$. Suppose there does not exist any $x \in \mathcal{X}$ such that $r(x) = +$. Then $r(s_r(x)) \equiv -$ for all $x \in \mathcal{X}$. Thus, $E(r) \subset [\mathbf{h}, 1]$. However, the receiver strategy $r'(x) \equiv -$ for all $x \in \mathcal{X}$ yields $E(r') = [\mathbf{h}, 1]$. Since r is a Stackelberg equilibrium, it follows that $\mathbb{E}(r) = [\mathbf{h}, 1]$. Now, suppose there exists $t \in \mathcal{X}$ such that $r(t) = +$. Since $\mathbf{u}_+ \geq 1$, it follows that for all $x \in [0, \mathbf{h})$, $r(s_r(x)) = +$. Therefore, $E(r) = [0, \mathbf{h})$. Comparing these two cases, it follows that $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label in the Stackelberg equilibrium. ■

3) $\mathbf{u}_-, \mathbf{u}_+ \geq 1$*:* For this case, we show that for any receiver strategy r whose range contains both $-$ and $+$ labels does not correctly classify any feature vector.

Theorem III.5. *The receiver strategy $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label is a Stackelberg equilibrium.*

Proof. Let r be a receiver Stackelberg equilibrium such that there exist $a, b \in \mathcal{X}$ with $r(a) = +$ and $r(b) = -$. Since $\mathbf{u}_+ > 1$, for all $x \in [0, \mathbf{h})$, the worst-case best response is $s_r(x) = a$, so $r(s_r(x)) = +$. Similarly, for $x \in [\mathbf{h}, 1)$, $s_r(x) = b$, so $r(s_r(x)) = -$. Therefore, $E(r) = \mathcal{X}$ and $\mathcal{E}(r) = 1$. However, for a receiver strategy $r'(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, we have $\mathcal{E}(r') < 0.5$. This contradicts the optimality of the Stackelberg equilibrium strategy r. Therefore, no such a, b exist. This implies that r must be a constant label strategy. Thus, $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, is a receiver Stackelberg equilibrium strategy. ■

4) $1 \geq \mathbf{u}_+ \geq 0$ *and* $|\mathbf{u}_-| < \mathbf{u}_+$*:* This case includes scenarios where the sender has a strictly non-uniform preference($\mathbf{u}_- \geq 0$) as well as scenarios where the sender has a uniform preference ($-\mathbf{u}_+ \leq \mathbf{u}_- < 0$). We show that in each of these cases, there exists a receiver Stackelberg equilibrium strategy that belongs to the family of strategies $\mathscr{R}$ defined as follows.

Let $\mathbf{h}^-, \mathbf{h}^+$ be such that $0 \leq \mathbf{h}^- \leq \mathbf{h} < \mathbf{h}^+ \leq 1$ and $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$. Since ϕ is a continuously increasing function, such $\mathbf{h}^-, \mathbf{h}^+$ exist. Let $r(\cdot; \mathbf{h}^-, \mathbf{h}^+)$ denote a receiver strategy defined as follows

$$r(x; \mathbf{h}^-, \mathbf{h}^+) := \begin{cases} h(x) & \text{if} \quad x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1] \\ \Delta & \text{if} \quad x \in (\mathbf{h}^-, \mathbf{h}^+) \end{cases}.$$

Let

$$\mathscr{R}_1 := \{r(\cdot; \mathbf{h}^-, \mathbf{h}^+) : 0 \leq \mathbf{h}^- \leq \mathbf{h} < \mathbf{h}^+ \leq 1, \phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+\}.$$

The strategy $r(\cdot, \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_1$ is identical to the true function h except in the interval $(\mathbf{h}^-, \mathbf{h}^+)$, where it maps the feature vectors to label Δ. Let

$$\mathscr{R}_2 := \{r(\cdot) : r(\cdot) \equiv -\} \cup \{r(\cdot) : r(\cdot) \equiv +\},$$

be the set of constant-label receiver strategies. Let $\mathscr{R} := \mathscr{R}_1 \cup \mathscr{R}_2$.

We define dominance between receiver strategies as follows: for a given probability law D on $\mathcal{X}$, a strategy r dominates another strategy $\tilde{r}$ if $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$. The next theorem shows that the family of receiver strategies $\mathscr{R}$ dominates every other receiver strategy. Hence, a receiver can restrict its search for a Stackelberg equilibrium to the set $\mathscr{R}$ itself.

Theorem III.6. *Let $D \in \mathfrak{D}$ be a law on $\mathcal{X}$ and let $h(x)$ be a true function. Let ϕ be a cost kernel and $\mathbf{u}_+ > 0$. Then their exists a Stackelberg equilibrium strategy $r(\cdot) \in \mathscr{R}$.*

Proof. The proof is given in the Proof VII-A in the Appendix. ■

5) $1 \geq \mathbf{u}_+ \geq 0$*:* and $\mathbf{u}_- \leq -\mathbf{u}_+$

This case corresponds to the sender having an uniform preference. The next theorem specifies a Stackelberg equilibrium.

Theorem III.7. *Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$ and let $h(x)$ be a true function. Let $\mathbf{h}^+ \in \mathcal{X}$ be such that $\phi(\mathbf{h}^+) - \phi(\mathbf{h}) = \mathbf{u}_+$. Then the receiver strategy*

$$r(x; \mathbf{h}, \mathbf{h}^+) := \begin{cases} - & \textit{for} \quad x \in [0, \mathbf{h}], \\ \Delta & \textit{for} \quad x \in (\mathbf{h}, \mathbf{h}^+), \\ + & \textit{for} \quad x \in [\mathbf{h}^+, 1], \end{cases}$$

is a Stackelberg equilibrium.

Proof. The proof is given in the Proof VII.2 in the Appendix. ■

C. Multi-dimensional decision problem

So far, we have analyzed a decision problem with a feature vector space $\mathcal{X} = [0, 1]$, and derived structural results for the Stackelberg equilibrium in this context. In this section, we generalize our analysis to a higher-dimensional feature space. We show that, under certain assumptions on the feature space, this higher-dimensional setting can be reduced to the one described in Section II.

We focus on decision problems where the class boundary is defined by a linear hyperplane. Such scenarios commonly arise when decisions are based on a weighted sum of features, for example, admission decisions in academic institutions often depend on weighted combinations of subject scores. Our objective is to compute the Stackelberg equilibrium for this multi-dimensional decision problem.

Let $w \in \mathbb{R}^n$ be a unit vector, and let $\mathcal{X} \subset \mathbb{R}^n$ be a bounded set denoting the set of feature vectors. Assume the distribution D on $\mathcal{X}$ is uniform. Let $\mathbf{u}_- = \mathbf{u}_+ = 2\mathbf{u} > 0$. Let the true function $h : \mathcal{X} \to \mathcal{Y}$ be given by

$$h(x) = \begin{cases} + & \text{if} \quad w^\top x > h, \\ - & \text{if} \quad w^\top x \leq h, \end{cases} \tag{4}$$

where $h \in \mathbb{R}$. Let $\mathcal{Z} := \{z \in \mathcal{X} : w^\top z = h\}$ denote the class boundary set. We assume that the feature space $\mathcal{X}$ is sufficiently wide along the direction of w, such that for any feature vector $z \in \mathcal{Z}$ on the class boundary, the feature vectors $z + 2\mathbf{u}w$ and $z - 2\mathbf{u}w$ remain within $\mathcal{X}$. Let the cost kernel be the Euclidean norm, i.e., $\phi(x, x') = \|x - x'\|$. All other assumptions from Section II remain unchanged.

Theorem III.8. *Let h given in* (4) *be the true function and let D be the uniform distribution on $\mathcal{X}$. Let $r : \mathcal{X} \to \mathcal{Y} \cup \Delta$ be the receiver's strategy. Then $\mathcal{E}(r) \geq 2\mathbf{u}\,\sigma(\mathcal{Z})$, where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D.*

Proof. The proof is given in Appendix VII.3. ■

We now show that the strategy r described in the following theorem is a Stackelberg equilibrium strategy.

Theorem III.9. *Let h be the true function, and let D be the uniform distribution on $\mathcal{X}$. Let the receiver's strategy r be defined as*

$$r(x) := \begin{cases} + & \textit{if} \quad w^\top x > \mathbf{h} + \mathbf{u} \\ - & \textit{if} \quad w^\top x < \mathbf{h} - \mathbf{u}\,. \\ \Delta & \textit{otherwise} \end{cases}$$

Then r is a Stackelberg equilibrium, and $\mathcal{E}(r) = 2\mathbf{u}\sigma(\mathcal{Z})$, where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D.

Proof. The proof is given in Appendix VII.4. ■

This theorem shows that if the class boundary is a hyperplane, then the decision boundaries of the receiver's Stackelberg equilibrium are also hyperplanes parallel to the class boundary hyperplane. We now show that this problem can be reformulated into our setup (Section III) such that the respective Stackelberg equilibria remain consistent with each other. Let $H_c := \{x \in \mathcal{X} : w^\top x = \mathbf{h}\}$ denote the class boundary, and let $H_+ := \{x \in \mathcal{X} : w^\top x = \mathbf{h} + \mathbf{u}\}$ and $H_- := \{x \in \mathcal{X} : w^\top x = \mathbf{h} - \mathbf{u}\}$ denote the decision boundaries of the corresponding Stackelberg equilibrium. Let

$$a_0 := \arg\min\{w^\top x : x \in \mathcal{X}\} \qquad \text{and} \qquad a_1 := \arg\max\{w^\top x : x \in \mathcal{X}\}.$$

It follows that $a_0 \leq \mathbf{h} \leq a_1$. Define the mapping $\mathcal{M} : \mathcal{X} \to [0, 1]$ as

$$\mathcal{M}(x) = \frac{w^\top x - w^\top a_0}{w^\top a_1 - w^\top a_0}.$$

This mapping projects all feature vectors onto the scalar interval $[0, 1]$ based on their position along the direction of the weight vector w. We now calculate the mapping of the feature vectors lying on the class boundary and the decision boundaries of the Stackelberg equilibrium.

$$\mathcal{M}(H_c) = \frac{\mathbf{h} - w^\top a_0}{w^\top (a_1 - a_0)},$$
$$\mathcal{M}(H_+) = \frac{\mathbf{h} + \mathbf{u} - w^\top a_0}{w^\top (a_1 - a_0)},$$
$$\mathcal{M}(H_-) = \frac{\mathbf{h} - \mathbf{u} - w^\top a_0}{w^\top (a_1 - a_0)}.$$

Now consider the decision problem outlined in Section III, with the class boundary at $\frac{\mathbf{h} - w^\top a_0}{w^\top (a_1 - a_0)}$ and $\mathbf{u}_- = \mathbf{u}_+ = \frac{2\mathbf{u}}{w^\top (a_1 - a_0)}$. From Theorem III.6, it follows that the decision boundaries of the Stackelberg equilibrium are at $\mathbf{h}^- = \frac{\mathbf{h} - \mathbf{u} + w^\top a_0}{w^\top (a_1 - a_0)}$ and $\mathbf{h}^+ = \frac{\mathbf{h} + \mathbf{u} - w^\top a_0}{w^\top (a_1 - a_0)}$. This precisely matches the image under $\mathcal{M}$ of the decision boundaries of the Stackelberg equilibrium derived in Theorem III.8, as claimed.

D. Stochastic strategies for the receiver

In this section, we allow the receiver to use a randomized classifier and characterize the Stackelberg equilibrium of the game. This work improves upon our previous study in [14] by refining key assumptions and proofs. Let the cost kernel $\phi(x)$ be $\phi(x) = x$, and let D, the probability law of X, be uniform. Let $\mathcal{P}(\mathcal{Y} \cup \Delta)$ be the set of probability laws on $\mathcal{Y} \cup \Delta$. A receiver strategy r is a randomized classifier if it is of the form $r : \mathcal{X} \to \mathcal{P}(\mathcal{Y} \cup \Delta)$. This means that for each x, $r(x)$ is a probability law on $\mathcal{Y} \cup \Delta$. In the strategic classification problem, the receiver observes the values of $\mathcal{X}$ as communicated by the sender. Let X' denote the random variable representing the receiver's observation of $\mathcal{X}$ that is communicated by the sender, and let D' denote the corresponding probability law. For a sender strategy $s \in \mathsf{S}$, the random variable X' is given by $X' = s(X)$, and the probability law D' is given by

$$D'_s(X' \in A) = D(X \in \{x : s(x) \in A\}) = \int_{\{x : s(x) \in A\}} dx, \qquad \text{for} \qquad A \subseteq \mathcal{X}.$$

We define a joint random variable (X', Y) over $(\mathcal{X} \times \mathcal{Y})$ whose probability law is given by

$$\mathbb{P}_r(X' \in [x_i, x_j], Y = y) := \int_{x_i}^{x_j} \mathbb{P}_r(Y = y | X' = x) dD'$$

where

$$\mathbb{P}_r(Y = y | X' = x) := r(x)(\{y\}), y \in \mathcal{Y} \cup \Delta.$$

We further constrain r such that

$$\exists x \in \mathcal{X} \quad \text{for which} \quad \mathbb{P}_r(\Delta \mid X' = x) = 0. \tag{5}$$

and the sets

$$\begin{aligned} &\{x : \mathbb{P}_r(- \mid X' = x) + \mathbb{P}_r(+ \mid X' = x) = 1, x \leq \mathbf{h}\} \quad \text{and} \\ &\{x : \mathbb{P}_r(- \mid X' = x) + \mathbb{P}_r(+ \mid X' = x) = 1, x > \mathbf{h}\} \quad \text{are closed.} \end{aligned} \tag{6}$$

The receiver's strategy set is defined as

$$\mathsf{R} := \{r \mid r : \mathcal{X} \to \mathcal{P}(\mathcal{Y} \cup \Delta),\ r \quad \text{satisfies} \quad (5) \quad \text{and} \quad (6)\}. \tag{7}$$

For a receiver strategy r, let

$$p(x) := \mathbb{P}_r(Y = -|X' = x) \quad \text{and} \quad q(x) := \mathbb{P}_r(\{Y = +|X' = x\})$$

be the probabilities of assigning labels $-$ and $+$ to the feature vector x, respectively. Consequently, the strategy $r(x)$ be represented as the triplet $(p(x), q(x), 1 - p(x) - q(x))$. Let

$$\mathfrak{U}(x; r, s) := \sum_{y \in \mathcal{Y} \cup \Delta} \mathbb{P}_r(Y = y|X' = s(x))U(y, x),$$

where U is defined in (2). Since r is a randomized classifier, the sender's payoff for the receiver's strategy r at the feature vector x is modified to

$$\mathbb{U}(x; r, s) := \mathfrak{U}(x; r, s) - |s(x), x|.$$

We further assume that $0 \leq \mathbf{u}_+ = \mathbf{u}_- = 2\mathbf{u} \ll \min\{\mathbf{h}, 1 - \mathbf{h}\}$. Let the error function $T(x; r, s, h) : \mathcal{X} \to [0, 1]$ be defined as:

$$T(x; r, s, h) := \mathbb{P}_r\{Y \neq h(x) \mid X' = s(x)\}. \tag{8}$$

The error function quantifies the error made at each feature vector x by the sender strategy s and the receiver strategy r under the true function h. Since r is a randomized classifier, the receiver's payoff is modified to

$$\mathcal{E}(r, s; h) := \int_{\mathcal{X}} T(x; r, s, h)\, dx.$$

Notation. We sometimes drop some of the arguments from $T(\cdot, \cdot, \cdot, \cdot)$ when it is apparent from the context.

For a receiver strategy r, the sender's worst-case lazy best response is defined as follows.

Definition III.2 (Worst-case Lazy Best Response)**.** Let r be a receiver strategy. Let the set $V(x; r)$ be given as

$$V(x; r) \in \arg\max_{x' \in \mathcal{X}} \Big(\sum_{y \in \mathcal{Y} \cup \Delta} \mathbb{P}_r(Y = y|X' = x')(U(y, x) - |x' - x|) \Big).$$

Let $\bar{B}(r)$ be the set of sender strategies s satisfying

$$s(x) = \arg \min_{x' \in V(x;r)} |x' - x| \qquad \text{for all} \quad x \in \mathcal{X}.$$

A sender strategy s_r is called a worst-case lazy best response to a receiver strategy r if

$$s_r := \arg \min_{s' \in \bar{B}(r)} \mathcal{E}(r, s'; h),$$

and there does not exist $s' \in \bar{B}(r)$ such that

$$T(x; r, s') \leq T(x; r, s_r) \quad \forall x \in \mathcal{X}.$$

Next, we define the Stackelberg equilibrium strategy.

Definition III.3. Let h be the true function. A strategy $(r_h^\star, s_{r_h^\star})$ is a Stackelberg equilibrium if

$$r_h^\star := \arg\max_{r \in \mathsf{R}} \mathcal{E}(r, s_r; h)$$

and $s_{r_h^\star}$ is its worst-case lazy best response.

Notation. Sometimes, we refer $r^\star$ itself as Stackelberg equilibrium if $s_{r^\star}$ and h are apparent from the context.

There are several reasons for considering the setup where the sender adopts the worst-case lazy best response. First, the existence of a Stackelberg equilibrium under the worst-case best response is hard to guarantee. This stands in stark contrast to the worst-case lazy best response, where we explicitly construct a Stackelberg equilibrium. Second, the error in the Stackelberg equilibrium under the worst-case lazy best response provides a lower bound on the error of the Stackelberg equilibrium (if it exists) under the worst-case best response. This, combined with other results, allows us to calculate the error of the Stackelberg equilibrium under the worst-case best response without having to explicitly construct the equilibrium itself. The following proposition and Corollary VII.1 illustrates these points.

Proposition III.1. Let $r \in \mathsf{R}$ be a receiver strategy such that its worst-case best response and worst-case lazy best response exist. Let s denote the worst-case best response to r, and let s' denote the worst-case lazy best response to r. Then, the receiver's error satisfies

$$\mathcal{E}(r, s; h) \geq \mathcal{E}(r, s'; h).$$

Proof. By definition, both s and s' are best responses to r. Suppose

$$\mathcal{E}(r, s'; h) > \mathcal{E}(r, s; h).$$

But this contradicts the assumption that s is the worst-case best response, as it implies there exists another strategy s' that results in a higher error for the receiver. Therefore, it follows that

$$\mathcal{E}(r, s; h) \geq \mathcal{E}(r, s'; h),$$

as claimed. ■

Notation. In this subsection, s_r denotes the worst-case lazy best response, not the worst-case best response unless mentioned otherwise.

The next theorem gives the structure of a Stackelberg equilibrium.

Theorem III.10. *Let h be the true function and let $\mathbf{h}^- := \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ := \mathbf{h} + \mathbf{u}$. Then the receiver strategy*

$$r(x; \mathbf{h}^-, \mathbf{h}^+) := \begin{cases} (1, 0, 0) & x \in [0, \mathbf{h}^-], \\ (1 - \frac{x - \mathbf{h}^-}{2\mathbf{u}}, \frac{x - \mathbf{h}^-}{2\mathbf{u}}, 0) & x \in (\mathbf{h}^-, \mathbf{h}^+), \\ (0, 1, 0) & x \in [\mathbf{h}^+, 1], \end{cases}$$

is a Stackelberg equilibrium and $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$.

Proof. From Lemma VII.6, we have $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$. Furthermore, from Lemma VII.10, we have $\mathcal{E}(r') \geq \frac{\mathbf{u}}{2}$ for all $r' \in \mathsf{R}$. Hence, r is a Stochastic Stackelberg equilibrium. ■

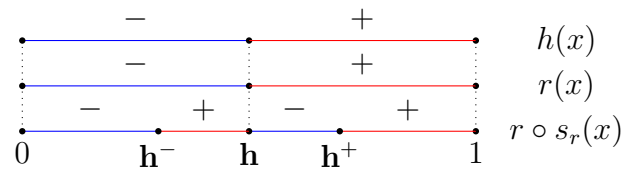

Fig. 1: Naive classifier.

E. Examples and comparison

A naive classifier is the one which uses the true function $h(x)$ for classification. In this subsection, we compare the error made by the naive strategy with the deterministic and the stochastic Stackelberg equilibrium. We consider a scenario where the true function is $h(x)$ given in (1), $\mathbf{u}_+ = \mathbf{u}_- := 2\mathbf{u}$ where $\mathbf{u}$ is sufficiently small and positive, the cost kernel is given by $\phi(x) = x$ and the probability law $D \in \mathfrak{D}$ is uniform.

In the first example, we calculate the error made by the naive classifier.

Example 1. The naive classifier $h(x)$ is given by

$$h(x) = \begin{cases} - & x \in [0, \mathbf{h}] \\ + & x \in (\mathbf{h}, 1]. \end{cases}$$

For a feature vector $x \in (\mathbf{h} - 2\mathbf{u}, \mathbf{h}]$, the sender can map it to a feature vector x', where x' is such that $\mathbf{h} < x' < x + 2\mathbf{u}$, and obtain a positive payoff. This is better than not modifying x, as in that case, the sender receives zero payoff. Similarly, for a feature vector $x \in (\mathbf{h}, \mathbf{h} + 2\mathbf{u}]$, the sender can map it to a feature vector x', where x' is such that $x - 2\mathbf{u} < x' \leq \mathbf{h}$, and obtain a positive payoff as opposed to a zero payoff when not modifying x. Hence $h(x)$ misclassifies the feature vectors belonging in the interval $(\mathbf{h} - 2\mathbf{u}, \mathbf{h} + 2\mathbf{u}]$ and $\mathcal{E}(h) = 4\mathbf{u}$.

In the next example, we calculate the error made by the deterministic Stackelberg equilibrium.

Example 2. Let r^* be a deterministic Stackelberg equilibrium obtained from Theorem III.6. Then r^* is given by

$$r^*(x) = \begin{cases} h(x) & x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \Delta & x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

where $\mathbf{h}^- = \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ = \mathbf{h} + \mathbf{u}$. The sender's worst-case best response is

$$s_{r^*}(x) = \begin{cases} x & x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \mathbf{h}^+ & x \in [\mathbf{h}^-, \mathbf{h}) \\ \mathbf{h}^- & x \in [\mathbf{h}, \mathbf{h}^+]. \end{cases}$$

Hence $E(r^*) = [\mathbf{h}^-, \mathbf{h}^+]$ and $\mathcal{E}(r^*) = 2\mathbf{u}$. This shows that ignoring feature vectors near the decision boundaries, which in any case will be misclassified under the naive decision rule, prevents other neighbouring feature vectors from getting misclassified.

In Section III-B, we analyzed the setup where the sender adopts the worst-case best response. In contrast, Section III-D examined the setup where the sender employs the worst-case lazy best response. To ensure a fair comparison between the deterministic and stochastic Stackelberg equilibria, the sender's style of play should remain fixed. Therefore, in the following example, we calculate the classification error made by a receiver strategy, assuming the sender adopts the worst-case best response.

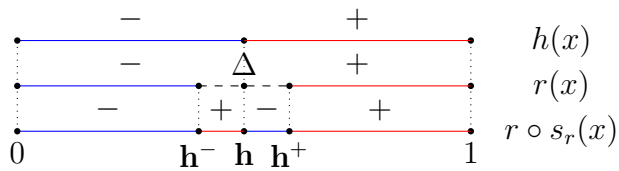

Fig. 2: Deterministic Stackelberg equilibrium.

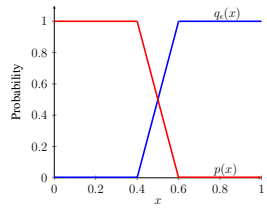

Fig. 3: Stochastic decision rule$r(x)$,$\mathbf{u} = 0.1$,$\mathbf{h} = 0.5$

Example 3. Let $\epsilon > 0$. Let $r_\epsilon \equiv (1 - q_\epsilon(x), q_\epsilon(x), 0)$ be stochastic receiver strategy where q_ϵ is defined as

$$q_\epsilon(x) = \begin{cases} 0 & \text{for} \quad x \in [0, \mathbf{h}_\epsilon^-) \\ \frac{(x-(\mathbf{h}_\epsilon^-)}{2(\mathbf{u}+\epsilon)} & \text{for} \quad x \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+] \\ 1 & \text{for} \quad x \in (\mathbf{h}_\epsilon^+, 1], \end{cases} \tag{9}$$

where $\mathbf{h}_\epsilon^- := \mathbf{h} - \mathbf{u} - \epsilon$ and $\mathbf{h}_\epsilon^+ := \mathbf{h} + \mathbf{u} + \epsilon$. It follows that the worst-case best response is $s_{r_\epsilon}(x) = x$ (the calculation is similar to what is done in Lemma VII.6 and is given in Appendix VII-D) and $\mathcal{E}(r_\epsilon)$ is $\frac{\mathbf{u}+\epsilon}{2}$. This strategy shows that randomizing classifications near the decision boundaries can correctly classify feature vectors close to the decision boundaries, albeit with a low but positive probability. This happens because randomization ensures that the cost increase is offset by the increase in utility (which takes a fractional value due to randomization).

It can be seen from these examples that both the deterministic and stochastic Stackelberg equilibria outperform the naive classifier. Furthermore, the stochastic classifier makes fewer errors than the deterministic classifier. The benefit of randomization in strategic interactions is well-established in game theory literature, and we now observe its advantages in the strategic classification as well.

One crucial point to note is that although the penalty label was useful in the deterministic classifier, it was not helpful in the stochastic classifier. Furthermore, the deterministic classifier is sub-optimal compared to the stochastic classifier. Thus, the randomization offers the receiver a richer source of leverage to make the sender behave appropriately than the penalty label. However, in many cases, such as university admissions, for transparency, the receivers are constrained to use only the deterministic classifier. Thus, in those cases, the penalty label can act as compensation for the lack of randomization.

The following example describes the deterministic Stackelberg equilibrium of the university admission problem introduced in the Section I.

Example 4. (Stackelberg equilibrium of the university admission problem):

Recall the example from the introduction. Suppose a university offers two programs: Theoretical Physics and Experimental Physics. Admission to these programs is determined by each student's performance on an entrance exam. Each prospective student has a unique combination of theoretical and experimental skills, and their *true program* corresponds to the program that aligns with their stronger skill. However, students often prefer a program that does not align with their true program. The university's goal is to design the entrance exam to maximize the number of students admitted to their true program.

We model this as a strategic classification problem as follows: The university acts as the receiver, and the student as the sender. Let $\mathcal{X} := [0, 1]$ represent the set of feature vectors, where $x \in \mathcal{X}$ denotes a student's true aptitude, with the interpretation that it has x theoretical skill and $(1 - x)$ experimental skill. We assume that true aptitudes are uniformly distributed across the student population; thus, D, the probability law on $\mathcal{X}$, is uniform. Let $\mathcal{Y} := \{-, +\}$ denote the label space, where $-$ indicates the Experimental Physics program and $+$ the Theoretical Physics program. The true function $h(x)$, which assigns each student to their true program, is given by the following equation:

$$h(x) = \begin{cases} + & \text{if} \quad x > 0.5, \\ - & \text{if} \quad x \leq 0.5. \end{cases}$$

This means $\mathbf{h} = 0.5$, and that for $x > 0.5$, a student's true program is Theoretical Physics, while for $x \leq 0.5$, it is Experimental Physics.

The university has access to a question bank containing questions of varying difficulties. We denote the difficulty level of a question by x, indicating that the question requires x theoretical knowledge and $(1 - x)$ experimental knowledge to solve. Note that there is a one-to-one correspondence between the feature vector set $\mathcal{X}$ and the question bank. Thus, a student may demonstrate an aptitude x' (not necessarily their true aptitude) by solving a question with difficulty level x'. The entrance exam comprises multiple questions selected from this bank, with each question associated with a program. A student can attempt only one question, and admission to the corresponding program is granted if they answer correctly.

We assume each student's preferred program differs from their true program. The student's utility function is given by

$$U(y, x) := \begin{cases} 0.04 & x \in [0, 0.5], y = + \\ 0.04 & x \in (0.5, 1], y = - \\ 0 & y = h(x) \\ -\infty & y = \Delta. \end{cases}$$

This means $\mathbf{u}_- = \mathbf{u}_+ = 0.04$, and a student gains a utility of 0.02 if assigned to their preferred program and 0 if assigned to their true program and $-\infty$ if not assigned any program. The cost kernel is given by $\phi(x) = x$, meaning that for a question with difficulty level x', a student with true aptitude x incurs a cost of $|x - x'|$ while solving it.

The university's strategy set includes all possible question papers for the entrance exam. For a given question paper, each student selects the question that maximizes their utility at the lowest cost, potentially demonstrating an aptitude x' different from their true aptitude to the university. The student's strategy set consists of mappings that specify which question to attempt for each question paper. We assume students answer their chosen question correctly. Consequently, the university, aware of students' strategic behaviour, must optimally design the question paper to maximize the number of students admitted to their true program.

According to Theorem III.6, questions with difficulty levels in the interval $[\mathbf{h} - \mathbf{u}_-, \mathbf{h} + \mathbf{u}_+]$ should not be included in the exam. Substituting the value for $\mathbf{h}, \mathbf{u}_-, \mathbf{u}_+$ for this example, this interval turns out to be $[0.48, 0.52]$. Furthermore, the students who correctly answers questions with difficulty levels less than 0.48 should be admitted to the Experimental Physics program, and the students who correctly answers the questions with difficulty levels greater than 0.52 should be admitted to the Theoretical Physics program. The intuition for excluding questions in the interval $[0.48, 0.52]$ is as follows: if a question of difficulty level, say 0.51 (corresponding to Theoretical Physics) is included, a student with true aptitude 0.49 (whose true program is Experimental Physics) could solve it with only effort 0.02, leading to misclassification. However, this approach has a drawback: students whose true aptitudes lie within the interval $[0.48, 0.52]$ must attempt questions of difficulty level outside this range. Thus, by incurring a cost, they attempt questions linked to their preferred program and get misclassified by the university. Nonetheless, the overall benefits of this strategy outweigh its drawbacks, making it the optimal option for the university.

F. Other Stackelberg equilibria

In this section, we only consider deterministic Stackelberg equilibria. In many scenarios, the strategic classification problem may admit more than one Stackelberg equilibrium. While all such Stackelberg equilibria guarantee the same classification error for the receiver, they are not equivalent from the sender's perspective. Here, we discuss various considerations that a receiver may take into account besides minimizing the classification error.

1) Participation constraint for the sender: Consider the following instance of the problem. Let D be a uniform distribution on $\mathcal{X}$, with $\mathbf{h} = 0.5$ and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. It can be shown that both r_1 and r_2, defined as

$$r_1(x) = \begin{cases} - & \text{if} \quad x \in [0, 0.45] \\ \Delta & \text{if} \quad x \in (0.45, 0.55) \\ + & \text{if} \quad x \in [0.55, 1] \end{cases}, \quad r_2(x) = \begin{cases} - & \text{if} \quad x = 0 \\ \Delta & \text{if} \quad x \in (0, 1) \\ + & \text{if} \quad x = 1 \end{cases}$$

are receiver Stackelberg equilibrium strategies. However, the situation differs from the sender's perspective. For $x \in \mathcal{X}$, the sender's payoff under r_1 and r_2 satisfies

$$\mathbb{U}(x; r_1, s_{r_1}) \geq \mathbb{U}(x; r_2, s_{r_2}) \quad \text{for all} \quad x \in \mathcal{X}.$$

That is, for each $x \in \mathcal{X}$, the payoff for the sender under r_1 is greater than or equal to that under r_2.

This scenario resembles a dilemma in the principal-agent problem, where the principal can impose an unfair contract on the agent, who ends up paying (instead of being paid) for their work. To remedy this in the principal-agent context, the agent is given the option to refuse the contract and receive a baseline payoff. This encourages the principal to offer a contract with a payoff greater than the baseline. This concept is known as the *participation constraint* for the agent.

Drawing a parallel to the principal-agent problem, the receiver can be viewed as the principal who offers a contract r, while the sender acts as the agent who, by making effort s on a feature vector x, gets a payoff of $\mathbb{U}(x; r, s)$. We allow the sender, after seeing the contract r, the option not to participate and instead receive a baseline payoff. There are many ways the sender can determine his decision to participate. Here, we assume that the sender does not participate if $\mathbb{U}(x; r, s_r)$ is lower than the baseline payoff for any $x \in \mathcal{X}$. We further assume that if the sender does not participate in the game, the receiver makes an error on the entire $\mathcal{X}$. This participation constraint of the sender would discourage the receiver from playing a degenerate strategy like r_2, since the sender would reject such a game. Given this constraint, strategies like r_1, with a Δ interval of length $\mathbf{u}$, are the only justifiable strategies, as we show in the following cases.

Case 1. Negative participation constraint
Suppose the sender incurs a fixed payoff $\nu \leq 0$ if it chooses not to participate in the game after seeing the receiver's strategy. This means the sender can opt-out with a small penalty upon encountering an unfavourable receiver strategy. Consequently, many degenerate receiver strategies like r_2 will not satisfy the participation constraint and will no longer be a receiver Stackelberg equilibrium.

Now consider the strategy r_1. For $x \in [0, 0.45) \cup (0.55, 1]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = 0$. For $x \in [0.45, 0.5]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = x - 0.45$ and for $x \in (0.55, 1]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = 0.55 - x$. Thus, $\min_{x \in \mathcal{X}} \mathbb{U}(x; r, s_r) = 0$. This satisfies the sender's participation constraint, and hence the sender will participate in the game. Therefore, r_1 is a Stackelberg equilibrium even under this constraint.

Case 2. Positive participation constraint:
Suppose the sender receives a fixed payoff $\nu > 0$ if it chooses not to participate in the game after seeing the receiver's strategy. We will show that every receiver strategy is a Stackelberg equilibrium strategy.

Let r be a receiver strategy. If $E(r, s_r, h) = \mathcal{X}$, then $\mathcal{E}(r) = 1$. If $E(r, s_r, h) \neq \mathcal{X}$, then there exists x such that $r(s_r(x)) = h(x)$. This implies $\mathbb{U}(x; r, s_r) = |s_r(x) - x| \leq 0$. But this violates the sender's participation constraint, and thus the sender will ignore this strategy r. Therefore, $\mathcal{E}(r) = 1$. Hence,

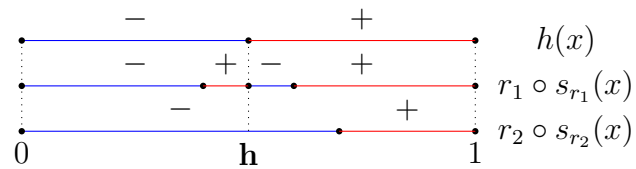

Fig. 4: Fairness and complexity tradeoff.

$\mathcal{E}(r) = 1$ for all $r \in \mathsf{R}$. Therefore, every receiver strategy $r \in \mathsf{R}$ is a receiver Stackelberg equilibrium strategy as claimed. More specifically, r_1 is a Stackelberg equilibrium even under this constraint.

2) Fairness and complexity of the classifier:: Let D be the uniform probability law on $[0, 1]$, $\mathbf{h} = 0.5$, and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. From Theorem III.6, any strategy $r(x; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}$ such that $\mathbf{h}^- \leq 0.5 \leq \mathbf{h}^+$ and $|\mathbf{h}^- - \mathbf{h}^+| = 0.2$ is a Stackelberg equilibrium. These equilibria yield the same receiver error but differ in how errors are distributed across each label class.

Consider two such receiver strategies, r_1 and r_2, with corresponding Δ-regions $(0.45, 0.55)$ and $(0.4, 0.5)$, respectively. The strategy r_1 appears fairer, as it balances the ratio of correctly classified labels, compared to r_2, which misclassifies only the $-$ class. However, the receiver strategy r_2 is simpler, with 2 decision boundaries in $r_2 \circ s_{r_2}$ compared to 3 in $r_1 \circ s_{r_1}$. This simplicity is an advantage for learnability; the VC dimension of a hypothesis class increases with the number of decision boundaries that functions in that class are allowed to have (see Proposition IV.2). This highlights a trade-off between fairness and efficient learnability. Depending on the application, one strategy may be preferable to the other.

IV. Learning the Stackelberg equilibrium

In this section, we address the learning problem, where the receiver is unaware of the true function and is limited to employing only deterministic strategies. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ denote a hypothesis class, and let

$$l(x, y; f) := \mathbb{1}_{f(x) \neq y}$$

be the *zero-one loss function*, where $x \in \mathcal{X}, y \in \mathcal{Y}$ and $f \in \mathcal{F}$. Let (X, Y) be a random variable over $(\mathcal{X}, \mathcal{Y})$ and D be its corresponding probability law. Let $L_D(f)$ denote the *risk function* corresponding to the zero-one loss function and be defined as

$$L_D(f) := \mathbb{E}_D[l(X, Y; f)]. \tag{10}$$

Suppose $Y = h(X)$, where h is the true function. Then we modify the definition of risk function given in (10). Let X random variable over $\mathcal{X}$ and D be it corresponding probability law. Then $L_D(f; h)$ denotes the *h-risk function* corresponding to the zero-one loss function and be defined as

$$L_D(f; h) := \mathbb{E}_D[l(X, h(X); f)].$$

In the absence of strategic interactions between the sender and the receiver, a hypothesis class is PAC (Probably Approximately Correct) learnable if it satisfies the following definition (which is the same as Definition 3.1 in [15]).

Definition IV.1. A hypothesis class $\mathcal{F}$ is PAC learnable if there exists a function $m_{\mathcal{F}} : (0, 1)^2 \to \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \to \mathcal{F}$ such that for every $(\epsilon, \delta) \in (0, 1)^2$, for every true function $h \in \mathcal{F}$ and every probability law D over $\mathcal{X}$, the following holds: Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ is a sample where x_i's are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}(\epsilon, \delta)$. Then

$$D^{\otimes m} \left(L_D(A(S); h) < \epsilon \right) > 1 - \delta.$$

There exists a more general notion of PAC learning called agnostic PAC learning which is defined in the following definition (which is the same as Definition 3.3 in [15]).

Definition IV.2. A hypothesis class $\mathcal{F}$ is agnostic PAC learnable if there exists a function $m_{\mathcal{F}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty}(\mathcal{X} \times \mathcal{Y})^n \to \mathcal{F}$ such that for every $(\epsilon, \delta) \in (0,1)^2$ and every probability law D over $\mathcal{X} \times \mathcal{Y}$, the following holds: Let $S = \{(x_i, y_i)\}_{i=1}^m$ be a sample where (x_i, y_i)'s are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}(\epsilon, \delta)$. Then

$$D^{\otimes m}\left(L_D(A(S)) - \inf_{f \in \mathcal{F}} L_D(f) < \epsilon\right) > 1 - \delta.$$

An algorithm A is a *successful PAC/agnostic PAC learner* of $\mathcal{F}$ if $\mathcal{F}$ is PAC/agnostic PAC learnable and A is the corresponding learning algorithm defined in Definition IV.1/IV.2. Consider the learning algorithm Empirical risk minimization(ERM)

$$\text{ERM} \quad : \cup_{n=1}^{\infty}(\mathcal{X} \times \mathcal{Y})^n \to \mathcal{F} \quad \text{defined as} \qquad \text{ERM} \quad (\{(x_i, y_i)_{i=1}^m\}) := \arg\min_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^{m} l(x_i, y_i; f).$$

The following theorem shows that a hypothesis class is PAC/agnostic PAC learnable if and only if it is learnable by an ERM algorithm.

Theorem IV.1 (Fundamental theorem of statistical learning)**.** *Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ be a hypothesis class. Then, the following statements are equivalent:*

a) $\mathcal{F}$ is agnostic PAC learnable.
b) $\mathcal{F}$ is PAC learnable
c) Any ERM rule is a successful PAC learner for $\mathcal{F}$.
d) $\mathcal{F}$ has finite VC dimension.

Furthermore, for a given $(\epsilon, \delta) \in (0,1)^2$ and VC dim$(\mathcal{F}) = d$, the sample complexity $m_{\mathcal{F}}$ is bounded as follows:

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2} \leq m_{\mathcal{F}} \leq C_2 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2},$$

where $C_1, C_2 > 0$ are constants.

Proof. Refer to Theorems 6.7 and 6.8 in [15]. ■

We now establish a similar learning criterion for a hypothesis class $\mathcal{F}$ in the context of strategic classification. Recall that for a true function $f \in \mathcal{F}$ and probability law $D \in \mathfrak{D}$, $r^{\star}_{f,D}$ denotes the Stackelberg equilibrium as defined in Definition II.3. Define $\widetilde{\mathcal{F}} := \{r^{\star}_{f,D} \circ s_{r^{\star}_{f,D}} : f \in \mathcal{F}, D \in \mathfrak{D}\}$, referred to as the *Stackelberg hypothesis class*, which represents the space of receiver Stackelberg equilibria derived from $\mathcal{F}$ and $\mathfrak{D}$. A hypothesis class $\mathcal{F}$ is strategically learnable if it satisfies the following definition.

Definition IV.3 (Strategically learnable class)**.** A hypothesis class $\mathcal{F}$ is strategically learnable if there exists a function $m_{\mathcal{F}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty}(\mathcal{X} \times \mathcal{Y})^n \to \widetilde{\mathcal{F}}$ such that for any true function $h \in \mathcal{F}$, any probability law $D \in \mathfrak{D}$, and any $(\epsilon, \delta) \in (0,1)^2$, the following holds: Let $S = \{(x_i, h(x_i))_{i=1}^m\}$ be a sample where x_i are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}(\epsilon, \delta)$, then

$$D^{\otimes m}\Big(\mathcal{E}(A(S); h, D) - \mathcal{E}(r^{\star}_{h,D}; h, D) < \epsilon\Big) > 1 - \delta.$$

The function $m_{\mathcal{F}}(\epsilon, \delta)$ is called the sample complexity of the algorithm.

Our definition of strategic learnability is different from [10] because we let the algorithm search over the Stackelberg equilibrium hypothesis class (i.e., over $r_f, f \in \mathcal{F}$) rather than the original hypothesis class (i.e., over $f \in \mathcal{F}$).

We propose two algorithms for learning a strategic classifier: *vanilla ERM* and *strategy aware ERM* (defined in Section IV-B and IV-C respectively). We show that a hypothesis class is strategically learnable if and only if it is learnable under the strategy-aware ERM algorithm. However, there exist strategically learnable hypothesis classes which are not learnable under the vanilla ERM algorithm.

Assuming the hypothesis class is learnable by each algorithm, we compare their ease of learning (compared in Section IV-D). This is controlled by the VC dimension of the hypothesis class, which determines the sample complexity in PAC learning. We observe that when the hypothesis class is complex (i.e., contains functions with many decision boundaries), strategy-aware ERM makes learning easier, as it can reduce the VC dimension of the effective learning space. Conversely, for simpler hypothesis classes, vanilla ERM is often the the more direct and efficient approach.

A. Some results on hypothesis class with multiple decision boundaries

A point is called a *decision boundary*, if it assigns different labels to its left and right neighbours. The hypothesis class we consider for our learning algorithms comprises of functions with multiple decision boundaries, requiring us to extend the results established in Theorem III.6 (which only had one decision boundary). To achieve this, we examine a generalized version of a decision problem where the true function has multiple decision boundaries. Let $\mathcal{X} = [0, 1]$ represent the feature vector space, and let $\mathcal{Y} = \{-1, +1\}$ be the label space, and $D \in \mathfrak{D}$ a probability law on $\mathcal{X}$. Decompose $\mathcal{X}$ as $\mathcal{X} = O_1 \cup O_2 \cup \cdots \cup O_{d-1}$, where each O_n is a disjoint interval defined as $O_n = [o_{(n)}, o^{(n)})$ for $n < d-1$, $O_{d-1} = [o_{d-1}, 1]$, and $o^{(n)} = o_{(n+1)}$. Define the true function $h(x)$ as follows:

$$h(x) = \begin{cases} - & \text{if} \quad x \in O_n,\ n \quad \text{is even} \quad , \\ + & \text{if} \quad x \in O_n,\ n \quad \text{is odd} \quad . \end{cases} \tag{11}$$

Let the cost kernel be $\phi(x) = x$, and let Δ denote the penalty label (as considered in Section II). Define the utility function $U(y, x)$ as:

$$U(y, x) := \begin{cases} \mathbf{u} & \text{if} \quad x \in O_n,\ y = +,\ n \quad \text{is even} \quad , \\ \mathbf{u} & \text{if} \quad x \in O_n,\ y = -,\ n \quad \text{is odd} \quad , \\ 0 & \text{if} \quad y = h(x), \\ -\infty & \text{if} \quad y = \Delta. \end{cases}$$

The structure of the Stackelberg equilibrium strategy generalizes Theorem III.6 (when $\mathbf{u}_+ = \mathbf{u}_-$) and is stated in the following proposition.

Proposition IV.1. Let D be a probability law on $\mathcal{X}$, and let $h(x)$ be the true function. For any n, let Θ_n be an interval such that $\Theta_n \subseteq O_n$. Define the receiver strategy r as follows:

$$r(x; \Theta_1, \cdots, \Theta_{d-1}) = \begin{cases} h(x) & x \in \cup_{n=1}^{d-1} \Theta_n, \\ \Delta & x \notin \cup_n \Theta_n. \end{cases} \tag{12}$$

Then their exist Θ_n's, such that r is a Stackelberg equilibrium, and $r \circ s_r$ is of the form:

$$r(s_r(x)) = \begin{cases} h(x) & x \in \cup_n \Theta_n, \\ h(x)^c & x \notin \cup_n \Theta_n. \end{cases}$$

Proof. The proof of the Proposition is given in Proof VII.5 in the Appendix. ■

Our next theorem establishes the VC dimension of a hypothesis class containing hypotheses with multiple decision boundaries.

Proposition IV.2. Let the hypothesis class $\mathcal{F} \subset \{f | f : \mathcal{X} \to \mathcal{Y}\}$ be a collection of all functions with at most $d-1$ decision boundaries. Then VC dim$(\mathcal{F}) = d$.

Proof. Suppose VC dim $(\mathcal{F}) < d$. This means that there exist $(x_1, y_1), \ldots, (x_d, y_d) \in \mathcal{X} \times \mathcal{Y}$ such that for each $f \in \mathcal{F}$, there exists an index i_f such that $f(x_{i_f}) \neq y_{i_f}$. Without loss of generality, we assume

$x_1 < x_2 < \cdots < x_d$. Let $J := \{j : y_j \neq y_{j+1}, j < d\}$. Let j_i denote the i-th smallest element of J. Now consider the following hypothesis f defined as

$$f(x) := \begin{cases} y_{j_1} & \text{if} \quad x \in [0, \frac{x_{j_1}+x_{j_2}}{2}), \\ y_{j_i} & \text{if} \quad x \in [\frac{x_{j_i}-x_{j_{i-1}}}{2}, \frac{x_{j_{i+1}}-x_{j_i}}{2}), \\ y_{j_{|J|}}, & \text{if} \quad x \in [\frac{j_{|J|-1}+j_{|J|}}{2}, 1]. \end{cases}$$

The hypothesis f has at most $d-1$ decision boundaries and hence $f \in \mathcal{F}$. Furthermore, $f(x_k) = y_k$ for all $k \in \{1, \ldots, d\}$ by construction. But this contradicts the assumption. Hence, VC dim $(\mathcal{F}) \geq d$. Now, consider $(x_1, y_1), \ldots, (x_{d+1}, y_{d+1})$, where $y_j \neq y_{j+1}$ for all $j \leq d$. This means that any function f' with the property that $f'(x_i) = y_i$ must have at least d decision boundaries. But functions in $\mathcal{F}$ have at most $d-1$ decision boundaries. Therefore, $\mathcal{F}$ cannot shatter $\{x_i\}_{i=1}^{d+1}$. Hence, VC dim$(\mathcal{F}) \ngeq d+1$. Therefore, VC dim $(\mathcal{F}) = d$.

■

B. Vanilla ERM

One intuitive algorithm for the receiver to learn the strategic classifier is to estimate the true function from the sample data. The receiver then plays the screening game under this estimated function, treating it as the true function, and outputs the corresponding Stackelberg equilibrium. The vanilla ERM algorithm expresses this idea. Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$, h be the true function and let $S = \{(x_i, h(x_i))_{i=1}^m\}$ be a sample where x_i are drawn i.i.d. from D. Let $D(S)$ be the empirical law of sample S defined as:

$$D(S)(B) := \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}_{x_i \in B}$$

where B is a Borel measurable set in $\mathcal{X}$.

Data: Hypothesis class $\mathcal{F}$, Sample $S = \{x_i, h(x_i)\}_{i=1}^m$,
Zero-one loss function $l_h(f, x) := \mathbb{1}_{\{f(x)=h(x)\}}(x)$
Result: Hypothesis $\hat{r}_{erm}$

`// Estimation`
$A(S) := \arg\min_{f \in \mathcal{F}} \sum_{i=1}^n \mathbb{1}_{\{f(x_i) \neq y_i\}}$
$D(S) :=$ Empirical law

`// Decision`
$\hat{r} := r^\star_{A(S),D(S)}$, i.e., Stackelberg equilibrium under true function $\hat{h}$ and probability law $D(S)$

Algorithm 1: Vanilla ERM algorithm

The Algorithm 1 is initialized by providing a hypothesis class $\mathcal{F}$ and a sample S. In the first step, the algorithm calculates $A(S)$, which estimates the true function h from the sample S via the ERM subroutine. In the second step, it calculates the empirical law $D(S)$ of sample S, which is an estimate of the input law D on $\mathcal{X}$ from which the x_i's in S were chosen. In the third step, it calculates the Stackelberg equilibrium $r^\star_{A(S),D(S)}$ (refer Definition II.3) under the (estimated) true function $A(S)$ and the probability law $D(S)$ on $\mathcal{X}$, as given in Theorem III.6.

We show that if the hypothesis class $\mathcal{F}$ has finite VC dimension, then it is strategically learnable under the Algorithm 1.

Theorem IV.2. *Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d-1$ decision boundaries, where $d < \infty$. Let $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \to \mathcal{F}$ denote the Algorithm 1 (vanilla ERM algorithm) and let $B : \mathcal{F} \to \widetilde{\mathcal{F}}$ map a hypothesis to its corresponding Stackelberg equilibrium. Then $\mathcal{F}$ is strategically learnable (cf. Definition IV.3) under $B \circ A$.*

We prove the theorem with the help of the following two lemmas. The first lemma states that for a fixed true function $f \in \mathcal{F}$, the classification errors of the Stackelberg equilibrium strategies $r^\star_{f,D}$ and $r^\star_{f,D(S)}$ are similar for all $D \in \mathfrak{D}$.

Lemma IV.3. *Let $(\epsilon, \delta) \in (0,1)^2$. Let $f \in \mathcal{F}$ be a true function with upto $d-1$ decision boundaries, and let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$. Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample of size m, where the x_i's are drawn i.i.d. according to a probability law D, and let $D(S)$ be the empirical law of the sample S. Let $r^\star_{D(S)}$ and $r^\star_D$ be the Stackelberg equilibrium under the true function f for probability law $D(S)$ and D respectively. Then, there exists an $\bar{m}(\epsilon, \delta)$ such that for all $m > \bar{m}(\epsilon, \delta)$ the following holds:*

$$D^{\otimes m}\big(\mathcal{E}(r^\star_{f,D(S)}; f, D) - \mathcal{E}(r^\star_{f,D}; f, D) < 2(d-1)\epsilon\big) > 1 - 2e^{-2m\epsilon^2}.$$

Proof. The proof is given in Proof VII.6 in the Appendix. ■

The next lemma states that for a fixed probability law $D \in \mathfrak{D}$, the classification errors of the Stackelberg equilibrium strategies $r^\star_{A(S),D}$ and $r^\star_{h,D}$ are similar.

Remark IV.1. *Notice that $r_{A(S),D}$ is the Stackelberg equilibrium when the estimated true function $A(S)$ is considered the true function. To avoid confusion with the actual true function h, which generated the samples, we will refer to h as the ground true function and $A(S)$ as the estimated true function whenever there is potential for ambiguity.*

Lemma IV.4. *Let $(\epsilon, \delta) \in (0,1)^2$. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d-1$ decision boundaries. Let $A : \cup_{n=1}^\infty (\mathcal{X} \times \mathcal{Y})^n \to \mathcal{F}$ denote the algorithm described in Algorithm 1 (vanilla ERM algorithm). For every ground true function $h \in \mathcal{F}$ and every probability law $D \in \mathfrak{D}$ on $\mathcal{X}$, the following holds. Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample of size m, where x_i are drawn i.i.d. according to D. Let $r^\star_{A(S),D}$ and $r^\star_{h,D}$ be the Stackelberg equilibria under D and the estimated true function $A(S)$ and the ground true function $h(x)$, respectively. Then there exists $\bar{m}(\epsilon, \delta)$ such that for all $m \geq \bar{m}(\epsilon, \delta)$, the following holds.*

(a) For any receiver strategy $r \in \mathsf{R}$ and (arbitrary) true function $f \in \mathcal{F}$ (s_r depends on f), we have

$$D^{\otimes m}\big(|\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| < \epsilon\big) > 1 - \delta,$$

(b) For receiver Stackelberg equilibria $r^\star_{A(S),D}$ and $r^\star_{h,D}$, we have

$$D^{\otimes m}\big(\mathcal{E}(r^\star_{A(S),D}; h, D) - \mathcal{E}(r^\star_{h,D}; h, D) < 2\epsilon\big) > 1 - \delta. \tag{13}$$

Furthermore, the sample complexity $\underline{m}(\epsilon, \delta)$ satisfies

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon} \leq \underline{m}(\epsilon, \delta) \leq C_2 \frac{d \log\left(\frac{1}{\epsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\epsilon} \quad .$$

Here $C_1, C_2 > 0$ are constants.

Proof. The proof is given in Proof VII.7 in the Appendix. ■

Combining Lemmas IV.4 and IV.3, we now prove our theorem

Proof of Theorem IV.2. Let $D \in \mathfrak{D}$ be the probability law on $\mathcal{X}$ and let $(\epsilon, \delta) \in (0,1)^2$. Let $h \in \mathcal{F}$ be the ground true function and let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a m-sized sample where x_i's are drawn i.i.d. from D. Let $m \geq m_{\mathcal{F}}, \bar{m}(\epsilon, \delta), \underline{\mathrm{m}}(\epsilon, \delta)$ where $m_{\mathcal{F}}, \bar{m}(\epsilon, \delta)$ and $\underline{\mathrm{m}}(\epsilon, \delta)$ are defined in Theorem IV.1, and Lemma IV.3 and IV.4 respectively. Let $D(S)$ be the corresponding empirical distribution of S. Let $r_1 := r^\star_{h,D}, r_2 := r^\star_{A(S),D}, r_3 := r^\star_{A(S),D(S)} = B(A(S))$.

Since VC dim$(\mathcal{F}) < \infty$, $\mathcal{F}$ is PAC learnable from Theorem IV.1. Furthermore, from Theorem IV.1 we have $A(S)$ as the estimated true function such that

$$D^{\otimes m}(L_D(A(S); h) < \epsilon) > 1 - \delta. \tag{14}$$

Let

$$\mathcal{S}_1 := \{S : L_D(A(S); h) < \epsilon\}$$

and

$$\mathcal{S}_2 := \{S : \sup_{x \in [0,1]} |\psi(x) - \psi_m(x)| < \epsilon\},$$

where $\psi(x)$ and $\psi_m(x)$ is the distribution function of D and $D(S)$ respectively. Then from (14), we get $D^{\otimes m}(\mathcal{S}_1) > 1 - \delta$ and from (66), we get $D^{\otimes m}(\mathcal{S}_2) > 1 - 2e^{-2m\epsilon^2}$. Let

$$\mathcal{S} := \mathcal{S}_1 \cap \mathcal{S}_2.$$

Then, from union bound, we have $D^{\otimes m}(\mathcal{S}) > 1 - 2e^{-2m\epsilon^2} - \delta$. Fix $S \in \mathcal{S}$. Using Lemma IV.4 (a) by identifying $r = r_3$ and $f = A(S)$ as the (estimated) true function, we get

$$\mathcal{E}(r_3; h, D) - \mathcal{E}(r_3; A(S), D) < \epsilon. \tag{15}$$

Using Lemma IV.3 by identifying $f = A(S)$ as the (estimated) true function and D as the probability law, we obtain

$$\mathcal{E}(r_3; A(S), D) - \mathcal{E}(r_2; A(S), D) < 2(d-1)\epsilon. \tag{16}$$

Using Lemma IV.4 (a) by identifying $r = r_2$ and $f = A(S)$ as the (estimated) true function, we get

$$\mathcal{E}(r_2; A(S), D) - \mathcal{E}(r_2; h, D) < \epsilon. \tag{17}$$

Finally, using Lemma IV.4 (b) by identifying D as the probability law, we have

$$\mathcal{E}(r_2; h, D) - \mathcal{E}(r_1; h, D) < \epsilon. \tag{18}$$

Thus combining (15), (17), (16), and (18), we get

$$\mathcal{E}(r_3; h, D) - \mathcal{E}(r_1; h, D) < (2d+1)\epsilon.$$

This equation holds for all $S \in \mathcal{S}$. But, we have $D^{\otimes m}(\mathcal{S}) > 1 - \delta - 2e^{-2m\epsilon^2}$. Let $\delta' = \delta + 2e^{-2m\epsilon^2}$. Thus we have

$$D^{\otimes m}(\mathcal{E}(r_3; h, D) - \mathcal{E}(r_1; h, D) < (2d+1)\epsilon) > 1 - \delta'.$$

The choice of D, (ϵ, δ) and h were arbitrary, therefore this shows that $\mathcal{F}$ is strategically learnable under $B \circ A$. ■

The previous theorem only provides sufficient conditions for strategic learnability. We will now present an algorithm that is both a necessary and sufficient condition for a hypothesis class to be strategically learnable.

C. Strategy aware ERM

The learning algorithm suggested in Algorithm 1 does not account for strategic behaviour during the testing phase. This is a potential source of inefficiency in determining the strategic learnability of a hypothesis class. An alternate algorithm which takes into account the strategic behaviour and which we show is both a necessary and sufficient condition for strategic learnability is described as follows. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ be the hypothesis class consisting of all functions with at most $d-1$ decision boundaries. Corresponding to this hypothesis class, consider the hypothesis class $\widetilde{\mathcal{F}}$ defined as

$$\widetilde{\mathcal{F}} := \{r \circ s_r \mid r := r^\star_{f,D}, D \in \mathfrak{D}, f \in \mathcal{F}\}.$$

We learn the Stackelberg equilibrium directly from $\widetilde{\mathcal{F}}$. The strategy-aware ERM algorithm expresses this idea.

Data: Hypothesis class $\mathcal{F}$, Sample $\{(x_i, y_i)\}_{i=1}^m$
Zero - one loss function
$l_h(r(s_r), x) := \mathbb{1}_{r(s_r(x))=h(x)}$
Result: Hypothesis $\hat{r}$

// Decision
$\widetilde{\mathcal{F}} := \{r \circ s_r \mid r := r^\star_{f,D}, D \in \mathfrak{D}, f \in \mathcal{F}\}$

// Estimation
$\hat{r}_{\text{serm}} \circ s_{\hat{r}_{\text{serm}}} := \arg\min_{r \circ s_r \in \widetilde{\mathcal{F}}} \sum_{i=1}^m \mathbb{1}_{\{r(s_r(x_i)) \neq y_i\}}$

Algorithm 2: Strategy aware ERM algorithm

The algorithm 2 is initialised with a hypothesis class $\mathcal{F}$ and a sample S. In the first step, we modify the hypothesis class $\mathcal{F}$ to $\widetilde{\mathcal{F}}$. In the second step, we run the ERM subroutine on the modified hypothesis class $\widetilde{\mathcal{F}}$ and sample S to get a hypothesis $\hat{r}_{\text{serm}} \circ s_{\hat{r}_{\text{serm}}}$. We now show that VC dim$(\widetilde{\mathcal{F}}) < \infty$ if and only if $\mathcal{F}$ is strategically learnable under Algorithm 2.

Theorem IV.5 (The fundamental theorem of strategic learning). *Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \to \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d-1$ decision boundaries. Let $A : (\mathcal{X} \times \mathcal{Y})^n \to \widetilde{\mathcal{F}}$ denote the algorithm described in Algorithm 2. The $\mathcal{F}$ is strategically learnable under A (refer Definition IV.3) iff VC dim$(\widetilde{\mathcal{F}}) < \infty$.*

Proof. Let $D \in \mathfrak{D}$ be the probability law on $\mathcal{X}$ and let $(\epsilon, \delta) \in (0,1)^2$. Let $h \in \mathcal{F}$ be the true function and let $S = \{(x_i, y_i)\}_{i=1}^m$ be the m-sized sample where x_i's are drawn i.i.d. from D. Let $\mathcal{E}(r, f) := \mathcal{E}(r, s_r, f, D)$. The Strategy aware ERM algorithm is similar to the vanilla ERM algorithm but applied to a different hypothesis class $\widetilde{\mathcal{F}}$ under the zero-one loss function $l_h(r \circ s_r, x) = \mathbb{1}_{\{r(s_r(x))=h(x)\}}$. We first prove the if part. Let $A(S)$ be the hypothesis obtained from the strategy aware ERM algorithm. Since VC dim$(\widetilde{\mathcal{F}}) < \infty$, we know that it is agnostic PAC learnable from Theorem IV.1. Furthermore, from Theorem IV.1 we have for $m \geq \bar{m}(\epsilon, \delta)$,

$$D^{\otimes m}(\mathcal{E}(A(S), h) - \min_{r \circ s_r \in \widetilde{\mathcal{F}}} \mathcal{E}(r, h) < \epsilon) > 1 - \delta. \tag{19}$$

Note that by construction of $\widetilde{\mathcal{F}}$, we have

$$\min_{f \in \mathcal{F}} \mathcal{E}(r_f, s_{r_f}, h, D) = \min_{r \circ s_r \in \widetilde{\mathcal{F}}} \mathcal{E}(r, s_r, h, D).$$

Substituting this in (19), we get with probability $1 - \delta$,

$$D^{\otimes m}(\mathcal{E}(A(S), h) - \min_{f \in \widetilde{\mathcal{F}}} \mathcal{E}(r, h) < \epsilon) > 1 - \delta. \tag{20}$$

The choice of D, (ϵ, δ) and h were arbitrary. Hence (20) shows that $\mathcal{F}$ is strategically learnable.
Next, we prove the only if part. If $\mathcal{F}$ is strategically learnable, there exists an algorithm $A' : (\mathcal{X} \times \mathcal{Y}) \to \widetilde{\mathcal{F}}$ such that

$$D^{\otimes m}\Big(\mathcal{E}(A'(S); h, D) - \mathcal{E}(r^\star_{h,D}; h, D) < \epsilon\Big) > 1 - \delta \tag{21}$$

holds. Note that from the definition of $r^\star_{h,D}$, $\mathcal{E}(r^\star_{h,D}; h, D) \leq \mathcal{E}(r; h, D)$ for all $r \in \widetilde{\mathcal{F}}$. Furthermore, from the definition of $\mathcal{E}(\cdot; h)$ and $L(\cdot; h)$, we have $\mathcal{E}(A(S), h) = L(A(S); h)$. Therefore, (21) reduces to

$$D^{\otimes m}\Big(L(A'(S); h) - \arg\min_{r \in \widetilde{\mathcal{F}}} L(r; h) < \epsilon\Big) > 1 - \delta.$$

This shows that $\widetilde{\mathcal{F}}$ is agnostic PAC learnable under $A'(S)$. Therefores from the Theorem IV.1 we have VC dim$(\widetilde{\mathcal{F}}) < \infty$. ■

Corollary IV.3. Let $\mathcal{F}$ be a hypothesis class used in Algorithm 1 and $\widetilde{\mathcal{F}}$ be the hypothesis in Algorithm 2 obtained from $\mathcal{F}$. Then

$$\text{VC dim} \quad (\mathcal{F}) < \infty \implies \quad \text{VC dim} \quad (\widetilde{\mathcal{F}}) < \infty.$$

We now demonstrate that certain hypothesis classes are not strategically learnable via Algorithm 1 but become strategically learnable via Algorithm 2. Specifically, we show the existence of a hypothesis class $\mathcal{F}$ such that VC dim$(\mathcal{F}) = \infty$ but VC dim$(\tilde{\mathcal{F}}) < \infty$. This establishes that the converse of Corollary IV.3 does not hold.

Example 5. Consider the deterministic decision problem described in Section II. Let h be the true function, D be the uniform probability law on $\mathcal{X}$ and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. Let $\mathbf{f} \in [0, 1)$, and let $F \subseteq ([\mathbf{f}, 1] \cap \mathbb{Q})$. Define the hypothesis $f_F : \mathcal{X} \to \mathcal{Y}$ as follows:

$$f_F(x) := \begin{cases} - & x \in [0, \mathbf{f}), \\ + & x \in F \setminus [0, \mathbf{f}), \\ - & x \in F^c \setminus [0, \mathbf{f}) \quad \text{where} \quad F^c = X \backslash F. \end{cases}$$

Now, consider the following hypothesis class:

$$\mathcal{F} := \{f_F : \mathbf{f} \in (0, 1),\ F \subseteq ([\mathbf{f}, 1] \cap \mathbb{Q})\}$$

Then VC dim$(\mathcal{F}) = \infty$ because $\mathcal{F}$ can shatter every countable subset of $[\mathbf{f}, 1] \cap \mathbb{Q}$. We now claim that VC dim$(\widetilde{\mathcal{F}}) = 0$. Let $f_F \in \mathcal{F}$ be the true function. The receiver strategy $r(x) \equiv -$ is a Stackelberg equilibrium. This is because $\mathcal{E}(r) = 0$. Thus, we have $\widetilde{\mathcal{F}} = \{r(\cdot) \equiv -\}$, implying that the VC dimension of $\widetilde{\mathcal{F}}$ is 0. In conclusion, while $\mathcal{F}$ is not strategically learnable by Algorithm 1 due to its infinite VC dimension, it becomes strategically learnable by Algorithm 2, as the hypothesis space is effectively reduced to one with finite VC dimension.

D. Comparison between algorithms

We now consider an example of a learning problem and show how strategy-aware ERM compares with standard ERM in various situations. This example guides our intuition on when to choose which algorithm.

Example 6. Consider the decision problem where the true function has d decision boundaries. Let the probability measure $\mathfrak{D}$ be uniform on $[0, 1]$ and let $\mathbf{u}_+ = \mathbf{u}_- = \mathbf{u} := 0.2$. To keep things simple, we additionally assume that each sub-interval O_i has the same width $k\mathbf{u}$, where k is a fraction (this implies $n \approx \frac{1}{\mathbf{u}k}$ assuming an exact division). From Proposition IV.1, we know the structure of the receiver Stackelberg strategy r. Because of symmetry, we further assume that the Δ-region is equally shared between two adjacent intervals I_i and I_{i+1}, except when the entire I_i is a Δ-region. Then, the structure of $r \circ s_r$ for different numbers of decision boundaries is given as follows:

Case 3. $k > 1$. In this case, the number of decision boundaries is $3n$.

Case 4. $k \leq 1$. In this case, the r assigns label Δ to the minority class. Hence, the number of decision boundaries is 1.

From the above example, we can see that as complexity increases, the number of decision boundaries increases to a point and then becomes 1. The threshold point, which is $\frac{1}{\mathbf{u}}$, can be considered a boundary separating the simple hypothesis from the complex hypothesis. This means that the complexity of a hypothesis is with respect to a given game structure, which here is characterised by $\mathbf{u}$.

Now consider F_n, a set of all hypotheses with a number of decision boundaries up to n. Clearly, VC dim$(F_n) \geq$ VC dim(F_m) if $n \geq m$. Now, if $\mathcal{F} = F_n$, then from Example 6, we have $\widetilde{\mathcal{F}} \subset F_n$ in Case 1 and $F_n \subset \widetilde{\mathcal{F}}$ in Case 2. Therefore, VC dim$(\mathcal{F}) \geq$ VC dim$(\widetilde{\mathcal{F}})$ and VC dim$(\mathcal{F}) \leq$ VC dim$(\widetilde{\mathcal{F}})$ respectively. Heuristically, it means that if the VC dim$(\mathcal{F})$ is large with respect to $\mathbf{u}$, then choose Algorithm 2, or else choose Algorithm 1.

V. Conclusions and Limitations

We studied the strategic classification problem where a strategic sender of data has a non-uniform preferences over labels that a receiver can assign. We analyzed two versions of this problem: a decision problem and a learning problem. For the decision problem, we showed that expanding the classifier's action space by either introducing a penalty label that ignores certain feature vectors or introducing a randomized classification rule, significantly improves classification accuracy against a strategic sender. Our results show that an optimal deterministic classifier equipped with a penalty label makes fewer errors than one without a penalty label. Furthermore, the optimal stochastic classifier makes fewer errors than the optimal deterministic classifier with a penalty label. For the learning problem, we show that incorporating the sender's best response structure can enhance the learning process. We demonstrated this by introducing and comparing two algorithms, vanilla ERM and strategy-aware ERM, which integrate game theory and learning theory in distinct ways. Vanilla ERM is preferable for simpler hypothesis classes and strategy-ware ERM is preferable for complex hypothesis classes. This opens avenues for exploring more nuanced combinations of game theory and learning, potentially leading to more powerful strategic learning algorithms. While our work is limited to the assumption of a scalar feature space or a vector feature space with a linear class boundary, and of simple utility and cost functions, the structural insights it provides offer guidance for understanding strategic classification in more complex settings.

VI. Acknowledgments

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VII. Appendix: Omitted Proofs

A. Omitted Proofs of Section III.B

We analyze the worst-case best response and error of a strategy $r \in \mathscr{R}_2$.

Lemma VII.1. *Let $c \in \mathcal{Y}$ be a label and $r \in \mathscr{R}_2$ be a constant-label receiver strategy $r(\cdot) \equiv c$. Then $s_r(x) = x$ and $E(r)^c = \{x : h(x) = c\}$ (here ^{c} denotes the set complement).*

Proof. Since the receiver uses a constant label strategy, for a given $x \in \mathcal{X}$, the sender's payoff is $U(r(s_r(x)), x) - C(s_r(x), x) = U(c, x) - C(s_r(x), x)$. Since, the utility term remains fixed irrespective of the sender strategy, the sender will focus on minimizing the cost term. Consequently, $s_r(x) = x$ incurs the least cost. The receiver strategy r correctly classify feature vectors whose true label is c, hence $E(r)^c = \{x : h(x = c)\}$. ■

We begin by analyzing the worst-case best response and error of a strategy r in $\mathscr{R}_1$.

Lemma VII.2. *Let $r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_1$.*

(a) If

$$\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}),$$

then $s_r(\cdot)$ is given by

$$s_r(x) := \begin{cases} x & \text{if} \quad x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^- & \text{if} \quad x \in (\mathbf{h}, \mathbf{g}), \\ \mathbf{h}^+ & \text{if} \quad x \in [\mathbf{h}^-, \mathbf{h}] \cup [\mathbf{g}, \mathbf{h}^+], \end{cases}$$

where $\mathbf{g} := \phi^{-1}\left(\frac{\phi(\mathbf{h}^-)+\phi(\mathbf{h}^+)+\mathbf{u}_-}{2}\right)$. Furthermore, $E(r) = [\mathbf{h}^-, \mathbf{g}]$.

(b) If

$$\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}),$$

then $s_r(\cdot)$ *is given by*

$$s_r(x) := \begin{cases} x & \textit{if} \quad x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^+ & \textit{if} \quad x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

Furthermore $E(r) = [\mathbf{h}^-, \mathbf{h}]$ *and* $\mathbf{u}_+ \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^-) \geq \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$.

Proof. We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$.
Case 1: $x \in [0, \mathbf{h}^-)$
The sender's payoff is given by $U(r(s_r(x)), x) - |\phi(s_r(x)) - \phi(x)|$. In the utility term, $r(s_r(x))$ can result in three labels: $-, +$ and Δ. The Δ label is unpreferable because it gives a $-\infty$ utility. Hence the sender will choose $s_r(x)$ such that $r(s_r(x)) = -$ or $+$. For $r(s_r(x)) = -$, from the definition of r, $s_r(x) \in [0, \mathbf{h}^-]$. Among these options, the feature vector which incurs the least cost to the sender is x. For $r(s_r(x)) = +$, from the definition of r, $s_r(x) \in [\mathbf{h}^+, 1]$. Among these options, the feature vector which incurs the least cost to the sender is $\mathbf{h}^+$. Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_+ + \phi(x) - \phi(s_r(x)) & \text{if} \quad s_r(x) = \mathbf{h}^+ \\ 0 & \text{if} \quad s_r(x) = x. \end{cases}$$

Since ϕ is an increasing function and $r \in \mathscr{R}_1$ satisfies $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+ > 0$, therefore for $x \in [0, \mathbf{h}^-)$, we have

$$\mathbf{u}_+ + \phi(x) - \phi(s_r(x)) < \mathbf{u}_+ + \phi(\mathbf{h}^-) - \phi(\mathbf{h}^+) = \mathbf{u}_+ - \mathbf{u}_+ = 0.$$

Thus the payoff for modifying x to $\mathbf{h}^+$ is negative and hence $s_r(x) = x$.

Case 2: $x \in (\mathbf{h}, \mathbf{h}^+]$.
Arguing as in the Case 1, the $s_r(x)$ can be equal to $\mathbf{h}^-$ or $\mathbf{h}^+$. Hence the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \phi(x) - \phi(\mathbf{h}^+) & \text{if} \quad s_r(x) = \mathbf{h}^+ \\ \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}^-. \end{cases}$$

If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x)$, then it follows that $\phi(x) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x)$. Hence $s_r(x) = \mathbf{h}^+$. If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then their exists $\mathbf{g} \in (\mathbf{h}, \mathbf{h}^+]$ (same $\mathbf{g}$ that is defined in the statement of this lemma) such that

$$\phi(\mathbf{g}) - \phi(\mathbf{h}^+) = \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x).$$

Hence, if $\phi(x) \geq \phi(\mathbf{g})$, then $s_r(x) = \mathbf{h}^+$, and if $\phi(x) < \phi(\mathbf{g})$, then $s_r(x) = \mathbf{h}^-$. Since ϕ is an increasing function, this implies that

$$s_r(x) = \begin{cases} \mathbf{h}^-, & \text{if} \quad x \in (\mathbf{h}, \mathbf{g}) \\ \mathbf{h}^+, & \text{if} \quad x \in [\mathbf{g}, \mathbf{h}^+]. \end{cases}$$

Case 3: $x \in [\mathbf{h}^-, \mathbf{h}]$.
Arguing as in the Case 1, the $s_r(x)$ can be equal to $\mathbf{h}^-$ or $\mathbf{h}^+$. Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_+ + \phi(x) - \phi(\mathbf{h}^+) & \text{if} \quad s_r(x) = \mathbf{h}^+ \\ \phi(\mathbf{h}^-) - \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}^+. \end{cases}$$

Since

$$\mathbf{u}_+ + \phi(x) - \phi(\mathbf{h}^+) \geq 0 \quad \text{and} \quad \phi(\mathbf{h}^-) - \phi(x) < 0,$$

therefore $s_r(x) = \mathbf{h}^+$.

Case 4: Let $x \in (\mathbf{h}^+, 1]$. Arguing as in the Case 1, the $s_r(x)$ can be equal to $\mathbf{h}^-$ or x. Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}^- \\ 0 & \text{if} \quad s_r(x) = x. \end{cases}$$

Since ϕ is an increasing function and $r \in \mathscr{R}_1$ satisfies $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+ > 0$, therefore for $x \in [\mathbf{h}^+, 1]$, we have

$$\mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) < \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}^+) < \mathbf{u}_- - \mathbf{u}_+ \leq 0.$$

Therefore $s_r(x) = x$.
Hence combining all the cases, we get that if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x)$, then

$$s_r(x) := \begin{cases} x & \text{for} \quad x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \mathbf{h}^- & \text{for} \quad x \in [\mathbf{h}^-, \mathbf{g}) \\ \mathbf{h}^+ & \text{for} \quad x \in [\mathbf{h}^-, \mathbf{h}] \cup [\mathbf{g}, \mathbf{h}^+], \end{cases}$$

and $E(r) = [\mathbf{h}^-, \mathbf{g}]$, where $g := \phi^{-1}(\frac{\phi(\mathbf{h}^-) + \phi(\mathbf{h}^+) + \mathbf{u}_-}{2})$ which proves part (a).
Similarly combining all the cases, we get if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x)$, then

$$s_r(x) := \begin{cases} x & \text{for} \quad x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \mathbf{h}^+ & \text{for} \quad x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

and $E(r) = [\mathbf{h}^-, \mathbf{h}^+]$ which proves part (b). ■

Lemma VII.3. *Let r be a receiver strategy. Let $h(x)$ be a true function, and let $x_1, x_2 \in E(r)^c$ (^{c} denote the set complement) be feature vectors such that $x_1 \leq \mathbf{h} \leq x_2$. Let s_r be the worst-case best response of r. Let $z_1, z_2 \in \mathcal{X}$ be such that $z_1 = s_r(x_1)$, $z_2 = s_r(x_2)$ and, $r(z_1) \neq r(z_2)$ and $r(z_1), r(z_2) \neq \Delta$. Now consider the following two sender strategies*

$$\hat{s}(t) := \begin{cases} z_2 & \textit{for} \quad t = x_1 \\ s_r(t) & \textit{otherwise} \end{cases}, \quad \bar{s}(t) := \begin{cases} z_1 & \textit{for} \quad t = x_2 \\ s_r(t) & \textit{otherwise} \end{cases}.$$

Then

$$\mathbb{U}(x_1; r, s_r) > \mathbb{U}(x_1; r, \hat{s}) \quad \textit{and} \quad \mathbb{U}(x_2; r, s_r) > \mathbb{U}(x_2; r, \bar{s}).$$

Proof. Since s_r is a worst-case best response to r, we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \hat{s}) \quad \text{and} \quad \mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \bar{s}) \quad \text{for all} \quad x \in \mathcal{X}.$$

Since $\hat{s}(t) = s_r(t)$ for $t \in \mathcal{X} \backslash \{x_1\}$ and $\bar{s}(t) = s_r(t)$ for $t \in \mathcal{X} \backslash \{x_2\}$, we have

$$\mathbb{U}(t; r, s_r) = \mathbb{U}(t; r, \hat{s}) \quad \text{for} \quad t \in \mathcal{X} \backslash \{x_1\} \quad \text{and} \quad \mathbb{U}(t; r, s_r) = \mathbb{U}(t; r, \bar{s}) \quad \text{for} \quad t \in \mathcal{X} \backslash \{x_2\}.$$

If $\mathbb{U}(x_1; r, s_r) = \mathbb{U}(x_1; r, \hat{s})$, then $\hat{s} \in B(r)$ (the best response set of r) and $E(r, \hat{s}) = E(r, s_r) \cup \{x_1\}$. This contradicts the definition of s_r. Hence, $\mathbb{U}(x_1; r, s_r) \neq \mathbb{U}(x_1; r, \hat{s})$. Similarly, if $\mathbb{U}(x_2; r, s_r) = \mathbb{U}(x_2; r, \bar{s})$, then $\bar{s} \in B(r)$ and $E(r, \hat{s}) = E(r, s_r) \cup \{x_2\}$. This contradicts the definition of s_r. Hence, $\mathbb{U}(x_2; r, s_r) \neq \mathbb{U}(x_2; r, \bar{s})$. Therefore, it follows that

$$\mathbb{U}(x_1; r, s_r) > \mathbb{U}(x_1; r, \hat{s}) \quad \text{and} \quad \mathbb{U}(x_2; \tilde{r}, s_r) > \mathbb{U}(x_2; r, \bar{s}).$$

which implies

$$-|\phi(z_1) - \phi(x_1)| > \mathbf{u}_+ - |\phi(z_2) - \phi(x_1)| \quad \text{and} \quad -|\phi(z_2) - \phi(x_2)| > \mathbf{u}_- - |\phi(z_1) - \phi(x_2)|.$$

■

We say a strategy r dominates another receiver strategy $\tilde{r}$ if $E(r, s_r, h, D) \subseteq E(\tilde{r}, s_{\tilde{r}}, h, D)$. We now show that the family of receiver strategies $\mathscr{R}$ dominates every other receiver strategy.

Lemma VII.4. *Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$ and let $h(x)$ be a true function. Let $\tilde{r}$ be any receiver strategy. Then there exists a strategy $r \in \mathscr{R}$ such that $\mathcal{E}(\tilde{r}) \geq \mathcal{E}(r)$.*

Proof. (Proof of Lemma VII.4)
Let $\mathbf{h}^- := \sup(E(\tilde{r})^c \cap [0, \mathbf{h}])$ and $\tilde{\mathbf{g}} := \inf(E(\tilde{r})^c \cap (\mathbf{h}, 1])$ (here ^{c} represents the set complement). It follows that $\mathbf{h}^-$ represents the supremum of all the feature vectors with true label $-$ that get correctly classified by $\tilde{r}$. Similarly, $\tilde{\mathbf{g}}$ represents the infimum of all the feature vectors with true label $+$ that get correctly classified by $\tilde{r}$. Hence $E(\tilde{r}) \supset (\mathbf{h}^-, \tilde{\mathbf{g}})$. Let $\mathbf{h}^+ \in \mathcal{X}$ be such that $\mathbf{h} < \mathbf{h}^+ \leq 1$ and $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$. Such a feature vector $\mathbf{h}^+$ exists due to the continuity of ϕ.
Case 1: $\mathbf{h}^- = 0$:
In this case, $E(\tilde{r}) \supset (0, \mathbf{h})$, i.e., it misclassifies all the feature vectors with the true label $-$. Hence, for a receiver strategy $r \in \mathscr{R}_2$ which assigns the majority label, $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case 2: $\tilde{\mathbf{g}} = 1$:
In this case, $E(\tilde{r}) \supset (\mathbf{h}, 1)$ i.e., it misclassifies all the feature vectors with the true label $+$. Hence, for a receiver strategy $r \in \mathscr{R}_2$ which assigns the majority label, $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case 3: $\mathbf{h}^- > 0, \tilde{\mathbf{g}} < 1$ and $\phi(\tilde{\mathbf{g}}) - \phi(\mathbf{h}^-) \geq \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$:
From the assumption on $\tilde{\mathbf{g}}$, it follows that $\mathbf{h} \leq \tilde{\mathbf{g}} \leq \mathbf{h}^+$. Now consider a receiver strategy $r(x; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_1$. If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then from Lemma VII.2 (a), $E(r) = [\mathbf{h}^-, \mathbf{g}]$, where $\mathbf{g} := \phi^{-1}\left(\frac{\phi(\mathbf{h}^-) + \phi(\mathbf{h}^+) + \mathbf{u}_-}{2}\right)$. Since $\phi(\mathbf{g}) - \phi(\mathbf{h}^-) = \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$, therefore $\mathbf{g} \leq \tilde{\mathbf{g}}$. Hence

$$E(r) = [\mathbf{h}^-, \mathbf{g}] \quad \text{and} \quad E(\tilde{r}) \supseteq (\mathbf{h}^-, \tilde{\mathbf{g}})$$

which implies $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$. If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then from Lemma VII.2 (b), $E(r) = [\mathbf{h}^-, \mathbf{h}]$. Since $\tilde{\mathbf{g}} \geq \mathbf{h}$, therefore we have

$$E(r) = [\mathbf{h}^-, \mathbf{h}] \quad \text{and} \quad E(\tilde{r}) \supset (\mathbf{h}^-, \tilde{\mathbf{g}}).$$

Hence $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case 4: $\mathbf{h}^- > 0, \tilde{\mathbf{g}} < 1$ and $\phi(\tilde{\mathbf{g}}) - \phi(\mathbf{h}^-) < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$:
We will show that this case does not occur. Let ϵ be such that $0 < 2\epsilon < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2} - \phi(\tilde{\mathbf{g}}) + \phi(\mathbf{h}^-)$. Let $\mathbf{h}^{-\prime}, \tilde{\mathbf{g}}' \in E(\tilde{r})^c$ be such that $\mathbf{h}^{-\prime} \leq \mathbf{h}^- \leq \mathbf{h} < \tilde{\mathbf{g}} \leq \tilde{\mathbf{g}}'$ and

$$\phi(\mathbf{h}^-) - \phi(\mathbf{h}^{-\prime}) < \epsilon \qquad \text{and} \qquad \phi(\tilde{\mathbf{g}}') - \phi(\tilde{\mathbf{g}}) < \epsilon. \tag{22}$$

Such feature vectors $\mathbf{h}^{-\prime}, \tilde{\mathbf{g}}'$ exist by the definition of $\mathbf{h}^-$ and $\tilde{\mathbf{g}}$ and the continuity of ϕ. Adding the inequalities in (22), and using the definition of ϵ, we get

$$\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-\prime}) < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}. \tag{23}$$

Let $y_1 := s_{\tilde{r}}(\mathbf{h}^{-\prime})$ and $y_2 := s_{\tilde{r}}(\tilde{\mathbf{g}}')$. It follows that

$$\tilde{r}(y_1) = h(\mathbf{h}^{-\prime}) = - \qquad \text{and} \qquad \tilde{r}(y_2) = h(\tilde{\mathbf{g}}') = +.$$

Furthermore, the sender's payoff at $\mathbf{h}^{-\prime}$ and $\tilde{\mathbf{g}}'$ under the worst-case best response $s_{\tilde{r}}$ is

$$\mathbb{U}(\mathbf{h}^{-\prime}; \tilde{r}, s_{\tilde{r}}) = U(\tilde{r}(y_1), \mathbf{h}^{-\prime}) - |\phi(y_1) - \phi(\mathbf{h}^{-\prime})| = -|\phi(y_1) - \phi(\mathbf{h}^{-\prime})| \leq 0 \tag{24}$$

and

$$\mathbb{U}(\tilde{\mathbf{g}}'; \tilde{r}, s_{\tilde{r}}) = U(\tilde{r}(y_2), \tilde{\mathbf{g}}') - |\phi(y_2) - \phi(\tilde{\mathbf{g}}')| = -|\phi(y_2) - \phi(\tilde{\mathbf{g}}')| \leq 0$$

respectively.
Additionally applying Lemma VII.3 by taking $r = \tilde{r}, x_1 = \mathbf{h}^{-\prime}$, $x_2 = \tilde{\mathbf{g}}'$, we get $z_1 = y_1, z_2 = y_2$ and,

$$-|\phi(y_1) - \phi(\mathbf{h}^{-\prime})| > \mathbf{u}_+ - |\phi(y_2) - \phi(\mathbf{h}^{-\prime})| \quad \text{and} \quad -|\phi(y_2) - \phi(\tilde{\mathbf{g}}')| > \mathbf{u}_- - |\phi(y_1) - \phi(\tilde{\mathbf{g}}')|. \tag{25}$$

Claim 1. $\tilde{r}(x) = \Delta$ for $x \in [\mathbf{h}^{-\prime}, \tilde{\mathbf{g}}']$.
Let $\mathbf{k}^+, \mathbf{k}^- \in \mathcal{X}$ be such that $\mathbf{k}^- < \mathbf{h}^{-\prime} < \mathbf{k}^+$, $\phi(\mathbf{k}^+) - \phi(\mathbf{h}^{-\prime}) = \mathbf{u}_+$ and $\phi(\mathbf{h}^{-\prime}) - \phi(\mathbf{k}^-) = \mathbf{u}_+$. Such feature vectors $\mathbf{k}^-, \mathbf{k}^+$ exist due to the continuity of ϕ. We now show that

$$\mathbf{k}^- < \mathbf{h}^{-\prime} < \tilde{\mathbf{g}}' < \mathbf{k}^+. \tag{26}$$

To show this, we only need to prove that $\tilde{\mathbf{g}}' < \mathbf{k}^+$. Recall from the assumption of **IV** that $\mathbf{u}_+ \geq |\mathbf{u}_-|$. From this, we have

$$\phi(\mathbf{k}^+) - \phi(\tilde{\mathbf{g}}') = \phi(\mathbf{k}^+) - \phi(\mathbf{h}^{-\prime}) + \phi(\mathbf{h}^{-\prime}) - \phi(\tilde{\mathbf{g}}') > \mathbf{u}_+ - \frac{\mathbf{u}_+ + \mathbf{u}_-}{2} = \frac{\mathbf{u}_+ - \mathbf{u}_-}{2} \geq 0, \tag{27}$$

where we have used the definition of $\mathbf{k}^+$ and (23) to lower bound the middle term. Hence, $\phi(\mathbf{k}^+) > \phi(\tilde{\mathbf{g}}')$. Since ϕ is strictly increasing, we have $\mathbf{k}^+ > \tilde{\mathbf{g}}'$ as required.
Now to prove the claim, we first show that $\tilde{r}(x) \neq +$ for $x \in [\mathbf{k}^-, \mathbf{k}^+]$. Suppose there exists $\mathbf{a} \in [\mathbf{k}^-, \mathbf{k}^+]$ such that $\tilde{r}(\mathbf{a}) = +$. Consider a sender strategy s defined as

$$s(t) = \begin{cases} \mathbf{a} & \text{for} \quad t = \mathbf{h}^{-\prime} \\ s_r(t) & \text{otherwise.} \end{cases}$$

Then, at the feature vector $\mathbf{h}^{-\prime}$, the sender can map $\mathbf{h}^{-\prime}$ to $s(\mathbf{h}^{-\prime}) = \mathbf{a}$ and obtain a payoff

$$\mathbb{U}(\mathbf{h}^{-\prime}; \tilde{r}, s) = U(\tilde{r}(s(\mathbf{h}^{-\prime})), \mathbf{h}^{-\prime}) - |\phi(s(\mathbf{h}^{-\prime})) - \phi(\mathbf{h}^{-\prime})| = U(+, \mathbf{h}^{-\prime}) - |\phi(\mathbf{a}) - \phi(\mathbf{h}^{-\prime})| \geq \mathbf{u}_+ - \mathbf{u}_+ = 0.$$

Using (24), it follows that the sender obtains a better payoff at $\mathbf{h}^{-\prime}$ by playing s instead of $s_{\tilde{r}}$. But this contradicts the fact that $s_{\tilde{r}}$ is the worst-case best response to $\tilde{r}$. Hence, such a sender strategy s cannot exist, which implies that

$$\tilde{r}(x) \neq + \qquad \text{for all} \quad x \in [\mathbf{k}^-, \mathbf{k}^+]. \tag{28}$$

We next show that $\tilde{r}(x) \neq -$ for $x \in [\mathbf{h}^{-\prime}, \tilde{\mathbf{g}}']$. Suppose there exists $\mathbf{b} \in [\mathbf{h}^{-\prime}, \tilde{\mathbf{g}}']$ such that $\tilde{r}(\mathbf{b}) = -$. Using (28) and $\tilde{r}(y_2) = +$, we have $y_2 \notin [\mathbf{k}^-, \mathbf{k}^+]$. Since $\tilde{\mathbf{g}}' \in [\mathbf{k}^-, \mathbf{k}^+]$ from (26), we have

$$|\phi(y_2) - \phi(\tilde{\mathbf{g}}')| \geq \min\{\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{k}^-), \phi(\mathbf{k}^+) - \phi(\tilde{\mathbf{g}}')\}.$$

We now lower bound the right-hand side. From (27), we have $\phi(\mathbf{k}^+) - \phi(\tilde{\mathbf{g}}') > \frac{\mathbf{u}_+ - \mathbf{u}_-}{2}$. Furthermore, from (26) and the definition of $\mathbf{k}^-$, we have $\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{k}^-) > \phi(\mathbf{h}^{-\prime}) - \phi(\mathbf{k}^-) = \mathbf{u}_+$. Therefore,

$$|\phi(y_2) - \phi(\tilde{\mathbf{g}}')| > \frac{\mathbf{u}_+ - \mathbf{u}_-}{2}. \tag{29}$$

Consider a sender strategy s' defined as

$$s'(t) = \begin{cases} \mathbf{b} & \text{for} \quad t = \tilde{\mathbf{g}}' \\ s_r(t) & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{U}(\tilde{\mathbf{g}}'; \tilde{r}, s') = U(\tilde{r}(s'(\tilde{\mathbf{g}}')), \tilde{\mathbf{g}}') - |\phi(s'(\tilde{\mathbf{g}}')) - \phi(\tilde{\mathbf{g}}')| = \mathbf{u}_- - \phi(\tilde{\mathbf{g}}') + \phi(\mathbf{b}).$$

Since $s_{\tilde{r}}$ is the worst-case best response, at the feature vector $\tilde{\mathbf{g}}'$, we have

$$\mathbb{U}(\tilde{\mathbf{g}}'; \tilde{r}, s') \leq \mathbb{U}(\tilde{\mathbf{g}}'; \tilde{r}, s_{\tilde{r}}),$$

which, using (29), gives

$$\mathbf{u}_- - \phi(\tilde{\mathbf{g}}') + \phi(\mathbf{b}) \leq -|\phi(y_2) - \phi(\tilde{\mathbf{g}}')| < \frac{\mathbf{u}_- - \mathbf{u}_+}{2}.$$

This implies

$$\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{b}) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}.$$

Since $\mathbf{b} \in [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$, and ϕ is increasing, we have $\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{b}) \leq \phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'})$. Hence

$$\frac{\mathbf{u}_+ + \mathbf{u}_-}{2} < \phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}).$$

But this contradicts (23). Hence,

$$\tilde{r}(x) \neq - \qquad \text{for all} \quad x \in [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']. \tag{30}$$

From (26), we have $[\mathbf{h}^{-'}, \tilde{\mathbf{g}}'] \subset [\mathbf{k}^+, \mathbf{k}^-]$. Hence, combining (28) with (30), it follows that $\tilde{r}(x) \neq +, -$ for $x \in [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$. Therefore, $\tilde{r}(x) = \Delta$ for $x \in [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$. As a consequence, we have that

$$y_1, y_2 \notin [\mathbf{h}^{-'}, \tilde{\mathbf{g}}'] \tag{31}$$

because it would result in a $-\infty$ payoff for the sender.

Claim 2. $y_1 < \mathbf{h}^{-'}$.
We will prove this claim by showing that $y_1 \not> \tilde{\mathbf{g}}'$. Combining this with (31), we will get $y_1 < \mathbf{h}^{-'}$. Assume that $y_1 > \tilde{\mathbf{g}}'$. If $y_2 > \tilde{\mathbf{g}}'$, then from (25) we get

$$\phi(y_1) < \phi(y_2) - \mathbf{u}_+ \quad \text{and} \quad \phi(y_2) < \phi(y_1) - \mathbf{u}_-$$

respectively. Combining these inequalities, we get

$$\phi(y_2) < \phi(y_2) - \mathbf{u}_+ - \mathbf{u}_-.$$

Since $\mathbf{u}_+ \geq |\mathbf{u}_-|$, this results in a contradiction. Therefore, either $y_1 \not> \tilde{\mathbf{g}}'$ or $y_2 \not> \tilde{\mathbf{g}}'$. If $y_1 \not> \tilde{\mathbf{g}}'$, then we have proved our claim. Hence, suppose $y_2 \not> \tilde{\mathbf{g}}'$ and $y_1 > \tilde{\mathbf{g}}'$. Since $y_2 \notin [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$ from (31), therefore suppose $y_2 < \mathbf{h}^{-'}$. From (25), we get

$$\phi(\mathbf{h}^{-'}) > \frac{\phi(y_1) + \phi(y_2) + \mathbf{u}_+}{2} \quad \text{and} \quad \phi(\tilde{\mathbf{g}}') < \frac{\phi(y_1) + \phi(y_2) - \mathbf{u}_-}{2}.$$

respectively. Since $\phi(\tilde{\mathbf{g}}') > \phi(\mathbf{h}^{-'})$, combining these inequalities yields

$$\frac{\phi(y_1) + \phi(y_2) - \mathbf{u}_-}{2} > \frac{\phi(y_1) + \phi(y_2) + \mathbf{u}_+}{2},$$

which implies

$$-\mathbf{u}_- > \mathbf{u}_+.$$

But this contradicts $\mathbf{u}_+ \geq |\mathbf{u}_-|$. Therefore, either $y_1 \not> \tilde{\mathbf{g}}'$ or $y_2 \not< \mathbf{h}^{-'}$. But $y_2 \in \mathcal{X}$, and previously we have supposed $y_2 \not> \tilde{\mathbf{g}}'$ and $y_2 \in [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$, therefore $y_1 \not> \tilde{\mathbf{g}}'$.

Claim 3. $y_2 > \tilde{\mathbf{g}}'$.
We will prove this claim by showing that $y_1 \not< \mathbf{h}^{-'}$. Combining this with (31), we will get $y_1 > \tilde{\mathbf{g}}'$. Assume that $y_2 < \mathbf{h}^{-'}$. If $y_1 > \tilde{\mathbf{g}}'$, then from (25), we get

$$\phi(y_2) > \mathbf{u}_- + \phi(y_1) \quad \text{and} \quad \phi(y_1) > \mathbf{u}_+ + \phi(y_2),$$

respectively. Combining these inequalities, we get

$$\phi(y_1) > \phi(y_1) + \mathbf{u}_+ + \mathbf{u}_-.$$

Since $\mathbf{u}_+ \geq |\mathbf{u}_-|$, this results in a contradiction. Therefore, either $y_1 \not> \tilde{\mathbf{g}}'$ or $y_2 \not< \mathbf{h}^{-'}$. If $y_2 \not< \mathbf{h}^{-'}$, then we have proved our claim. Hence, suppose $y_1 \not> \tilde{\mathbf{g}}'$ and $y_2 < \mathbf{h}^{-'}$. Since $y_1 \notin [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$ from (31), therefore suppose $y_2 < \mathbf{h}^{-'}$. Then (25) reduces to

$$\phi(\tilde{\mathbf{g}}') < \frac{\phi(y_1) + \phi(y_2) - \mathbf{u}_-}{2} \quad \text{and} \quad \phi(\mathbf{h}^{-'}) > \frac{\phi(y_1) + \phi(y_2) + \mathbf{u}_+}{2},$$

respectively. Since $\phi(\tilde{\mathbf{g}}') > \phi(\mathbf{h}^{-'})$, combining these inequalities gives

$$\frac{\phi(y_1) + \phi(y_2) - \mathbf{u}_-}{2} > \frac{\phi(y_1) + \phi(y_2) + \mathbf{u}_+}{2},$$

which implies

$$-\mathbf{u}_- > \mathbf{u}_+.$$

But this contradicts $\mathbf{u}_+ \geq |\mathbf{u}_-|$. Therefore, either $y_1 \not< \mathbf{h}^{-'}$ or $y_2 \not< \mathbf{h}^{-'}$. But $y_1 \in \mathcal{X}$, and previously we have supposed $y_1 \not> \tilde{\mathbf{g}}'$ and $y_1 \notin [\mathbf{h}^{-'}, \tilde{\mathbf{g}}']$, therefore $y_2 \not< \mathbf{h}^{-'}$.

Claim 4. $\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$.
From Claims 1-3, we have

$$y_1 < \mathbf{h}^{-'} < \mathbf{h} < \tilde{\mathbf{g}}' < y_2. \tag{32}$$

Using (32) in (25), we get

$$\phi(\mathbf{h}^{-'}) < \frac{\phi(y_1) + \phi(y_2) - \mathbf{u}_+}{2} \quad \text{and} \quad \phi(\tilde{\mathbf{g}}') > \frac{\phi(y_1) + \phi(y_2) + \mathbf{u}_-}{2}.$$

Thus,

$$\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$$

respectively. This result contradicts the assumption made in Case 4. Hence, Case 4 cannot occur.

We now complete the proof of Theorem III.6.

Proof VII.1. (Proof of Theorem III.6).
From Lemma VII.4, every receiver strategy $\tilde{r}$ is dominated by a receiver strategy $r(\cdot) \in \mathscr{R}$. Hence, the Stackelberg equilibrium is a strategy in $\mathscr{R}$ with the least error. Recall that $\mathscr{R} = \mathscr{R}_1 \cup \mathscr{R}_2$. Let's further partition the subset $\mathscr{R}_1$ into two parts, $\mathscr{R}_{10}$ and $\mathscr{R}_{11}$, i.e., $\mathscr{R}_1 := \mathscr{R}_{10} \cup \mathscr{R}_{11}$, where

$$\mathscr{R}_{10} := \{r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \mid r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_1, \phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})\},$$
$$\mathscr{R}_{11} := \{r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \mid r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_1, \phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})\}.$$

Thus, $\mathscr{R} = \mathscr{R}_{10} \cup \mathscr{R}_{11} \cup \mathscr{R}_2$. We show that there exists a Stackelberg equilibrium in $\mathscr{R}$ by showing that each subset, $\mathscr{R}_{10}$, $\mathscr{R}_{11}$, and $\mathscr{R}_2$, contains a strategy that yields the least error within its respective subset. Thus, the Stackelberg equilibrium will be a strategy with the least error among the three strategies that have the least error in their subsets.
For $r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_{10}$, using Lemma VII.2(a), the error set is $E(r(\cdot; \mathbf{h}^-, \mathbf{h})) = [\mathbf{h}^-, \mathbf{h}]$. Here, $\mathbf{u}_+ \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^-) \geq \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$. Now consider the strategy $r(\cdot; \mathbf{h}^{-*}, \mathbf{h})$, where $\mathbf{h}^{-*}$ is such that $\phi(\mathbf{h}) - \phi(\mathbf{h}^{-*}) = \frac{\mathbf{u}_- + \mathbf{u}_+}{2}$. Such a feature vector $\mathbf{h}^{-*}$ exists due to the continuity of ϕ. The error set of $r(\cdot; \mathbf{h}^{-*}, \mathbf{h})$ is $E(r(\cdot; \mathbf{h}^{-*}, \mathbf{h})) = [\mathbf{h}^{-*}, \mathbf{h}]$. Since $\phi(\mathbf{h}) - \phi(\mathbf{h}^-) \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^{-*})$, and ϕ is increasing, we have $\mathbf{h}^- \leq \mathbf{h}^{-*}$, and hence, $[\mathbf{h}^{-*}, \mathbf{h}] \subset [\mathbf{h}^-, \mathbf{h}]$. Since $r(\cdot; \mathbf{h}^-, \mathbf{h})$ was an arbitrary strategy, $r(\cdot; \mathbf{h}^{-*}, \mathbf{h})$ is the strategy with the least error in $\mathscr{R}_{10}$.
For $r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathscr{R}_{11}$, using Lemma VII.2(b), it follows that the error set of the strategy is $E(r(\cdot; \mathbf{h}^-, \mathbf{h}^+)) = [\mathbf{h}^-, \tilde{\mathbf{g}}]$. Since we seek a strategy with the least error, we choose $\mathbf{h}^{-*}$ such that the following holds:

$$\mathbf{h}^{-*} := \arg \inf_{\mathbf{h}^- \in [\phi^{-1}(\phi(\mathbf{h}) - \mathbf{u}_+), \mathbf{h}]} D(\{x : x \in [\mathbf{h}^-, \mathbf{g}]\}).$$

We now show that the infimum of the above optimization problem does indeed exist. Recall that $\mathbf{g}$ is completely determined by $\mathbf{h}^-$ and $\mathbf{u}_+$. Since the probability law D has a continuous (in fact differentiable) density function $\dot{\psi}$ except at finitely many points, the map

$$\mathbf{h}^- \mapsto D(\{x : x \in [\mathbf{h}^-, \tilde{\mathbf{g}}]\}) = \int_{\mathbf{h}^-}^{\tilde{\mathbf{g}}} \frac{d\psi(x)}{dx}\, dx$$

is (finite) piecewise continuous for $\mathbf{h}^- \in [\phi^{-1}(\phi(\mathbf{h}) - \mathbf{u}_+), \mathbf{h}]$. Because we are optimizing a (finite) piecewise continuous function over a compact set, the infimum is indeed attained. For $\mathscr{R}_2$, which contains only two elements, the majority class (w.r.t. D) strategy is the one with the least error. ■

Based on all the cases, we can see that there always exists a receiver strategy $r \in \mathscr{R}$ which dominates $\tilde{r}$. Therefore, the family of receiver strategies $\mathscr{R}$ dominates all other receiver strategies. ■

Proof VII.2. (Proof of Theorem III.7)

We will show that $E(r) = \emptyset$, which will prove that $r(\cdot; \mathbf{h}, \mathbf{h}^+)$ is a Stackelberg equilibrium. Note that such a $\mathbf{h}^+$ exists due to the continuity of ϕ. We now calculate s_r. The calculation is similar to Lemma VII.2.

Case 1. $x \in [0, \mathbf{h})$.
The sender's payoff is given by $U(r(s_r(x)), x) - |\phi(s_r(x)) - \phi(x)|$. Based on the structure of r, $s_r(x)$ can either be x or $\mathbf{h}^+$. Therefore, the sender's payoff is

$$\mathbb{U}(x; r, s_r) = \begin{cases} 0 & \text{if} \quad s_r(x) = x, \\ \mathbf{u}_+ - \phi(\mathbf{h}^+) + \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}^+. \end{cases}$$

We have $-\mathbf{u}_+ = \phi(\mathbf{h}) - \phi(\mathbf{h}^+) < \phi(x) - \phi(\mathbf{h}^+)$. Thus $\mathbf{u}_+ - \phi(x) - \phi(\mathbf{h}^+) < 0$. Hence, $s_r(x) = x$.

Case 2. $x \in (\mathbf{h}^+, 1]$.
The sender's payoff is given by $U(r(s_r(x)), x) - |\phi(s_r(x)) - \phi(x)|$. Based on the structure of r, $s_r(x)$ can either be x or $\mathbf{h}$. The sender's payoff is

$$\mathbb{U}(x; r, s_r) = \begin{cases} 0 & \text{if} \quad s_r(x) = x, \\ -\phi(\mathbf{h}) + \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}. \end{cases}$$

Since $x \geq \mathbf{h}$, we have $\phi(\mathbf{h}) - \phi(x) < 0$. Hence, $s_r(x) = x$.

Case 3. $x \in [\mathbf{h}, \mathbf{h}^+]$.
The sender's payoff is given by $U(r(s_r(x)), x) - |\phi(s_r(x)) - \phi(x)|$. Based on the structure of r, $s_r(x)$ can either be $\mathbf{h}$ or $\mathbf{h}^+$. Therefore, the sender's payoff is

$$\mathbb{U}(x; r, s_r) = \begin{cases} \mathbf{u}_- - \phi(x) + \phi(\mathbf{h}) & \text{if} \quad s_r(x) = \mathbf{h}, \\ -\phi(\mathbf{h}^+) + \phi(x) & \text{if} \quad s_r(x) = \mathbf{h}^+. \end{cases}$$

Since $\mathbf{h}^+ \geq x \geq \mathbf{h}$, we have $\phi(x) - \phi(\mathbf{h}^+) \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^+) = -\mathbf{u}_+$, $\mathbf{u}_- \leq -\mathbf{u}_+ \leq 0$, and $\phi(\mathbf{h}) - \phi(x) \leq 0$. Hence, $\mathbf{u}_- - \phi(x) + \phi(\mathbf{h}) \leq 0 - \phi(\mathbf{h}^+) + \phi(x)$ for all $x \in [\mathbf{h}, \mathbf{h}^+]$, and therefore we have $s_r(x) = \mathbf{h}^+$.

Combining all the cases, we have

$$s_r(x) := \begin{cases} x & \text{for} \quad x \in [0, \mathbf{h}) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^+ & \text{for} \quad x \in [\mathbf{h}, \mathbf{h}^+]. \end{cases}$$

Thus, $E(r) = \{x \mid r(s_r(x)) = h(x)\} = \emptyset$. ■

B. Omitted Proofs of Section III.C

Proof VII.3. (Proof of Theorem III.8)
Let r be a receiver strategy and let s_r be its worst-case best response. Define

$$K_+ = \{x : w^\top x > \mathbf{h} \quad \text{and} \quad r(s_r(x)) = h(x) = +\}, \quad K_- = \{x : w^\top x \leq \mathbf{h} \quad \text{and} \quad r(s_r(x)) = h(x) = -\}.$$

For each $z \in \mathcal{Z}$, define

$$f(z) := \min\{t > 0 : z + tw \in K_+,\, 2\mathbf{u}\}, \quad g(z) := \max\{t \leq 0 : z + tw \in K_-,\, -2\mathbf{u}\}.$$

We claim that

$$f(z) - g(z) > 2\mathbf{u} \qquad \text{for all} \qquad z \in \mathcal{Z}.$$

Suppose there exists $z \in \mathcal{Z}$ such that

$$f(z) - g(z) \leq 2\mathbf{u}. \tag{33}$$

This then rules out the possibility that either $f(z) = 2\mathbf{u}$ or $g(z) = -2\mathbf{u}$. Define

$$\bar{x} := z + f(z)w, \quad \underline{x} := z + g(z)w.$$

Note that this implies $\bar{x} \in K_+$ and $\underline{x} \in K_-$. Then

$$\|\bar{x} - \underline{x}\| = |f(z) - g(z)|\|w\| \leq 2\mathbf{u}.$$

Suppose $\|\bar{x} - s_r(\bar{x})\| < \|\underline{x} - s_r(\underline{x})\|$ (similar reasoning applies for the opposite inequality). Then the payoff for $\underline{x}$ under s_r is

$$-\|\underline{x} - s_r(\underline{x})\|.$$

Now consider a new sender strategy s defined by

$$s(x) := \begin{cases} s_r(x) & \text{if} \quad x \neq \underline{x} \\ s_r(\bar{x}) & \text{if} \quad x = \underline{x} \end{cases}.$$

The payoff for $\underline{x}$ under s is

$$\mathbf{u} - \|\underline{x} - s_r(\bar{x})\| \geq \mathbf{u} - \|\underline{x} - \bar{x}\| - \|\bar{x} - s_r(\bar{x})\| \geq -\|\underline{x} - s_r(\underline{x})\|,$$

which contradicts the optimality of s_r. Hence, the assumption in (33) is false, and therefore

$$f(z) - g(z) > 2\mathbf{u}$$

for all $z \in \mathcal{Z}$. By the definitions of f and g, for $z \in \mathcal{Z}$, all points $z + tw$, with $t \in (g(z), f(z))$, are misclassified by r. Thus,

$$E(r) \supseteq \{x : x = z + tw,\, z \in \mathcal{Z},\, t \in (g(z), f(z))\}$$

and

$$\mathcal{E}(r) = \int_{\mathcal{X}} \mathbb{1}_{E(r)}\, dx = \int_E dx.$$

We now evaluate this integral by using a change of variables. Let $\Phi : \mathcal{Z} \times \mathbb{R} \to \mathbb{R}^n$ be the change of variable map given by $\Phi(z, t) = z + wt$. It follows that $\det |J_\Phi| = \|w\| = 1$. Thus,

$$\int_E dx = \int_{\mathcal{Z}} \int_{g(z)}^{f(z)} \det |J_\Phi|\, d\sigma(z)\, dt = \int_{\mathcal{Z}} \int_{g(z)}^{f(z)} d\sigma(z)\, dt = \int_{\mathcal{Z}} (f(z) - g(z))\, d\sigma(z) \geq 2\mathbf{u}\, \sigma(\mathcal{Z}),$$

where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D. ■

Proof VII.4. (Proof of Theorem III.9)
For $z \in \mathcal{Z}$, let

$$K_+ := \{x : w^\top x > \mathbf{h} + \mathbf{u}\} \qquad \text{and} \qquad K_- := \{x : w^\top x < \mathbf{h} - \mathbf{u}\}.$$

Let $x \in K_+$. We now calculate the worst-case best response for x. Let $s_r(x) = t$. If $t \in K_-$, then the payoff for the sender is $2\mathbf{u} - \|x - t\| < 0$ because K_+ and K_- are separated by $2\mathbf{u}$ units. Since $r(x) = \Delta$ for $x \in \mathcal{X} \setminus (K_+ \cup K_-)$, it follows that $t \notin \mathcal{X} \setminus (K_+ \cup K_-)$. Thus, $t = x$, as the sender gets 0 payoff,

which is the maximum it could get. Hence, $s_r(x) = x$ for $x \in K_+$. Using similar reasoning for $x \in K_-$, we get $s_r(x) = x$ for $x \in K_-$ as well. Therefore,

$$E(r) \subset \{x : x = z + tw,\, z \in \mathcal{Z},\, t \in (-\mathbf{u}, \mathbf{u})\}. \quad \text{and} \quad \mathcal{E}(r) = \int_{E(r)} dx.$$

Evaluating this integral by using a change of variables from x to (z, t), we get

$$\int_{E(r)} dx \leq \int_{\mathcal{Z}} \int_{-\mathbf{u}}^{\mathbf{u}} dt\, d\sigma(z) = \int_{\mathcal{Z}} 2\mathbf{u}\, d\sigma(z) = 2\mathbf{u}\, \sigma(\mathcal{Z}),$$

where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D. Theorem III.8 establishes that the receiver's error is lower-bounded by $2\mathbf{u}\, \sigma(\mathcal{Z})$. We conclude that $\mathcal{E}(r) = 2\mathbf{u}\, \sigma(\mathcal{Z})$ and r is a Stackelberg equilibrium. ■

C. Omitted Proofs of Section III.D

The next Corollary show that if a Stackelberg equilibrium exist under worst-case best response setup, then its error will be $\frac{\mathbf{u}}{2}$.

Corollary VII.1. Suppose there exists a Stackelberg equilibrium $(r^\star, s_{r^\star})$ under the setup where the sender plays the worst-case best response. Then, $\mathcal{E}(r^\star, s_{r^\star}) = \frac{\mathbf{u}}{2}$.

Proof. Let s' be the worst-case lazy best response to $r^\star$. Let $(\bar{r}^\star, s_{\bar{r}^\star})$ be the Stackelberg equilibrium under the worst-case lazy best response. From Proposition III.1, we know that

$$\mathcal{E}(r^\star, s_{r^\star}) \geq \mathcal{E}(r^\star, s').$$

Additionally, by the definition of $\bar{r}^\star$, we have

$$\mathcal{E}(r^\star, s') \geq \mathcal{E}(\bar{r}^\star, s_{\bar{r}^\star}).$$

From Theorem III.10, we know that

$$\mathcal{E}(\bar{r}^\star, s_{\bar{r}^\star}) = \frac{\mathbf{u}}{2}.$$

Combining these results, we conclude that

$$\mathcal{E}(r^\star, s_{r^\star}) \geq \frac{\mathbf{u}}{2}.$$

Furthermore, from Example 3 (which appears in Section III-E), we know that for every $\epsilon > 0$,

$$\mathcal{E}(r_\epsilon, s_{r_\epsilon}) = \frac{\mathbf{u} + \epsilon}{2}.$$

Thus, if a Stackelberg equilibrium $r^\star$ exists, then it follows that

$$\mathcal{E}(r^\star, s_{r^\star}) = \frac{\mathbf{u}}{2}.$$

■

From the definition of R in (7), every receiver strategy $r \in \mathsf{R}$ has the structure $r(x) \equiv (p(x), q(x), 1 - p(x) - q(x))$. The first lemma shows that without loss of generality, we can assume the structure of r at each x to be either $r(x) \equiv (1 - q(x), q(x), 0)$ or $r(x) \equiv (0, 0, 1)$.

Lemma VII.5. *Let $r \equiv (p, q, 1-p-q) \in \mathsf{R}$ be a receiver strategy. Let $\Theta := \{x' : x' \in \mathcal{X}, p(x') + q(x') < 1\}$. Now consider another receiver strategy $\tilde{r} \equiv (\tilde{p}, \tilde{q}, 1 - \tilde{p} - \tilde{q})$ defined as*

$$\tilde{r}(x) = \begin{cases} r(x) & \textit{for} \quad x \in \mathcal{X} \setminus \Theta, \\ (0, 0, 1) & \textit{for} \quad x \in \Theta. \end{cases}$$

Then $\mathcal{E}(r) = \mathcal{E}(\tilde{r})$.

Proof. Let $x \in \mathcal{X}$. We first show that $s_r(x), s_{\tilde{r}}(x) \notin \Theta$. From the definition of p, q and Θ, we have

$$1 - p(x') - q(x') = \mathbb{P}_r(\Delta | X' = x') > 0, \quad \text{for all} \quad x' \in \Theta.$$

If $s_r(x) \in \Theta$, then

$$\mathbb{P}_r(y = \Delta | X'_{s_r} = s_r(x)) > 0$$

and the sender's payoff is $\mathbb{U}(x; r, s_r) = -\infty$. Similarly, if $s_{\tilde{r}}(x) \in \Theta$, then

$$\mathbb{P}_{\tilde{r}}(y = \Delta | X'_{s_{\tilde{r}}} = s_{\tilde{r}}(x)) = 1$$

and the sender's payoff is $\mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) = -\infty$. But from (5), the feature vector set $\mathcal{X} \setminus \Theta \neq \emptyset$. Therefore, there exists $t \in \mathcal{X} \setminus \Theta$ such that

$$\mathbb{P}_r(Y = \Delta | X' = t) = \mathbb{P}_{\tilde{r}}(Y = \Delta | X' = t) = 0.$$

Hence, for the sender strategy $s(x) = t$ for all $x \in \mathcal{X}$, we have

$$\mathbb{P}_r(Y = \Delta | X' = t) = \mathbb{P}_{\tilde{r}}(Y = \Delta | X' = t)) = 0$$

and the sender's payoff at x for receiver strategies r and $\tilde{r}$ will be $\mathbb{U}(x, r, s) < \infty$ and $\mathbb{U}(x, \tilde{r}, s) < -\infty$ respectively. But this contradicts that s_r and $s_{\tilde{r}}$ are the worst-case lazy best responses, as $\mathbb{U}(x, r, s) < \mathbb{U}(x, r, s_r)$ and $\mathbb{U}(x, \tilde{r}, s) < \mathbb{U}(x, \tilde{r}, s_{\tilde{r}})$. Hence,

$$s_r(x), s_{\tilde{r}}(x) \in \mathcal{X} \setminus \Theta \quad \text{for all} \quad x \in \mathcal{X}. \tag{34}$$

From the construction of $\tilde{r}$, we know that $r(x) = \tilde{r}(x)$ for $x \in \mathcal{X} \setminus \Theta$. Therefore, combining it with (34), we get

$$r(s_r(x)) = \tilde{r}(s_r(x)) \quad \text{and} \quad r(s_{\tilde{r}}(x)) = \tilde{r}(s_{\tilde{r}}(x)) \quad \text{for all} \quad x \in \mathcal{X}. \tag{35}$$

We next show that s_r and $s_{\tilde{r}}$ are best responses to both r and $\tilde{r}$. Since s_r and $s_{\tilde{r}}$ are worst-case lazy best responses of r and $\tilde{r}$, respectively, we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, s_{\tilde{r}}) \quad \text{and} \quad \mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) \geq \mathbb{U}(x; \tilde{r}, s_r) \quad \text{for all} \quad x \in \mathcal{X}. \tag{36}$$

Substituting (35) in (36) and expanding it, we get

$$\begin{aligned} &\mathfrak{U}(x; r, s_r) - |s_r(x) - x| \geq \mathfrak{U}(x; r, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| = \mathfrak{U}(x; \tilde{r}, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| \qquad \text{and} \\ &\mathfrak{U}(x; \tilde{r}, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| \geq \mathfrak{U}(x; \tilde{r}, s_r) - |s_r(x) - x| = \mathfrak{U}(x; r, s_r) - |s_r(x) - x|. \end{aligned}$$

which implies

$$|s_{\tilde{r}}(x) - x| - |s_r(x) - x| \geq \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x) - \mathfrak{U}(r(s_r(x)), x) \geq |s_{\tilde{r}}(x) - x| - |s_r(x) - x|$$

or

$$\mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x) - \mathfrak{U}(r(s_r(x)), x) = |s_{\tilde{r}}(x) - x| - |s_r(x) - x|. \tag{37}$$

Thus, substituting (37) back in (36), we get

$$\mathbb{U}(x; r, s_r) = \mathbb{U}(x; r, s_{\tilde{r}}) \quad \text{and} \quad \mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) = \mathbb{U}(x; \tilde{r}, s_r). \tag{38}$$

This shows that $s_{\tilde{r}}$ and s_r are best responses to both r and $\tilde{r}$. Furthermore, since s_r and $s_{\tilde{r}}$ are both worst-case lazy best responses to r and $\tilde{r}$, respectively, using the laziness constraint on (38), we have

$$|s_r(x) - x| \leq |s_{\tilde{r}}(x) - x| \quad \text{and} \quad |s_{\tilde{r}}(x) - x| \leq |s_r(x) - x| \quad \text{which implies} \quad |s_r(x) - x| = |s_{\tilde{r}}(x) - x|.$$

As an additional consequence, (37) reduces to

$$\mathfrak{U}(r(s_r(x)), x) = \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x). \tag{39}$$

Next, we show that s_r and $s_{\tilde{r}}$ are worst-case lazy best responses to both r and $\tilde{r}$. Since s_r and $s_{\tilde{r}}$ are worst-case lazy best responses of r and $\tilde{r}$, respectively, we have (using the worst-case constraint)

$$\begin{aligned} \mathcal{E}(r, s_r) &= \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{\mathcal{X}} T(x; r, s_{\tilde{r}})\, dx = \mathcal{E}(r, s_{\tilde{r}}) \qquad \text{and} \\ \mathcal{E}(\tilde{r}, s_{\tilde{r}}) &= \int_{\mathcal{X}} T(x; \tilde{r}, s_{\tilde{r}})\, dx \geq \int_{\mathcal{X}} T(x; \tilde{r}, s_r)\, dx = \mathcal{E}(\tilde{r}, s_r). \end{aligned} \tag{40}$$

Recall from (8) that T is defined as $T(x; r, s) = \mathbb{P}_r(Y \neq h(x) \mid X' = s(x))$. Furthermore, if $s(x) \in \mathcal{X} \setminus \Theta$, then

$$\mathfrak{U}(x; r, s) = 2\mathbf{u}\mathbb{P}_r(Y \neq h(x) \mid X' = s(x)) = 2\mathbf{u}T(x; r, s).$$

Hence, using this in (40), we get

$$\int_{\mathcal{X}} \mathfrak{U}(r(s_r(x)), x)\, dx \geq \int_{\mathcal{X}} \mathfrak{U}(r(s_{\tilde{r}}(x)), x)\, dx \qquad \text{and} \qquad \int_{\mathcal{X}} \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x)\, dx \geq \int_{\mathcal{X}} \mathfrak{U}(\tilde{r}(s_r(x)), x)\, dx.$$

But combining (35) with (39), we get

$$\mathfrak{U}(\tilde{r}(s_r(x)), x) = \mathfrak{U}(r(s_r(x)), x) = \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x) = \mathfrak{U}(r(s_{\tilde{r}}(x)), x).$$

Thus,

$$\int_{\mathcal{X}} T(x; r, s_r)\, dx = \int_{\mathcal{X}} T(x; r, s_{\tilde{r}})\, dx \qquad \text{and} \qquad \int_{\mathcal{X}} T(x; \tilde{r}, s_{\tilde{r}})\, dx = \int_{\mathcal{X}} T(x; \tilde{r}, s_r)\, dx.$$

Hence, $s_{\tilde{r}}$ and s_r are also worst-case lazy best responses of r and $\tilde{r}$, respectively. Therefore, $\mathcal{E}(r) = \mathcal{E}(\tilde{r})$ as claimed. ■

The next lemma calculates the worst-case lazy best response and classification error of a specific receiver strategy r, which will later be shown to be a Stackelberg equilibrium.

Lemma VII.6. *Let $r \equiv (1-q, q, 0)$, where*

$$q(x) = \begin{cases} 0 & \textit{for} \quad x \in [0, \mathbf{h}^-], \\ \frac{x - \mathbf{h}^-}{2\mathbf{u}} & \textit{for} \quad x \in (\mathbf{h}^-, \mathbf{h}^+), \\ 1 & \textit{for} \quad x \in [\mathbf{h}^+, 1] \end{cases} \tag{41}$$

and, $\mathbf{h}^- = \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ = \mathbf{h} + \mathbf{u}$. Then the worst-case lazy best response of r is $s_r(x) = x$, and the error function $T(x; r, s_r)$ is

$$T(x; r, s_r) = \begin{cases} 0 & \textit{for} \quad x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1], \\ q(x) & \textit{for} \quad x \in (\mathbf{h}^-, \mathbf{h}], \\ 1 - q(x) & \textit{for} \quad x \in (\mathbf{h}, \mathbf{h}^+). \end{cases}$$

Furthermore, $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$.

Proof. We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$ and show that $s_r(x) = x$ in all of them. Let $s_r(x) = x'$.

Case 1. $x \in [0, \mathbf{h}^-]$.
We now calculate $\mathbb{U}(x; r, s_r)$ and show that in the current case, $x = x'$.

$$\mathbb{U}(x;r,s_r)=\begin{cases}-|x-x'|=0 & \text{if} \quad x'=x,\\ -|x-x'|<0 & \text{if} \quad x'\neq x \quad \text{and} \quad x'\in[0,\mathbf{h}^-],\\ 2\mathbf{u}q(x')-x'+x\leq 0 & \text{if} \quad x'\in(\mathbf{h}^-,\mathbf{h}^+),\\ 2\mathbf{u}-x'+x\leq 0 & \text{if} \quad x'\in[\mathbf{h}^+,1].\end{cases}$$

Here we have used (41) to expand $q(x')$ and to show that $2\mathbf{u}q(x')-x'+x\leq 0$, and have used $\mathbf{h}^+-\mathbf{h}^-=2\mathbf{u}$ to show that $2\mathbf{u}-x'+x\leq 0$. Thus for $x'\neq x$, we have $\mathbb{U}(x;r,s_r)\leq 0$ whereas for $x'=x$, $\mathbb{U}(x;r,s_r)=0$. Furthermore, $x'=x$ is also the laziest option for s_r. Hence $s_r(x)=x$.

Case 2. $x\in(\mathbf{h}^-,\mathbf{h}]$.
We now calculate $\mathbb{U}(x;r,s_r)$ and show that in the current case, $x=x'$.

$$\mathbb{U}(x;r,s_r)=\begin{cases}2\mathbf{u}q(x')-x+x'=x-\mathbf{h}^- & \text{if} \quad x'=x,\\ -x+x'<0. & \text{if} \quad x'\in[0,\mathbf{h}^-],\\ 2\mathbf{u}q(x')-x+x'\leq x-\mathbf{h}^- & \text{if} \quad x'\in(\mathbf{h}^-,x] \quad \text{and} \quad x'\neq x,\\ 2\mathbf{u}q(x')-x'+x<x-\mathbf{h}^- & \text{if} \quad x'\in(x,\mathbf{h}^+),\\ 2\mathbf{u}-x'+x\leq x-\mathbf{h}^- & \text{if} \quad x'\in[\mathbf{h}^+,1].\end{cases}$$

Here we have used (41) to expand $q(x')$ wherever it appears and have used $2\mathbf{u}=\mathbf{h}^+-\mathbf{h}^-$ to show that $2\mathbf{u}-x'+x\leq x-\mathbf{h}^-$. Note that $x-\mathbf{h}^-\geq 0$. Thus for $x'\neq x$, we have $\mathbb{U}(x;r,s_r)\leq x-\mathbf{h}^-$ whereas for $x'=x$, $\mathbb{U}(x;r,s_r)\leq x-\mathbf{h}^-$. Furthermore, $x'=x$ is also the laziest option for s_r. Hence $s_r(x)=x$.

Case 3. $x\in[\mathbf{h},\mathbf{h}^+)$.
We now calculate $\mathbb{U}(x;r,s_r)$ and show that in the current case, $x=x'$.

$$\mathbb{U}(x;r,s_r)=\begin{cases}2\mathbf{u}(1-q(x'))-x'+x=\mathbf{h}^+-x & \text{if} \quad x'=x,\\ 2\mathbf{u}-x+x'\leq\mathbf{h}^+-x & \text{if} \quad x'\in[0,\mathbf{h}^-],\\ 2\mathbf{u}(1-q(x'))-x+x'<\mathbf{h}^+-x & \text{if} \quad x'\in(\mathbf{h}^-,x],\\ 2\mathbf{u}(1-q(x'))-x'+x\leq\mathbf{h}^+-x & \text{if} \quad x'\in(x,\mathbf{h}^+) \quad \text{and} \quad x'\neq x,\\ -x'+x\leq-\mathbf{h}^++x & \text{if} \quad x'\in[\mathbf{h}^+,1].\end{cases}$$

Here we have used (41) to expand $q(x')$ wherever it appears and have used $2\mathbf{u}=\mathbf{h}^+-\mathbf{h}^-$ to show that $2\mathbf{u}-x+x'\leq\mathbf{h}^+-x$. Note that $\mathbf{h}^+-x\geq 0$. Thus for $x'\neq x$, we have $\mathbb{U}(x;r,s_r)\leq\mathbf{h}^+-x$ whereas for $x'=x$, $\mathbb{U}(x;r,s_r)=\mathbf{h}^+-x$. Furthermore, $x'=x$ is also the laziest option for s_r. Hence $s_r(x)=x$.

Case 4. $x\in[\mathbf{h}^+,1]$.
We now calculate $\mathbb{U}(x;r,s_r)$ and show that in the current case, $x=x'$.

$$\mathbb{U}(x;r,s_r)=\begin{cases}-|x'-x|=0 & \text{if} \quad x'=x,\\ 2\mathbf{u}-x+x'\leq\mathbf{h}^+-x\leq 0 & \text{if} \quad x'\in[0,\mathbf{h}^-],\\ 2\mathbf{u}(1-q(x'))-x+x'<\mathbf{h}^+-x<0 & \text{if} \quad x'\in(\mathbf{h}^-,\mathbf{h}^+),\\ -|x'-x|<0 & \text{if} \quad x'\in[\mathbf{h}^+,1] \quad \text{and} \quad x'\neq x.\end{cases}$$

Here we have used (41) to expand $q(x')$ to show that $2\mathbf{u}(1-q(x'))-x+x'<\mathbf{h}^+-x<0$. Thus for $x'\neq x$, we have $\mathbb{U}(x;r,s_r)\leq 0$ whereas for $x'=x$, $\mathbb{U}(x;r,s_r)=0$. Furthermore, $x'=x$ is also the laziest option for s_r. Hence $s_r(x)=x$.

Hence, by combining all the cases, we conclude that $s_r(x)=x$. Therefore,

$$T(x;r,s_r)=\mathbb{P}_r\{Y\neq h(x)\mid X'=s_r(x)\}=\begin{cases}0 & \text{for} \quad x\in[0,\mathbf{h}^-]\cup[\mathbf{h}^+,1],\\ q(x) & \text{for} \quad x\in(\mathbf{h}^-,\mathbf{h}],\\ 1-q(x) & \text{for} \quad x\in(\mathbf{h},\mathbf{h}^+).\end{cases}$$

Since $\mathcal{E}(r) = \int_{\mathcal{X}} T(x)\,dx$, it follows that $\mathcal{E}(r) = \frac{\mathfrak{u}}{2}$. ■

We next show that if a feature vector x' is the worst-case lazy best response of some other feature vector x and both have the same true label, then j is also the worst-case lazy best response of itself.

Lemma VII.7. *Let $r \in \mathsf{R}$ be a receiver strategy, and let $x, x' \in \mathcal{X}$ be such that $s_r(x) = x'$. Then, for all $\bar{x} \in [\min(x, x'), \max(x, x')] \setminus \{x\}$ such that $h(\bar{x}) = h(x)$, we have $s_r(\bar{x}) = x'$.*

Proof. Let $\bar{x} \in [\min(x, x'), \max(x, x')] \setminus \{x\}$ such that $h(\bar{x}) = h(x)$. Let $s_r(\bar{x}) = z$. We will assume $x' < x$. A similar argument will hold if $x' \geq x$. Consider the sender strategies $s(t) \equiv x'$ and $\hat{s}(t) \equiv z$ for all $t \in \mathcal{X}$. Since s_r is the worst-case lazy best response of r, we have

$$\mathbb{U}(t, r; s_r) \geq \mathbb{U}(t, r; s) \quad \text{and} \quad \mathbb{U}(r; s_r) \geq \mathbb{U}(r; \hat{s}) \quad \text{for all} \quad t \in \mathcal{X}.$$

This implies for $t = \bar{x}$ and $t = x$, we have

$$\mathbb{U}(\bar{x}; r, s_r) \geq \mathbb{U}(\bar{x}; r, s) \qquad \text{and} \qquad \mathbb{U}(x; r, s_r) \geq \mathbb{U}(x, r, \hat{s}) \quad \text{respectively} \quad .$$

Recall that from the definition of U given in (2), $U(k, \bar{x}) = U(k, x)$ for all $k \in \mathcal{Y} \cup \Delta$ because $h(\bar{x}) = h(x)$. As a consequence, for a fixed receiver strategy $\mathfrak{r}$ and sender strategy $\mathfrak{s}$, we have $\mathfrak{U}(x; \mathfrak{r}, \mathfrak{s}) = \mathfrak{U}(\bar{x}; \mathfrak{r}, \mathfrak{s})$. Hence, this implies

$$\begin{aligned} \mathfrak{U}(x; r, s_r) - |\bar{x} - z| = \mathfrak{U}(\bar{x}; r, s_r) - |\bar{x} - z| \geq \mathfrak{U}(\bar{x}; r, s) - \bar{x} + x' = \mathfrak{U}(x; r, s) - \bar{x} + x' \qquad \text{and} \\ \mathfrak{U}(x; r(x'), s) - x + x' \geq \mathfrak{U}(x; r, s_r) - |x - z|. \end{aligned} \tag{42}$$

We show that $z = x'$ by the method of contradiction. Suppose $z \neq x'$. Then, consider the following cases.
Case 1. $z \geq x$.
We show that this case does not happen. From the case assumption, (42) reduces to

$$\mathfrak{U}(x; r, s_r) \geq \mathfrak{U}(x; r, s) - 2\bar{x} + x' + z \qquad \text{and} \qquad \mathfrak{U}(x; r, s) \geq \mathfrak{U}(x; r, s_r) + 2x - x' - z.$$

Combining them gives $\mathfrak{U}(x; r, s_r) \geq \mathfrak{U}(x; r, s_r) + 2(x - \bar{x})$, which gives rise to a contradiction because $x - w > 0$. Hence, this case does not happen as claimed.
Case 2. $x \geq z > \bar{x} \geq x'$.
We show that this case does not happen. From the case assumption, (42) reduces to

$$\mathfrak{U}(x, r, s_r) \geq \mathfrak{U}(x; r, s) - 2\bar{x} + z + x' \qquad \text{and} \qquad \mathfrak{U}(x; r, s) > \mathfrak{U}(x; r, s_r) + z - x'.$$

The second part has a strict inequality because $|x - z| < |x - x'|$. The reason is that if there were equality, then z would have been a low-cost (lazy) option for the sender at x as compared to x', which would contradict that x' is the worst-case lazy best response of r at x. Combining them gives $\mathfrak{U}(r(z), x) > \mathfrak{U}(r(z), x) + 2(z - \bar{x})$, which is a contradiction because $z > \bar{x}$. Hence, this case does not happen as claimed.
Case 3. $x > \bar{x} \geq z > x'$.
We show that this case does not happen. From the case assumption, (42) reduces to

$$\mathfrak{U}(x; r, s_r) \geq \mathfrak{U}(x; r, s) + x' - z \qquad \text{and} \qquad \mathfrak{U}(x; r, s) > \mathfrak{U}(x; r, s_r) + z - x'.$$

The second part has a strict inequality because $|x - z| < |x - x'|$. The same reasons as mentioned in Case 2. Combining them gives $\mathfrak{U}(x; r, s_r) > \mathfrak{U}(x; r, s_r)$, which is a contradiction. Hence, this case does not happen as claimed.
Case 4. $z < x'$.
We show that this case does not happen. From the case assumption, (42) reduces to

$$\mathfrak{U}(x; r, s_r) > \mathfrak{U}(x; r, s) + x' - z \qquad \text{and} \qquad \mathfrak{U}(x; r, s) \geq \mathfrak{U}(x; r, s_r) + z - x'.$$

The first part has a strict inequality because $|\bar{x} - x'| < |\bar{x} - z|$. The reason is the same as mentioned in Case 2. This further reduces to $\mathfrak{U}(x; r, s_r) > \mathfrak{U}(x; r, s_r)$, which is a contradiction. Hence, combining all the cases, we get a contradiction. Therefore, $z = x'$. ■

Let $r \equiv (p, q, 1-p-q) \in \mathsf{R}$ be the receiver strategy. For feature vectors x_1 and $x_2 \in \mathcal{X}$ such that $x_1 \leq \mathbf{h} < x_2$, we calculate the contribution of error from the intervals $[x_1, x_1 - 2\mathbf{u}q(x_1)]$ and $[x_2, x_2 + 2\mathbf{u}q(x_2)]$.

Lemma VII.8. *Let $r \equiv (p, q, 1-p-q) \in \mathsf{R}$ be a receiver strategy. Let $x_1, x_2 \in \mathcal{X}$ be such that $x_1 \leq \mathbf{h} < x_2$, and $p(x_i) + q(x_i) = 1$ for $i = 1, 2$. Then*

$$\int_{x_1 - 2\mathbf{u}q(x_1)}^{x_1} T(x; r, s_r)\, dx \geq \mathbf{u}q(x_1)^2, \qquad \textit{and} \qquad \int_{x_2}^{x_2 + 2\mathbf{u}(1-q(x_2))} T(x; r, s_r)\, dx \quad \geq \mathbf{u}(1-q(x_2))^2.$$

Proof. Let $q_1 := q(x_1)$ and $q_2 := q(x_2)$. Recall that from (8), T is defined as $T(x; r, s_r) = \mathbb{P}(r(s_r(x)) \neq h(x) | X' = s_r(x) = q(s_r(x))$. We upper bound $q(s_r(x))$ (or $1 - q(x)$ depending upon x) to get a upper bound on $T(x; r, s_r)$ and eventually over $\int_{x_1 - 2\mathbf{u}q(x_1)}^{x_1} T(x; r, s_r) dx$ and $\int_{x_2}^{x_2 + 2\mathbf{u}(1-q(x_2))} T(x; r, s_r) dx$. Let $x \in [x_1 - 2\mathbf{u}q_1, x_1]$. Consider a sender strategy $s(x) = x_1$. Then, from the definition of the worst-case lazy best response, we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, s)$$

which implies

$$2\mathbf{u}q(s_r(x)) - |s_r(x) - x| \geq 2\mathbf{u}q_1 - x_1 + x \qquad \text{or} \qquad 2\mathbf{u}q(s_r(x)) \geq 2\mathbf{u}q_1 - x_1 + x. \tag{43}$$

Hence, from (43), we get $T(x; r, s_r) \geq q_1 - \frac{x_1 - x}{2\mathbf{u}}$. Therefore,

$$\int_{x_1 - 2\mathbf{u}q_1}^{x_1} T(x; r, s_r)\, dx \geq \int_{x_1 - 2\mathbf{u}q_1}^{x_1} \left(q_1 - \frac{x_1 - x}{2\mathbf{u}} \right) dx = \mathbf{u}q_1^2. \tag{44}$$

Similarly, let $x \in [x_2, x_2 + 2\mathbf{u}(1 - q_2)]$. Consider a sender strategy $\bar{s}(x) = x_2$. Then, from the definition of the worst-case lazy best response, we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \bar{s})$$

which implies

$$2\mathbf{u}(1 - q(s_r(x))) - |s_r(x) - x| \geq 2\mathbf{u}(1 - q_2) - x + x_2 \qquad \text{or} \qquad 2\mathbf{u}(1 - q(s_r(x))) \geq 2\mathbf{u}(1 - q_2) - x + x_2.$$

Hence, $T(x; r, s_r) = 1 - q(s_r(x)) \geq (1 - q_2) - \frac{x - x_2}{2\mathbf{u}}$. Therefore,

$$\int_{x_2}^{x_2 + 2\mathbf{u}(1-q_2)} T(x; r, s_r)\, dx \geq \int_{x_2}^{x_2 + 2\mathbf{u}(1-q_2)} \left((1 - q_2) - \frac{x - x_2}{2\mathbf{u}} \right) dx = \mathbf{u}(1 - q_2)^2. \tag{45}$$

Hence, combining (44) and (45), we get

$$\int_{x_1 - 2\mathbf{u}q(x_1)}^{x_1} T(x; r, s_r)\, dx \geq \mathbf{u}q(x_1)^2, \qquad \text{and}$$
$$\int_{x_2}^{x_2 + 2\mathbf{u}(1-q(x_2))} T(x; r, s_r)\, dx \geq \mathbf{u}(1 - q(x_2))^2.$$

■

We can improve upon the result of Lemma VII.8 if we have an additional constraint that $x_2 - x_1 \leq 2\mathbf{u}(q(x_2) - q(x_1))$.

Lemma VII.9. *Let $r \equiv (p, q, 1-p-q) \in \mathsf{R}$ be a receiver strategy. Let $x_1, x_2 \in \mathcal{X}$ be such that $x_1 \leq \mathbf{h} < x_2$, $p(x_i) + q(x_i) = 1$ for $i = 1, 2$, and $x_2 - x_1 \leq 2\mathbf{u}(q(x_2) - q(x_1))$. Then*

$$\int_{x_2 - 2\mathbf{u}q(x_2)}^{x_1} T(x; r, s_r)\, dx \geq \mathbf{u}\Big(q(x_2) - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2 \quad \textit{and}$$
$$\int_{x_2}^{x_1 + 2\mathbf{u}(1-q(x_1))} T(x; r, s_r)\, dx \geq \mathbf{u}\Big(1 - q(x_1) - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2.$$

Proof. Let $q_1 := q(x_1)$ and $q_2 := q(x_2)$. We upper bound $q(s_r(x))$ (or $1-q(x)$ depending upon x) to get a upper bound on $T(x; r, s_r)$ and eventually over $\int_{x_2 - 2\mathbf{u}q(x_2)}^{x_1} T(x; r, s_r) dx$ and $\int_{x_2}^{x_1 + 2\mathbf{u}(1-q(x_1))} T(x; r, s_r) dx$. Let $x \in [x_2 - 2\mathbf{u}q_2, x_1]$. Consider the sender strategy $\tilde{s}(t) = x_2$. From the definition of the worst-case lazy best response, we have $\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \tilde{s})$, which implies

$$2\mathbf{u}q(s_r(x)) - |s_r(x) - x| \geq 2\mathbf{u}q_2 - x_2 + x \qquad \text{or} \qquad 2\mathbf{u}q(s_r(x)) \geq 2\mathbf{u}q_2 - x_2 + x.$$

Therefore,

$$\int_{x_2 - 2\mathbf{u}q_2}^{x_1} T(x; r, s_r)\, dx \geq \int_{x_2 - 2\mathbf{u}q_2}^{x_1} \left(q_2 - \frac{x_2 - x}{2\mathbf{u}}\right) dx = \mathbf{u}\Big(q_2 - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2. \tag{46}$$

Similarly, let $x \in [x_2, x_1 + 2\mathbf{u}(1-q_1)]$ and consider the sender strategy $\hat{s}(t) = x_1$. Then, from the definition of the worst-case lazy best response, we have $\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \hat{s})$, which implies

$$2\mathbf{u}(1 - q(s_r(x))) - |s_r(x) - x| \geq 2\mathbf{u}(1 - q_1) - x + x_1 \qquad \text{or} \qquad 2\mathbf{u}(1 - q(s_r(x))) \geq 2\mathbf{u}(1 - q_1) - x + x_1.$$

Hence, $T(x; r, s_r) = 1 - q(s_r(x)) \geq (1 - q_1) - \frac{x - x_1}{2\mathbf{u}}$. Therefore,

$$\int_{x_2}^{x_1 + 2\mathbf{u}(1-q_1)} T(x; r, s_r)\, dx \geq \int_{x_2}^{x_1 + 2\mathbf{u}(1-q_1)} \left((1 - q_1) - \frac{x - x_1}{2\mathbf{u}}\right) dx = \mathbf{u}\Big((1 - q_1) - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2. \tag{47}$$

Hence, combining (46) and (47), we get

$$\int_{x_2 - 2\mathbf{u}q(x_2)}^{x_1} T(x; r, s_r)\, dx \geq \mathbf{u}\Big(q(x_2) - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2 \qquad \text{and}$$
$$\int_{x_2}^{x_1 + 2\mathbf{u}(1-q(x_1))} T(x; r, s_r)\, dx \geq \mathbf{u}\Big(1 - q(x_1) - \frac{x_2 - x_1}{2\mathbf{u}}\Big)^2.$$

■

In the next lemma, we show that any arbitrary receiver strategy $r \in \mathsf{R}$ will make at least $\frac{\mathbf{u}}{2}$ errors.

Lemma VII.10. *Let $r \in \mathsf{R}$ be a receiver strategy. Then $\mathcal{E}(r) \geq \frac{\mathbf{u}}{2}$.*

Proof. Let $r \in \mathsf{R}$ be a receiver strategy. Then, from Lemma VII.5, r is of the form $r(x) \equiv (p(x), q(x), 1 - p(x) - q(x))$, where either $p(x) + q(x) = 1$ or $p(x) + q(x) = 0$. Let $k_1 := \sup\{x : p(x) + q(x) = 1, x \leq \mathbf{h}\}$ and $k_2 := \inf\{x : p(x) + q(x) = 1, x > \mathbf{h}\}$. From (6), both the infimum and supremum are attained in the set. Let $s_r(k_1) := l_1$ and $s_r(k_2) := l_2$. Let $q(k_1) := q_1$, $q(k_2) := q_2$, $q(l_1) := \bar{q}_1$, and $q(l_2) := \bar{q}_2$. We now calculate the receiver errors for various cases.

Case 1. $q_1 > q_2$.

From the definition of $\mathcal{E}(r)$, the following inequality holds:

$$\mathcal{E}(r) = \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{k_1 - 2\mathbf{u}q_1}^{k_1} T(x; r, s_r)\, dx + \int_{k_2}^{k_2 + 2\mathbf{u}(1-q_2)} T(x; r, s_r)\, dx.$$

Using Lemma VII.8 by taking $x_1 = k_1$ and $x_2 = k_2$, we get

$$\mathcal{E}(r) \geq \mathbf{u}(q_1^2 + (1 - q_2)^2).$$

Since from the case assumption, $q_1 > q_2$, therefore, we have

$$\mathcal{E}(r) \geq \mathbf{u}(q_1^2 + (1-q_1)^2).$$

The minimum value of $q_1^2 + (1-q_1)^2$ is $\frac{1}{2}$ at $q_1 = \frac{1}{2}$. Hence, we have $\mathcal{E}(r) \geq \frac{\mathbf{u}}{2}$.
Case 2. $q_2 > q_1$ and $k_2 - k_1 < \mathbf{u}(q_2 - q_1)$.
From the definition of $\mathcal{E}(r)$, the following inequality holds:

$$\mathcal{E}(r) = \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{k_2 - 2\mathbf{u}q_2}^{k_1} T(x; r, s_r)\, dx + \int_{k_2}^{k_1 + 2\mathbf{u}(1-q_1)} T(x; r, s_r)\, dx.$$

Using Lemma VII.9 with $x_1 = k_1$ and $x_2 = k_2$, and using $k_2 - k_1 < \mathbf{u}(q_2 - q_1)$, we get

$$\mathcal{E}(r) \geq \mathbf{u}((1-q_1)^2 + q_2^2).$$

Since $q_2 > q_1$, therefore,

$$\mathcal{E}(r) \geq \mathbf{u}((1-q_2)^2 + q_2^2).$$

The minimum value of $(1-q_2)^2 + q_2^2$ is $\frac{1}{2}$ at $q_2 = \frac{1}{2}$. Hence, $\mathcal{E}(r) \geq \frac{\mathbf{u}}{2}$.
Case 3. $q_2 \geq q_1$, $2\mathbf{u}(q_2 - q_1) \geq k_2 - k_1 \geq \mathbf{u}(q_2 - q_1)$, and $l_2 \leq k_1 < k_2 \leq l_1$.
We now estimate $T(x; r, s_r)$ for $x \in [k_1, \mathbf{h}]$. Let $x \in [k_1, \mathbf{h}]$. It follows that $h(x) = h(k_1) = -$. Using Lemma VII.7 with $i = k_1$, $j = l_1$, and $w = x$, we have $s_r(x) = l_1$. Consider a sender strategy $s(x) := k_2$. Since s_r is the worst-case lazy best response, we have

$$2\mathbf{u}q(s_r(x)) - l_1 + k_1 \geq 2\mathbf{u}q_2 - k_2 + k_1.$$

Since $l_1 \geq k_2$, therefore we have $q(s_r(x)) \geq q_2$. This implies

$$T(x; r, s_r) \geq q_2 \qquad \text{for} \qquad x \in [k_1, \mathbf{h}]. \tag{48}$$

Similarly, we estimate $T(x; r, s_r)$ for $x \in (\mathbf{h}, k_2]$. Let $x \in (\mathbf{h}, k_2]$. It follows that $h(x) = h(k_2) = +$. Using Lemma VII.7 with $i = k_2$, $j = l_2$, and $w = x$, we have $s_r(x) = l_2$. Consider a sender strategy $\bar{s}(i) := k_2$. Since s_r is the worst-case lazy best response, we have

$$2\mathbf{u}(1 - q(s_r(x))) - k_2 - l_2 \geq 2\mathbf{u}(1 - q_1) - k_2 + k_1.$$

Since $l_2 \geq k_1$, therefore we have $1 - q(s_r(x)) \geq 1 - q_1$. This implies

$$T(x; r, s_r) \geq 1 - q_1 \qquad \text{for} \qquad x \in (\mathbf{h}, k_2). \tag{49}$$

From the definition of $\mathcal{E}(r)$, we have

$$\mathcal{E}(r) := \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{k_2 - 2\mathbf{u}q_2}^{k_1} T(x; r, s_r)\, dx + \int_{k_2}^{k_1 + 2\mathbf{u}(1-q_1)} T(x; r, s_r)\, dx + \int_{k_1}^{k_2} T(x; r, s_r)\, dx. \tag{50}$$

Using Lemma VII.9 with $x_1 = k_1$, $x_2 = k_2$, and substituting (48) and (49) into (50), we get

$$\mathcal{E}(r) \geq \mathbf{u}\left(q_2 - \frac{k_2 - k_1}{2\mathbf{u}}\right)^2 + \mathbf{u}\left((1-q_1) - \frac{k_2 - k_1}{2\mathbf{u}}\right)^2 + (1-q_1)(\mathbf{h} - k_1) + q_2(k_2 - \mathbf{h}).$$

Let $k_2 - k_1 := w\mathbf{u}(q_2 - q_1)$, where $1 \leq w \leq 2$. Suppose $1 - q_1 > q_2$ (a similar argument will hold when $1 - q_1 \leq q_2$). Then

$$(1-q_1)(\mathbf{h} - k_1) + q_2(k_2 - \mathbf{h}) \geq q_2(k_2 - k_1) = \mathbf{u}w(q_2 - q_1)q_2,$$

which implies

$$\mathcal{E}(r) \geq \mathbf{u}\left(q_2 - w\frac{q_2 - q_1}{2}\right)^2 + \mathbf{u}\left((1-q_1) - w\frac{q_2 - q_1}{2}\right)^2 + \mathbf{u}w(q_2 - q_1)q_2.$$

The minimum value of

$$\mathbf{u}\left(q_2 - w\frac{q_2-q_1}{2}\right)^2 + \mathbf{u}\left((1-q_1) - w\frac{q_2-q_1}{2}\right)^2 + \mathbf{u}w(q_2-q_1)q_2$$

is $\frac{1}{2}$ at $q_1 = \frac{1}{2}, q_2 = \frac{1}{2}$, and $w = 2$. Hence, $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$.
Case 4. $q_2 \geq q_1$, $k_2 - k_1 > 2\mathbf{u}(q_2 - q_1)$, and $l_2 \leq k_1 < k_2 \leq l_1$.
Using the same argument used to derive (48) and (49) in Case 3, we get

$$T(x; r, s_r) \geq q_2 \qquad \text{for} \quad x \in [k_1, \mathbf{h}] \tag{51}$$

and

$$T(x; r, s_r) \geq (1 - q_1) \qquad \text{for} \quad x \in (\mathbf{h}, k_2]. \tag{52}$$

From the definition of $\mathcal{E}(r)$, we have

$$\mathcal{E}(r) := \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{k_1 - 2\mathbf{u}q_1}^{k_1} T(x; r, s_r)\, dx + \int_{k_2}^{k_2 + 2\mathbf{u}(1-q_2)} T(x; r, s_r)\, dx + \int_{k_1}^{k_2} T(x; r, s_r)\, dx. \tag{53}$$

Using Lemma VII.8 with $x_1 = k_1$, $x_2 = k_2$, and substituting (51) and (52) into (53), we get

$$\mathcal{E}(r) \geq \mathbf{u}q_1^2 + \mathbf{u}(1-q_2)^2 + (1-q_1)(\mathbf{h} - k_1) + q_2(k_2 - \mathbf{h}).$$

Suppose $q_2 > 1 - q_1$ (a similar argument will hold when $1 - q_1 \leq q_2$). Then

$$(1-q_1)(\mathbf{h} - k_1) + q_2(k_2 - \mathbf{h}) \geq (1-q_1)|k_2 - k_1| > 2\mathbf{u}(q_2 - q_1)(1 - q_1),$$

which implies

$$\mathcal{E}(r) > \mathbf{u}\left(q_1^2 + (1-q_2)^2 + 2(q_2 - q_1)(1 - q_1)\right).$$

The minimum value of $q_1^2 + (1-q_2)^2 + 2(q_2 - q_1)(1 - q_1)$ is $\frac{1}{2}$ at $q_1 = q_2 = \frac{1}{2}$. Hence, $\mathcal{E}(r) > \frac{\mathbf{u}}{2}$.
Case 5. $q_2 \geq q_1$, $k_2 - k_1 > \mathbf{u}(q_2 - q_1)$, and $l_1 \leq k_1 < k_2 \leq l_2$.
Here we will bound $T(x; r, s_r)$, To do so, we first show that $\bar{q}_1 > q_1$ and $\bar{q}_2 < q_2$. From Lemma VII.7 with $i = k_1$, $j = l_1$, it follows that $s_r(l_1) = l_1$ and $s_r(l_2) = l_2$. Recall from the notation mentioned at the start of the proof that $q(l_1) := \bar{q}_1$ and $q(l_2) := \bar{q}_2$. We now show that $l_2 - l_1 \geq 2\mathbf{u}(\bar{q}_2 - \bar{q}_1)$. Let $s(i) := l_2$ be a sender strategy. Since s_r is the worst-case lazy best response, we have $\mathbb{U}(l_1, r, s_r) \geq \mathbb{U}(l_1, r, s)$. This implies

$$2\mathbf{u}\bar{q}_1 \geq 2\mathbf{u}\bar{q}_2 - l_2 + l_1 \qquad \text{or} \qquad 2\mathbf{u}(\bar{q}_2 - \bar{q}_1) \leq l_2 - l_1$$

as claimed. Now let $\bar{s}(i) := k_1$ be another sender strategy. Then it follows that $\mathbb{U}(k_1, r, s_r) \geq \mathbb{U}(k_1, r, \bar{s})$, which implies

$$2\mathbf{u}\bar{q}_1 - k_1 + l_1 > 2\mathbf{u}q_1 \qquad \text{or} \qquad \bar{q}_1 > q_1. \tag{54}$$

This is a strict inequality since k_1 is the cheaper option for k_1 compared to l_1. A similar reasoning shows that

$$\bar{q}_2 < q_2. \tag{55}$$

Next, we bound $s_r(x)$ for $x \in [m_1, m_2]$ where m_1 and m_2 are defined as:

$$m := \frac{k_1 + k_2}{2}, \quad m_1 := m - \mathbf{u}(q_2 - \bar{q}_1), \quad m_2 := m + \mathbf{u}(\bar{q}_2 - q_1).$$

Claim. $m_1 \geq k_1$ and $m_2 < k_2$.
Let $m \leq \mathbf{h}$ (a similar argument holds when $m > \mathbf{h}$). We now show that $m_1 \geq k_1$. Let $x \in (m_1, \mathbf{h}]$. Suppose $s_r(x) \leq x$, then $s_r(x) \leq k_1$. This is because from the definition of k_1, we have $r(x) = \Delta$ for $x \in [k_1, \mathbf{h}]$. But $h(x) = h(s_r(x)) = -$, therefore $s_r(s_r(x)) = s_r(x)$ from Lemma VII.7. Furthermore, we have $h(l_1) = h(k_1) = h(s_r(x)) = -$. Therefore, from Lemma VII.7 with $i = k_1$, $j = l_1$, and $w = s_r(x)$,

we have $s_r(s_r(x)) = s_r(x) = l_1$. Now consider the sender strategy $s(x) := k_2$. From the definition of the worst-case lazy best response, we have $\mathbb{U}(x, r, s_r) \geq \mathbb{U}(x, r, s)$, which implies

$$2\mathbf{u}\bar{q}_1 - x + l_1 \geq 2\mathbf{u}q_2 - k_2 + x.$$

Since $x > m_1$, we have

$$2\mathbf{u}\bar{q}_1 - x + l_1 < 2\mathbf{u}\bar{q}_1 - m_1 + l_1 \qquad \text{and} \qquad 2\mathbf{u}q_2 - k_2 + x > 2\mathbf{u}q_2 - k_2 + m_1.$$

Hence, we have

$$2\mathbf{u}\bar{q}_1 - m_1 + l_1 > 2\mathbf{u}q_2 - k_2 + m_1,$$

which implies

$$2\mathbf{u}(\bar{q}_1 - q_2) > 2m_1 - l_1 - k_2.$$

Substituting for m_1 and simplifying, we get

$$2\mathbf{u}(\bar{q}_1 - q_2) > k_1 + k_2 - 2\mathbf{u}(q_2 - \bar{q}_1) - l_1 - k_2,$$

which implies

$$0 > k_1 - l_1.$$

This is a contradiction since $k_1 \geq l_1$. Therefore, our assumption that $s_r(x) \leq x$ is false. Hence,

$$s_r(x) > x \qquad \text{for} \qquad x \in (m_1, \mathbf{h}]. \tag{56}$$

If $m_1 < k_1$, then by taking $x = k_1$, we get $s_r(x) = l_1 > k_1$, which is false as $l_1 \leq k_1$ by case assumption. Hence, $m_1 \geq k_1$. A similar reasoning shows

$$s_r(x) < x \qquad \text{for} \qquad x \in (\mathbf{h}, m_2)$$

and that $m_2 \leq k_2$.

Next, we calculate the error in the interval (m_1, m_2). Let $x \in (m_1, \mathbf{h}]$. Consider the sender strategy $s(x) := k_2$. Since s_r is the worst-case lazy best response, we have $\mathbb{U}(x, r, s_r) \geq \mathbb{U}(x, r, s)$, which implies

$$2\mathbf{u}q(s_r(x)) - s_r(x) + x \geq 2\mathbf{u}q_2 - k_2 + x \qquad \text{or} \qquad q(s_r(x)) \geq q_2,$$

where we have used (56) and the fact that $r(x) = \Delta$ for $x \in (\mathbf{h}, \mathbf{h}^+)$ (which implies $s_r(x) \geq k_2$). Thus,

$$T(x; r, s_r) \geq q_2 \qquad \text{for} \qquad x \in (m_1, \mathbf{h}]. \tag{57}$$

Using similar reasoning for $x \in (\mathbf{h}, m_2)$, we have $q(s_r(x)) \leq q_1$, and hence

$$T(x; r, s_r) \geq 1 - q_1 \qquad \text{for} \qquad x \in (\mathbf{h}, m_2]. \tag{58}$$

From the definition of $\mathcal{E}(r)$, we have

$$\mathcal{E}(r) := \int_{\mathcal{X}} T(x; r, s_r)\, dx \geq \int_{l_1 - 2\mathbf{u}q_1}^{l_1} T(x; r, s_r)\, dx + \int_{l_2}^{l_2 + 2\mathbf{u}(1-q_2)} T(x; r, s_r)\, dx + \int_{m_1}^{m_2} T(x; r, s_r)\, dx. \tag{59}$$

Using Lemma VII.8 with $x_1 = l_1$, $x_2 = l_2$, and substituting (57) and (58) in (59), we get

$$\mathcal{E}(r) \geq \mathbf{u}\bar{q}_1^2 + \mathbf{u}(1 - \bar{q}_2)^2 + (\mathbf{h} - m_1)q_2 + (m_2 - \mathbf{h})(1 - q_1). \tag{60}$$

Suppose $q_2 < 1 - q_1$ (a similar argument will hold when $1 - q_1 < q_2$). Then, we have

$$(\mathbf{h} - m_1)q_2 + (m_2 - \mathbf{h})(1 - q_1) \geq (m_2 - m_1)q_2 = \mathbf{u}(\bar{q}_2 - \bar{q}_1 + q_2 - q_1)q_2. \tag{61}$$

Using (54), (55), and (61) in (60), and using $q_2 - q_1 \geq 0$, we obtain

$$\mathcal{E}(r) \geq \mathbf{u}\bar{q}_1^2 + \mathbf{u}(1 - \bar{q}_2)^2 + 2\mathbf{u}(\bar{q}_2 - \bar{q}_1)\bar{q}_2.$$

The minimum of the expression $\mathbf{u}\bar{q}_1^2 + \mathbf{u}(1-\bar{q}_2)^2 + 2\mathbf{u}(\bar{q}_2 - \bar{q}_1)\bar{q}_2$ is $\frac{1}{2}$, and it occurs when $\bar{q}_1 = \bar{q}_2 = \frac{1}{2}$. Hence, we conclude that

$$\mathcal{E}(r) = \frac{\mathbf{u}}{2}.$$

Case 6. $q_2 \geq q_1$, $k_2 - k_1 > \mathbf{u}(q_2 - q_1)$, and either $k_2 \leq \min\{l_1, l_2\}$ or $k_1 \geq \max\{l_1, l_2\}$.
We will show that this case does not arise. We will prove this for $k_1 \geq \max\{l_1, l_2\}$. The proof for $k_2 \leq \min\{l_1, l_2\}$ is similar. Consider the following sender strategies:

$$\bar{s}(i) := l_1, \quad s'(i) := l_2.$$

Since s_r is the worst-case lazy best response, we have

$$\mathbb{U}(k_1; r, s_r) \geq \mathbb{U}(k_1; r, s') \qquad \text{and} \qquad \mathbb{U}(k_2; r, s_r) \geq \mathbb{U}(k_2; r, \bar{s}),$$

which implies

$$2\mathbf{u}\bar{q}_1 - k_1 + l_1 \geq 2\mathbf{u}\bar{q}_2 - k_1 + l_2 \qquad \text{or} \qquad 2\mathbf{u}(\bar{q}_1 - \bar{q}_2) \geq l_2 - l_1 \tag{62}$$

and

$$2\mathbf{u}(1-\bar{q}_2) - k_2 + l_2 \geq 2\mathbf{u}(1-\bar{q}_1) - k_2 + l_1 \qquad \text{or} \qquad 2\mathbf{u}(\bar{q}_1 - \bar{q}_2) \geq l_1 - l_2, \tag{63}$$

respectively. Combining (62) and (63), we get $l_1 = l_2$. Now, consider the strategy $s(i) = k_1$. Since s_r is the worst-case lazy best response, we have

$$\mathbb{U}(k_1; r, s_r) > \mathbb{U}(k_1; r, s) \qquad \text{and} \qquad \mathbb{U}(k_2; r, s_r) > \mathbb{U}(k_2; r, s).$$

The inequality is strict for the sender, k_1 is the less costly (lazy) option compared to l_1 and l_2 at k_1 and k_2, respectively. This implies

$$2\mathbf{u}\bar{q}_1 - k_1 + l_1 > 2\mathbf{u}q_1 \qquad \text{or} \qquad \bar{q}_1 > q_1 \tag{64}$$

and

$$2\mathbf{u}(1-\bar{q}_1) - k_2 + l_2 > 2\mathbf{u}(1-q_1) - k_2 + k_1 \qquad \text{or} \qquad 2\mathbf{u}(q_1 - \bar{q}_1) > k_1 - l_2 = k_1 - l_1 \qquad \text{or} \qquad q_1 > \bar{q}_1. \tag{65}$$

Combining (65) and (64), we get $q_1 > q_1$, which is a contradiction. Hence, this case does not arise. ■

D. Omitted Calculations in Section III.E

Calculation of Example 3: We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$.
Case 1: $x \in [0, \mathbf{h}_\epsilon^-)$.
Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = -|x - x'| = 0.$$

If $x' \neq x$ and $x' \in [0, \mathbf{h}_\epsilon^-)$, then

$$\mathbb{U}(x; r, s_r) = -|x - x'| < 0.$$

If $x' \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x' + x = \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x' - \mathbf{h}_\epsilon^-) - x' + x < x' - \mathbf{h}_\epsilon^- - x' + x = -\mathbf{h}_\epsilon^- + x \leq 0,$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x' + x < \mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- - x' + x < -\mathbf{h}_\epsilon^- + x < 0,$$

where we have used the fact that $\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- > 2\mathbf{u}$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Hence $s_r(x) = x$.
Case 2: $x \in [\mathbf{h}_\epsilon^-, \mathbf{h}]$.
Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x + x' = \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x' - \mathbf{h}_\epsilon^-) - x + x' = \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in [0, \mathbf{h}_\epsilon^-)$, then

$$\mathbb{U}(x; r, s_r) = -x + x' < 0.$$

If $x' \in [\mathbf{h}_\epsilon^-, x]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x + x' < \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (x, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x' + x < \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x' + x < x - \mathbf{h}_\epsilon^- - 2\epsilon < \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we used the fact that $\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- = 2\mathbf{u} + 2\epsilon$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-)$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-)$. Hence $s_r(x) = x$.
Case 3: $x \in [\mathbf{h}, \mathbf{h}_\epsilon^+]$.
Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x' + x = \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-) + 2\mathbf{u},$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in [0, \mathbf{h}_\epsilon^-)$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x + x' \leq 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-).$$

If $x' \in (\mathbf{h}_\epsilon^-, x]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x + x' < 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (x, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x' + x < \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-) + 2\mathbf{u},$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = -x' + x < 0.$$

Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 2\mathbf{u} - \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-)$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 2\mathbf{u} - \frac{\mathbf{u}}{\mathbf{u}+\epsilon}(x - \mathbf{h}_\epsilon^-)$. Hence $s_r(x) = x$.
Case 4: $x \in (\mathbf{h}_\epsilon^+, 1]$.
Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = -|x' - x| = 0.$$

If $x' \in [0, \mathbf{h}_\epsilon^-)$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - \mathbf{h}_\epsilon^+ + \mathbf{h}_\epsilon^- \leq -2\epsilon < 0$$

If $x' \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x + x' = 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x' - \mathbf{h}_\epsilon^-) - x + x'$$
$$< 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(x' - \mathbf{h}_\epsilon^- + x - x') \leq 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u}+\epsilon}(\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^-) = 0.$$

where we have used (9) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = -|x' - x| < 0.$$

Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Hence $s_r(x) = x$.

Hence, by combining all the cases, we conclude that $s_r(x) = x$. Therefore,

$$T(x; r, s_r) = \mathbb{P}_r\{Y \neq h(x) \mid X' = s_r(x)\} = \begin{cases} 0 & \text{for} \quad x \in [0, \mathbf{h}_\epsilon^-) \cup (\mathbf{h}_\epsilon^+, 1], \\ q_\epsilon(x) & \text{for} \quad x \in [\mathbf{h}_\epsilon^-, \mathbf{h}], \\ 1 - q_\epsilon(x) & \text{for} \quad x \in (\mathbf{h}, \mathbf{h}_\epsilon^+]. \end{cases}$$

Since $\mathcal{E}(r) = \int_{\mathcal{X}} T(x)\, dx$, it follows that $\mathcal{E}(r) = \frac{\mathbf{u}+\epsilon}{2}$.

E. Omitted Proofs of Section IV.A

Proof VII.5. (Proof of Proposition IV.1) Let $r^\star$ be a Stackelberg equilibrium, and define:

$$\theta^{(n)} := \sup(E(r^\star)^c \cap O_n),$$
$$\theta_{(n)} := \inf(E(r^\star)^c \cap O_n),$$
$$\Theta_n := [\theta_{(n)}, \theta^{(n)}].$$

This implies $E(r^\star)^c \subseteq \cup_{n=1}^{d-1} \Theta_n$. We demonstrate that with this choice of Θ_n, the receiver strategy $r(x; \Theta_1, \cdots, \Theta_{d-1})$ given in (12) is a Stackelberg equilibrium. Consider the following cases for Θ_n's.

Case 1. All Θ_n's are empty.
This case will not happen because if all Θ_n's are empty, then $r^\star$ misclassifies all feature vectors. However, a majority-label (described in Section II) receiver strategy will correctly classify at least half of them, contradicting that $r^\star$ is a Stackelberg equilibrium. Thus, at least one Θ_n is non-empty.

Case 2. For all odd (or even) n, Θ_n is empty.
Suppose $\Theta_n = \emptyset$ for all odd n. For even n, denote the corresponding Θ_n as Θ_{2k}, where $k \in \left\{1, \ldots, \left\lfloor \frac{d-1}{2} \right\rfloor\right\}$. Since $r(x) = \Delta$ for all $x \in \mathcal{X} \setminus \bigcup_k \Theta_{2k}$, it follows that for $x \in \mathcal{X}$, $s_r(x) \in \bigcup_k \Theta_{2k}$. Thus, $r(s_r(x)) \equiv h(s_r(x))$ from (12). From (11), we have $h(t) = -$ for $t \in \cup_k O_{2k}$. Therefore, $r(s_r(x)) \equiv -$ and $r(s_r(x)) = h(x)$ for $x \in \cup_k O_{2k}$. Hence

$$E(r)^c \supseteq \bigcup_{2k} O_k \supseteq \bigcup_{2k} \Theta_k.$$

Since $r^\star$ is a Stackelberg equilibrium, it follows that $E(r^\star) \subseteq \bigcup_k \Theta_{2k}$ by construction. Consequently, $E(r)^c = E(r^\star) = \bigcup_k O_{2k}$, and r is a Stackelberg equilibrium. A similar argument applies to show that r is a Stackelberg equilibrium when n for which $\Theta_n = \emptyset$ are all even.

Case 3. Their exist an odd k and even l such that $\Theta_k, \Theta_l \neq \emptyset$.
We will first show that $\theta_{(\max\{l,k\})} - \theta^{(\min\{l,k\})} \geq \mathbf{u}$. Suppose $k < l$ (similar reasoning holds when $k > l$). Suppose $\theta_{(l)} - \theta^{(k)} < \mathbf{u}$. Let $\epsilon > 0$ be such that it satisfies $0 < 2\epsilon < \mathbf{u} - (\theta_{(l)} - \theta^{(k)})$. Let $\theta^{(k')}, \theta_{(l')} \in \mathcal{X}$ be feature vectors such that $\theta^{(k')}, \theta_{(l')} \in E(r^\star)^c$, $0 \leq \theta_{(l')} - \theta_{(l)} < \epsilon$, and $0 \leq \theta^{(k)} - \theta^{(k')} < \epsilon$. Such feature vectors exist by the definition of $\theta^{(k)}$ and $\theta_{(l)}$. It follows that

$$\theta_{(l')} - \theta^{(k')} < 2\epsilon + \theta_{(l)} - \theta^{(k)} < \mathbf{u}.$$

Note that $h(\theta_{(l')}) = -$ and $h(\theta^{(k')}) = +$. Suppose $\mathbb{U}(\theta^{(k')}; r^\star, s_{r^\star}) \leq \mathbb{U}(\theta_{(l')}; r^\star, s_{r^\star})$ (similar arguments applies when $\mathbb{U}(\theta^{(k')}; r^\star, s_{r^\star}) > \mathbb{U}(\theta_{(l')}; r^\star, s_{r^\star})$). Let s be a sender strategy defined as $s(x) := s_{r^\star}(\theta_{(l')})$ for all $x \in \mathcal{X}$. Then

$$\mathbb{U}(\theta^{(k')}; r^\star, s) = \mathbf{u} - |\theta^{(k')} - s_{r^\star}(\theta_{(l')})| \geq \mathbf{u} - |\theta_{(l')} - \theta^{(k')}| - |\theta_{(l')} - s_{r^\star}(\theta_{(l')})| > -|\theta_{(l')} - s_{r^\star}(\theta_{(l')})| = \mathbb{U}(\theta_{(l')}; r^\star, s_{r^\star}).$$

This implies $\mathbb{U}(\theta^{(k')}; r^\star, s) > \mathbb{U}(\theta^{(k')}; r^\star, s_{r^\star})$, which contradicts the fact that $s_{r^\star}$ is the worst-case best response. Therefore, $\theta_{(l)} - \theta^{(k)} \geq \mathbf{u}$, proving the claim.

Now consider the receiver strategy $r(x; \{\Theta_n\})$. We calculate its worst-case best response s_r. Let $t \in \Theta_i$. Since $r(x) = \Delta$ for $x \notin \cup_n \Theta_n$, we have $s_r(x) \in \cup_n \Theta_n$. Suppose i is even (similar arguments hold for odd i) and $s_r(t) \geq t$ (a similar arguments hold for $s_r(t) < t$). The sender's payoff is:

$$\mathbb{U}(t; r, s_r) = \begin{cases} -s_r(t) + t & \text{if} \quad s_r(t) \in \Theta_j,\ j \quad \text{even} \quad , \\ \mathbf{u} - s_r(t) + t & \text{if} \quad s_r(t) \in \Theta_j,\ j \quad \text{odd} \quad . \end{cases}$$

If j is even, $\mathbb{U}(t; r, s_r) = -s_r(t) + t \leq 0$, with the maximum value 0 at $s_r(t) = t$. If j is odd, then using $\theta_{(l)} - \theta^{(k)} \geq \mathbf{u}$, we get $s_r(t) - t \geq \mathbf{u}$, and hence $\mathbb{U}(t; r, s_r) \leq 0$. Thus $s_r(t) = t$ gives the maximum payoff to the sender. Therefore, $r(s_r(t)) = h(t)$ for all $t \in \cup_n \Theta_n$. This gives $E(r)^c \supseteq \cup_n \Theta_n \supseteq E(r^\star)$, proving that r is a Stackelberg equilibrium.

■

F. Omitted Proofs of Section IV.B

Proof VII.6. (Proof of Lemma IV.3) Let $r_1 := r^\star_{f,D},\ r_2 := r^\star_{f,D(S)}$. Let $\psi(x)$ be the distribution function of the probability law D and $\psi_m(x)$ be the distribution function of the empirical probability law $D(S)$ for a sample S, i.e.,

$$\psi_m(x) := \frac{1}{m} \sum_{x_i \in S} \mathbb{1}_{[0,x]}.$$

According to the Dvoretzky–Kiefer–Wolfowitz inequality ([16]), for any $(\epsilon, \delta) \in (0,1)^2$, there exists an $\bar{m}(\epsilon, \delta)$ such that for all $m > \bar{m}(\epsilon, \delta)$, we have:

$$D^{\otimes m}\big(\sup_{x \in \mathcal{X}} |\psi(x) - \psi_m(x)| < \epsilon\big) > 1 - 2e^{-2m\epsilon^2}. \tag{66}$$

Let

$$\mathcal{S} := \{S : \sup_{x \in \mathcal{X}} |\psi(x) - \psi_m(x)| < \epsilon\}. \tag{67}$$

Then from (66), we know that $D^{\otimes m}(\mathcal{S}) > 1 - 2e^{-2em\epsilon^2}$. Fix a $S \in \mathcal{S}$ and let ψ_m be the corresponding empirical distribution function. Consider the receiver strategies r_1 and r_2. From Proposition IV.1, we have $r_1(x) \equiv r_1(x; \Theta_1^{(1)}, \cdots, \Theta_{d-1}^{(1)})$ and $r_2(x) \equiv r_2(x; \Theta_1^{(2)}, \cdots, \Theta_{d-1}^{(2)})$ for some $\Theta_n^{(i)}$, where $i = 1, 2$ and $n \in \{0, 1, \cdots, d-1\}$, and where $\Theta_n^{(i)} := [\theta_{(i,n)}, \theta^{(i,n)}]$ is a closed interval. Let $E_n^{(i)} := (\theta^{(i,n)}, \theta_{(i,n+1)})$. From Lemma IV.1, $E_n^{(i)}$ is the set of feature vectors misclassified by r_i near the decision boundary n. From the definition of ψ_m and ψ, we have

$$D(S)(E_n^{(i)}) = \psi_m(\theta^{(i,n+1)}) - \psi_m(\theta_{(i,n)}), \quad D(E_n^{(i)}) = \psi(\theta^{(i,n+1)}) - \psi(\theta_{(i,n)}), \quad i \in \{1,2\}, n \in \{1, \cdots, d-1\}.$$

Since $S \in \mathcal{S}$, using (67), we get

$$|D(S)(E_n^{(i)}) - D(E_n^{(i)})| \leq |\psi_m(\theta^{(i,t+1)}) - \psi(\theta^{(i,n+1)})| + |\psi(\theta_{(i,n)}) - \psi_m(\theta_{(i,n)})| < 2\epsilon,$$
$$\text{for all} \quad i \in \{1,2\}, n \in \{1, \cdots, d-1\}.$$

Summing over all $d-1$ decision boundaries, and noting that $\mathcal{E}(r_i; f, D) = \sum_n D(E_n^{(i)})$ and $\mathcal{E}(r_i; f, D(S)) = \sum_n D(S)(E_n^{(i)})$, we get

$$|\mathcal{E}(r_2; f, D) - \mathcal{E}(r_1; f, D)| < 2(d-1)\epsilon.$$

This holds true for all $S \in \mathcal{S}$. But we have $D^{\otimes m}(\mathcal{S}) > 1 - 2e^{-2m\epsilon^2}$. Therefore, we get

$$D^{\otimes m}(|\mathcal{E}(r_2; f, D) - \mathcal{E}(r_1; f, D)| < 2(d-1)\epsilon) > 1 - 2e^{-2m\epsilon^2}.$$

as required.

■

Proof VII.7. (Proof of Lemma IV.4) Let $r_1 := r^\star_{h,D}$, $r_2 := r^\star_{A(S),D}$, and $E(r, f) := E(r, s_r, f)$. Since VC dim$(\mathcal{F}) < \infty$, $\mathcal{F}$ is PAC learnable from Theorem IV.1. Furthermore, from Theorem IV.1 we have

$$D^{\otimes m}(L_D(A(S); h) < \epsilon) > 1 - \delta, \tag{68}$$

where $m \geq \underline{\mathrm{m}}(\epsilon, \delta)$ and $\underline{\mathrm{m}}(\epsilon, \delta)$ satisfies

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon} \leq \underline{\mathrm{m}}(\epsilon, \delta) \leq C_2 \frac{d \log\left(\frac{1}{\epsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\epsilon}.$$

Here $C_1, C_2 \geq 0$ are constants and d is the VC dimension of $\mathcal{F}$. Let

$$\mathcal{S} := \{S : L_D(A(S); h) < \epsilon\}. \tag{69}$$

From (68), we know that $D^{\otimes m}(\mathcal{S}) > 1 - \delta$. Fix a $S \in \mathcal{S}$. Define $W(S) := \{x \mid h(x) \neq A(S)(x)\}$. For a receiver strategy r and (an arbitrary) true function f (recall from Definition II.1 that s_r will depends upon the true function f v.i.a. utility function), the classification error set $E(r; A(S))$ (defined in (3)) can be decomposed as

$$E(r; A(S)) = (E(r; A(S)) \cap W(S)) \cup (E(r; A(S)) \cap W(S)^c).$$

Since $S \in \mathcal{S}$, using (69), we get $L(A(S); h) < \epsilon$, and therefore it follows that $D(W(S)) < \epsilon$. Hence, we get

$$\mathcal{E}(r; A(S), D) < \epsilon + D(E(r; A(S)) \cap W(S)^c).$$

From the definition of $W(S)$, $h(x) = A(S)(x)$ for all $x \in W(S)^c$. Therefore,

$$\begin{aligned} E(r; A(S)) \cap W(S)^c &= \{x : r(s_r(x)) = A(S)(x)\} \cap \{x : h(x) = A(S)(x)\} = \{x : r(s_r(x)) = A(S)(x) = h(x)\} \\ &= \{x : h(x) = A(S)(x)\} \cap \{x : r(s_r(x)) = h(x)\} = E(r; h) \cap W(S)^c, \end{aligned}$$

and thus

$$\mathcal{E}(r; A(S), D) < \epsilon + D(E(r; h)) = \epsilon + \mathcal{E}(r; h, D). \tag{70}$$

Similarly, we can decompose the classification error set $E(r; h)$ as

$$E(r; h) := (E(r; h) \cap W(S)) \cup (E(r; h) \cap W(S)^c).$$

Repeating the argument used for deriving (70), we obtain

$$\mathcal{E}(r; h, D) < \epsilon + \mathcal{E}(r; A(S), D). \tag{71}$$

Combining (70) and (71), we get

$$|\mathcal{E}(r; h, D) - \mathcal{E}(r, A(S), D)| < \epsilon.$$

This holds true for all $S \in \mathcal{S}$. But we have $D^{\otimes m}(\mathcal{S}) > 1 - \delta$, therefore we get

$$D^{\otimes m}(|\mathcal{E}(r; h) - \mathcal{E}(r; A(S))| < \epsilon) > 1 - \delta \quad \text{for all receiver strategy} \quad r \quad \text{and true function} \quad f.$$

This proves the (a) part. Using part (a), we now establish

$$D^{\otimes m}(\mathcal{E}(r_2;h)) - \mathcal{E}(r_1;h) < 2\epsilon) > 1-\delta,$$

where h is the ground true function of r_1 and $A(S)$ is the estimated true function of r_2. This is the same as the claim (given in (13)) we want to prove. Substituting $r = r_1$ and $f = h$ (ground true function) in (70), we get

$$\mathcal{E}(r_1; A(S), D) < \epsilon + \mathcal{E}(r_1; h, D). \tag{72}$$

Substituting $r = r_2$ and $f = A(S)$ (estimated true function) in (71), we get

$$\mathcal{E}(r_2; h, D) < \epsilon + \mathcal{E}(r_2; A(S), D). \tag{73}$$

Consider the following:

$$\mathcal{E}(r_2;h,D) - \mathcal{E}(r_1;h,D) = \underbrace{\big(\mathcal{E}(r_2;h,D) - \mathcal{E}(r_2;A(S),D)\big)}_{a} + \underbrace{\big(\mathcal{E}(r_2;A(S),D) - \mathcal{E}(r_1;A(S),D)\big)}_{b} + \underbrace{\big(\mathcal{E}(r_1;A(S),D) - \mathcal{E}(r_1;h,D)\big)}_{c}.$$

From the definition of r_2, we know $b = \mathcal{E}(r_2;A(S),D) - \mathcal{E}(r_1;A(S),D) \leq 0$ as $\mathcal{E}(r_2;A(S),D)$ is the Stackelberg error. From (73), we have $a < \epsilon$ and using (72), we have $c < \epsilon$. Therefore we get

$$\mathcal{E}(r_2;h,D) - \mathcal{E}(r_1;h,D) < 2\epsilon.$$

This holds true for all $S \in \mathcal{S}$. But we have $D^{\otimes m}\mathcal{S} > 1-\delta$. Therefore we get

$$D^{\otimes m}(\mathcal{E}(r_2;h,D) - \mathcal{E}(r_1;h,D)) < 2\epsilon$$

as claimed in part (b).

■

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